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Carbajal

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(54) **SYSTEMS, METHODS, AND DEVICES FOR ELECTRONIC SPECTRUM MANAGEMENT FOR IDENTIFYING OPEN SPACE**

(58) **Field of Classification Search**

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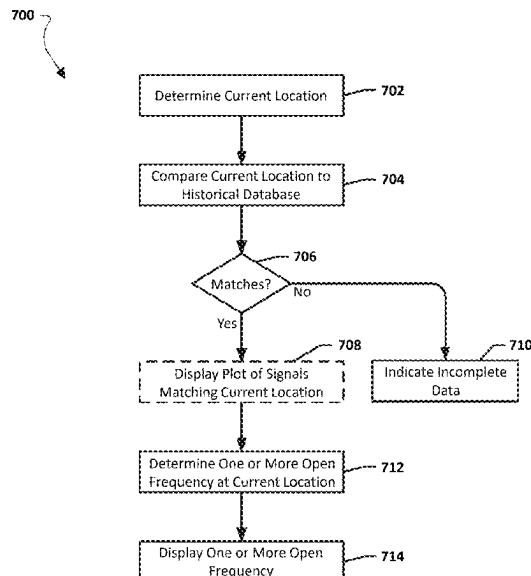
ABSTRACT

Systems, methods, and apparatus are provided for automated identification of open space in a wireless communications spectrum, by identifying sources of signal emission in the spectrum by automatically detecting signals, analyzing signals, comparing signal data to historical and reference data, creating corresponding signal profiles, and determining information about the open space based upon the measured and analyzed data in near real-time.

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- (58) **Field of Classification Search**
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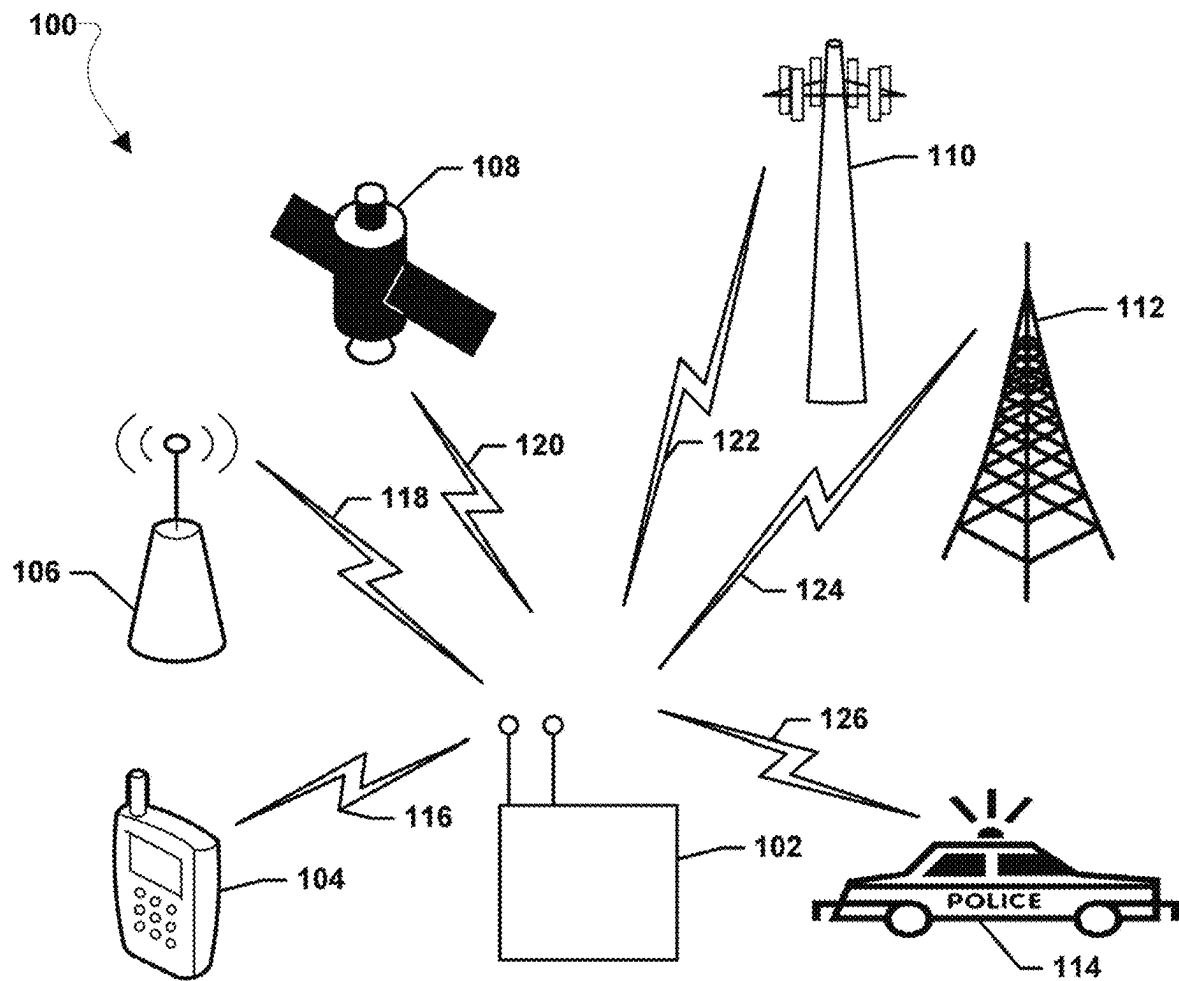


FIG. 1

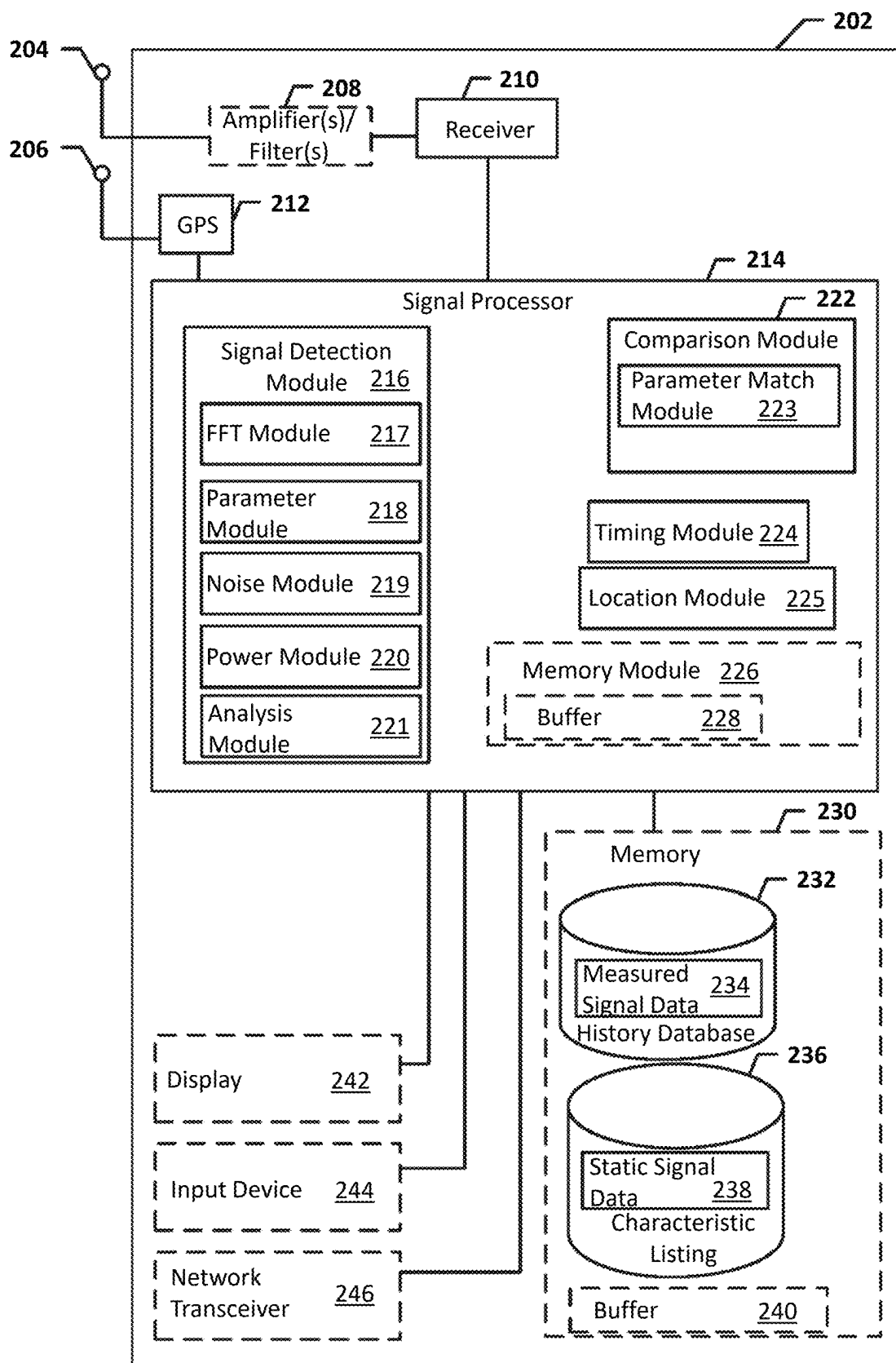


FIG. 2A

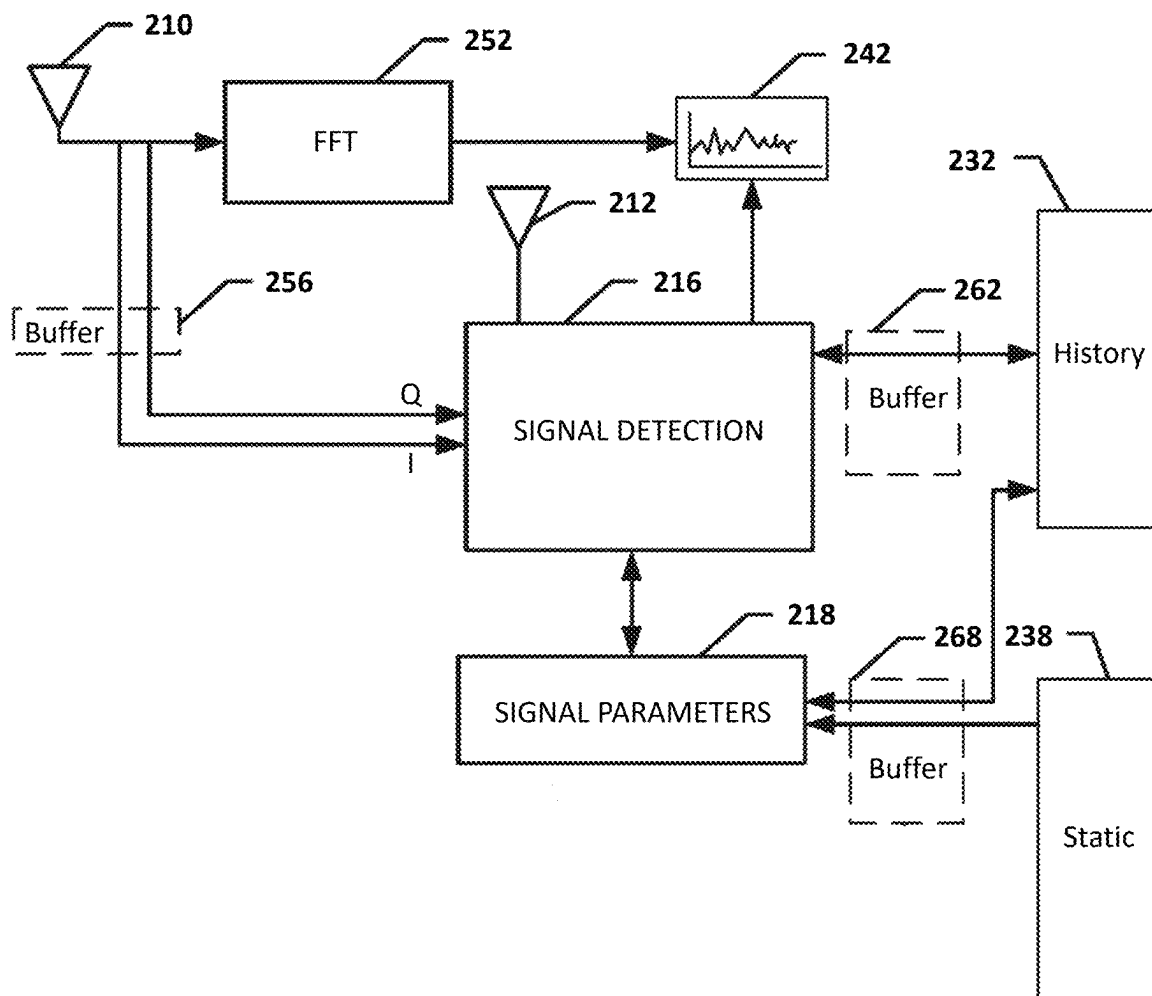


FIG. 2B

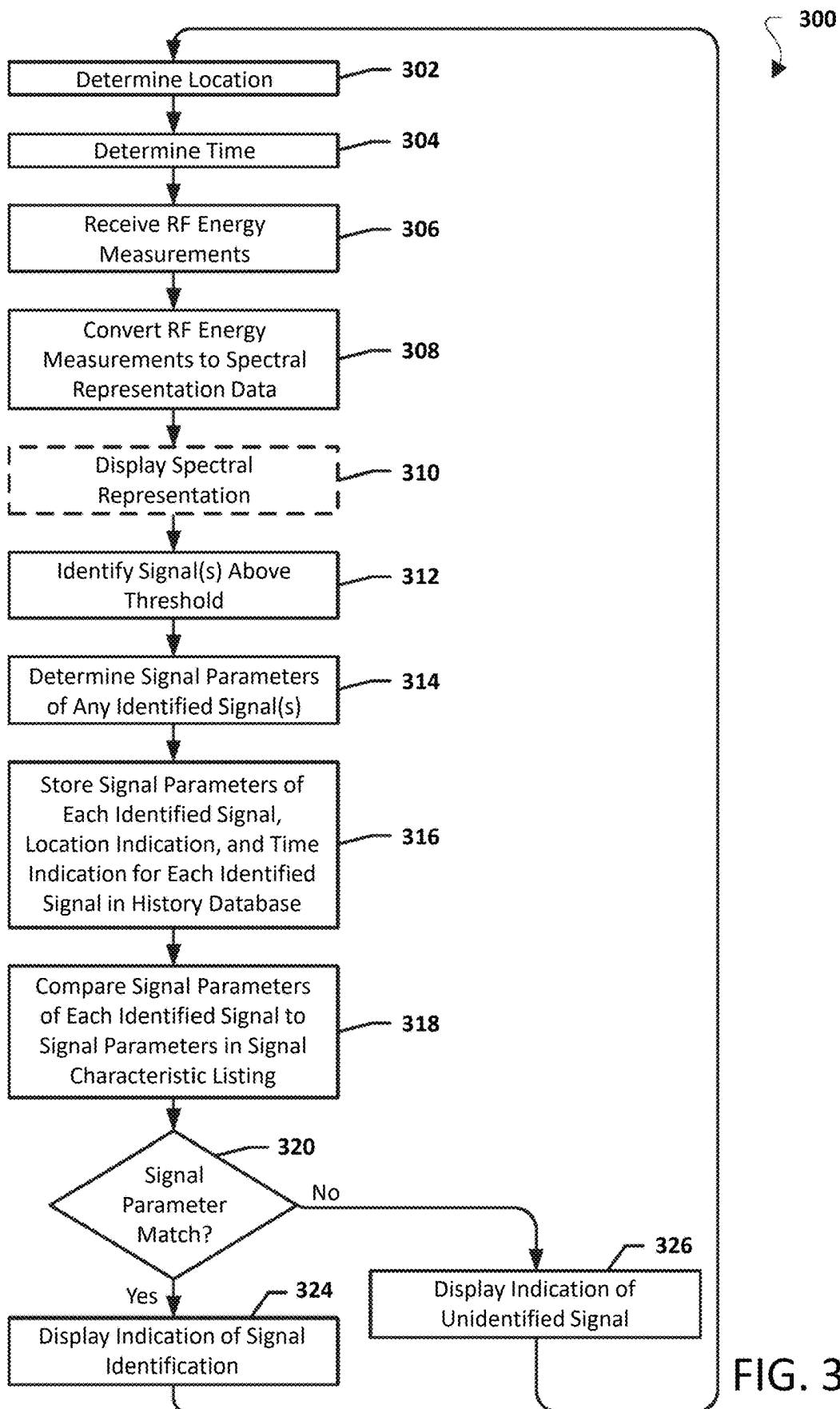
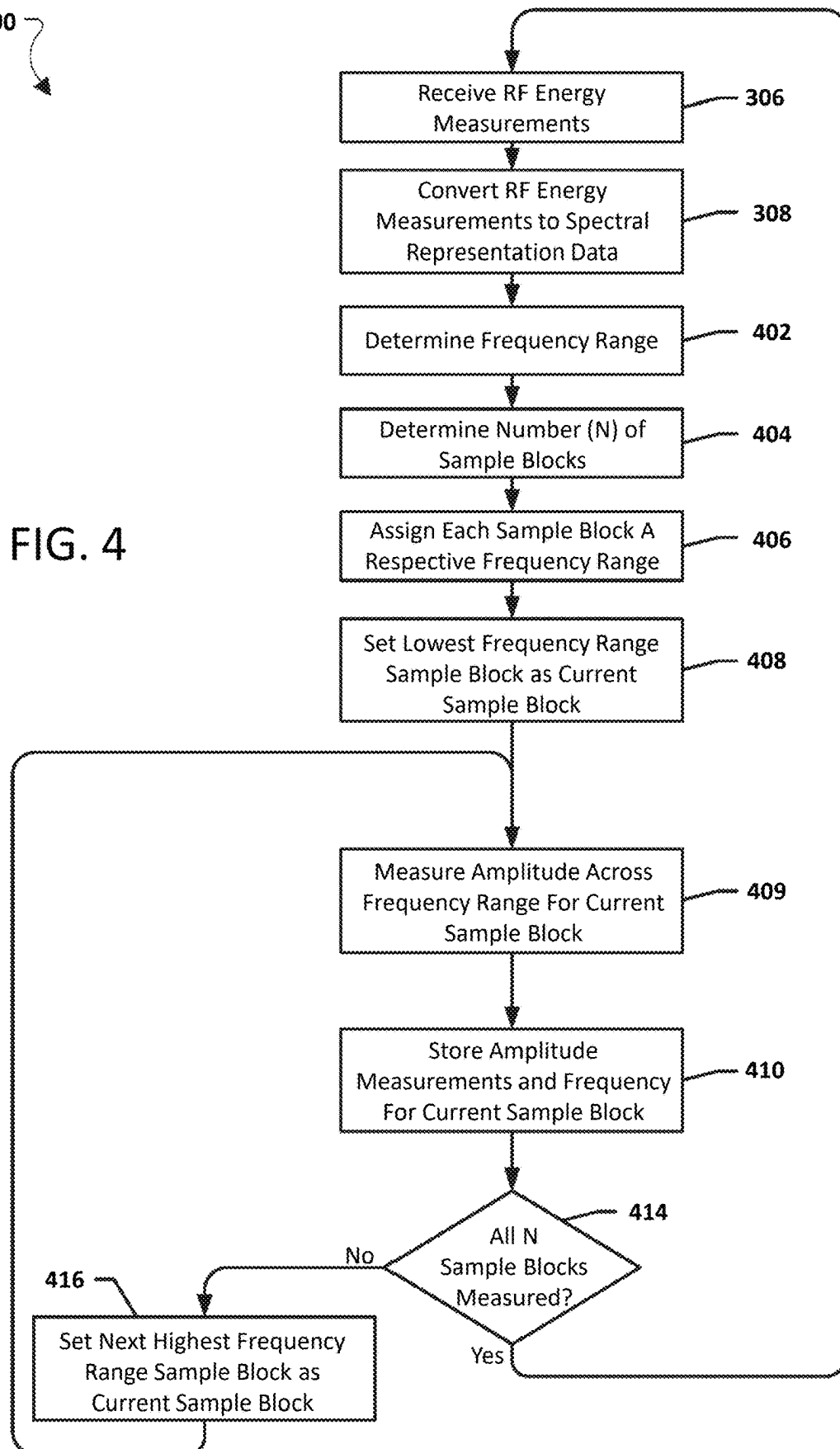
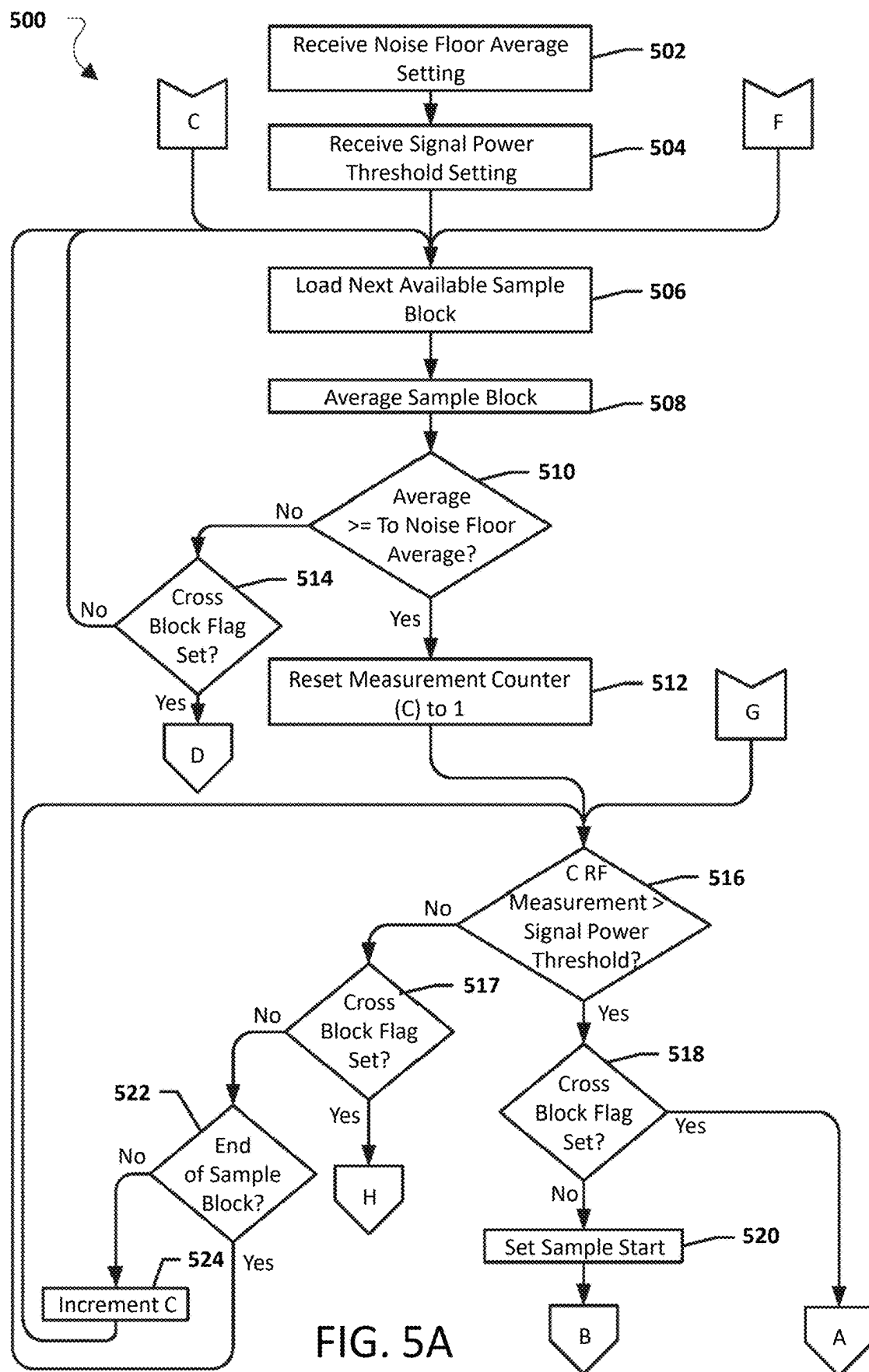


FIG. 3





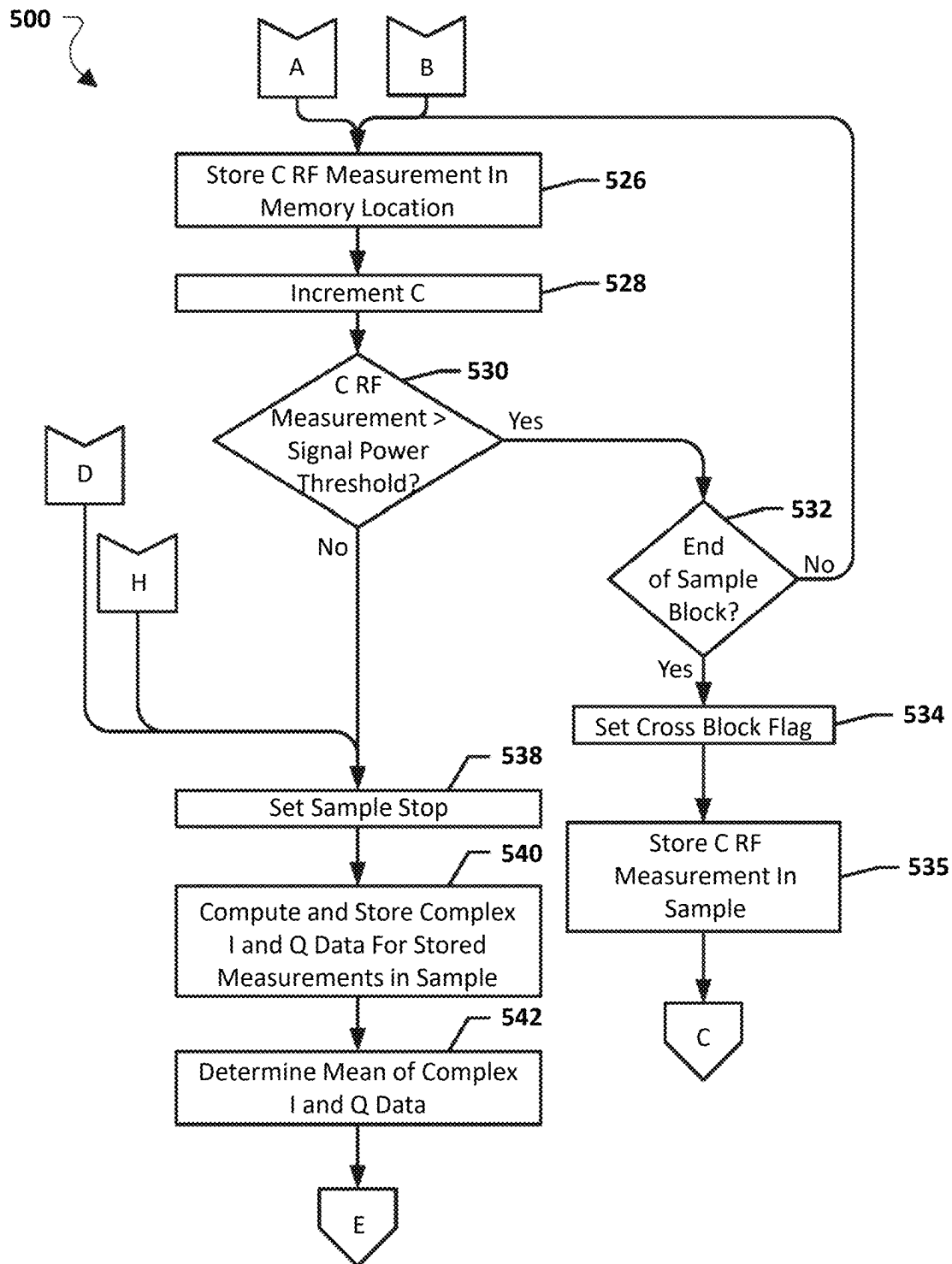


FIG. 5B

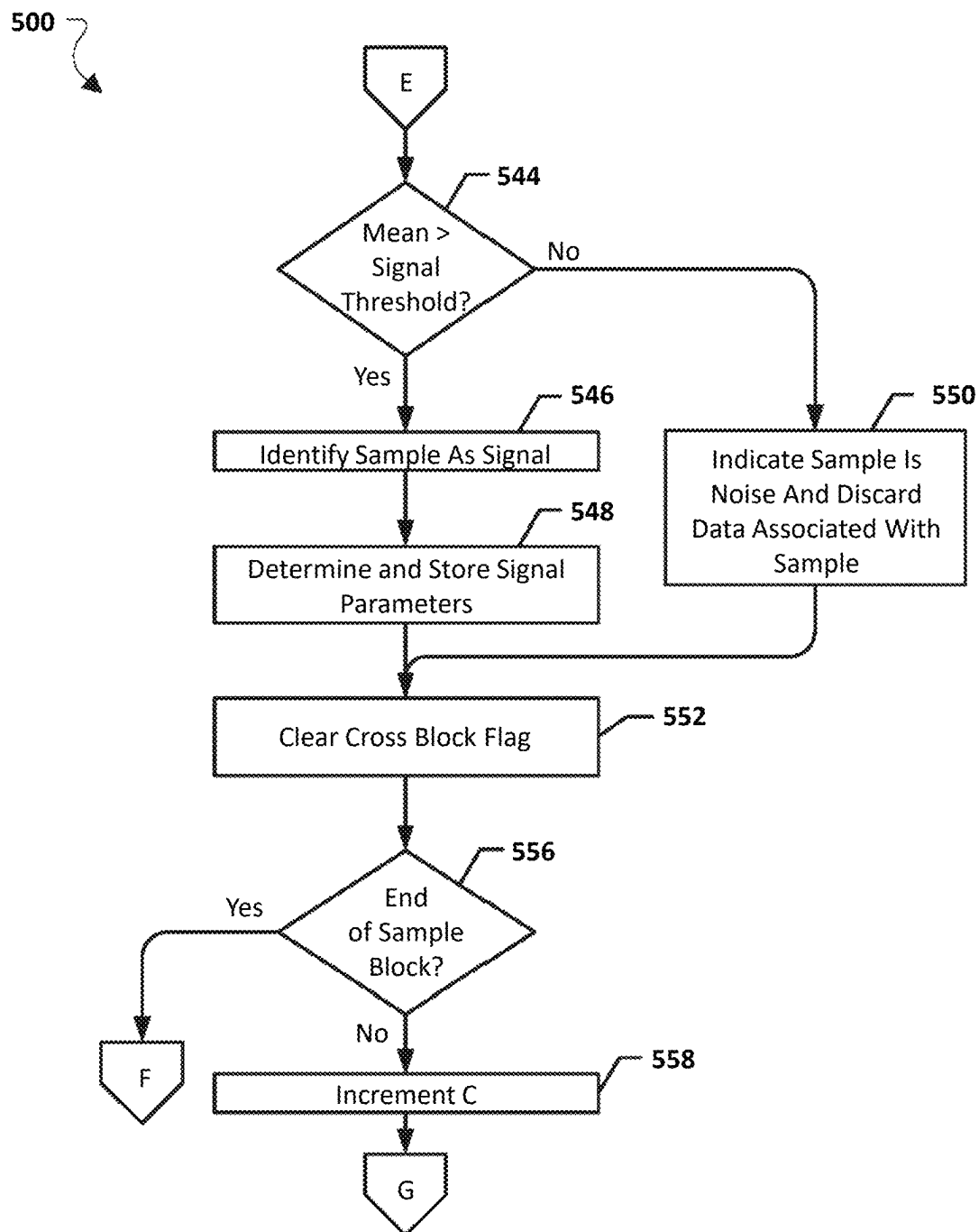
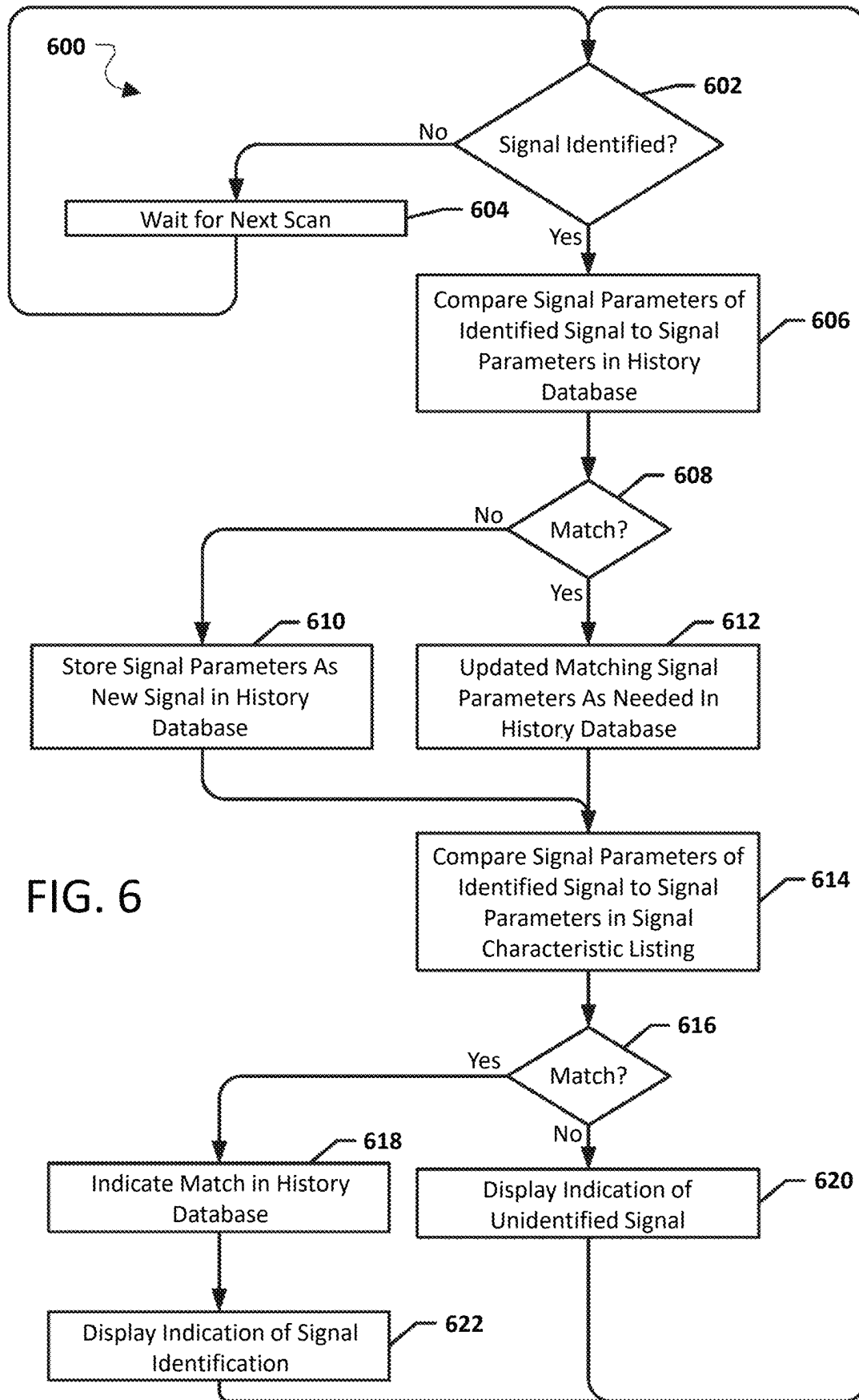


FIG. 5C



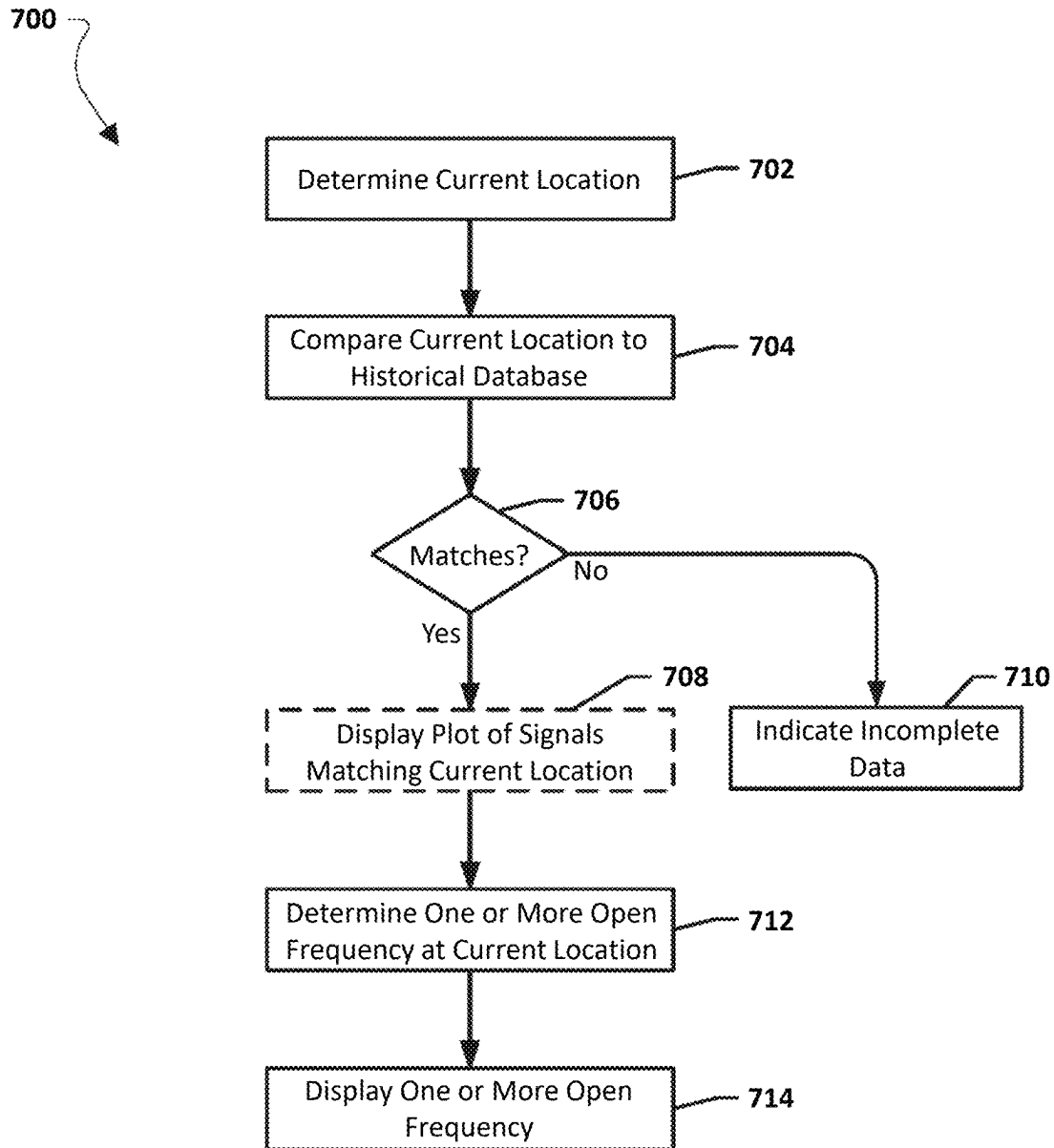


FIG. 7

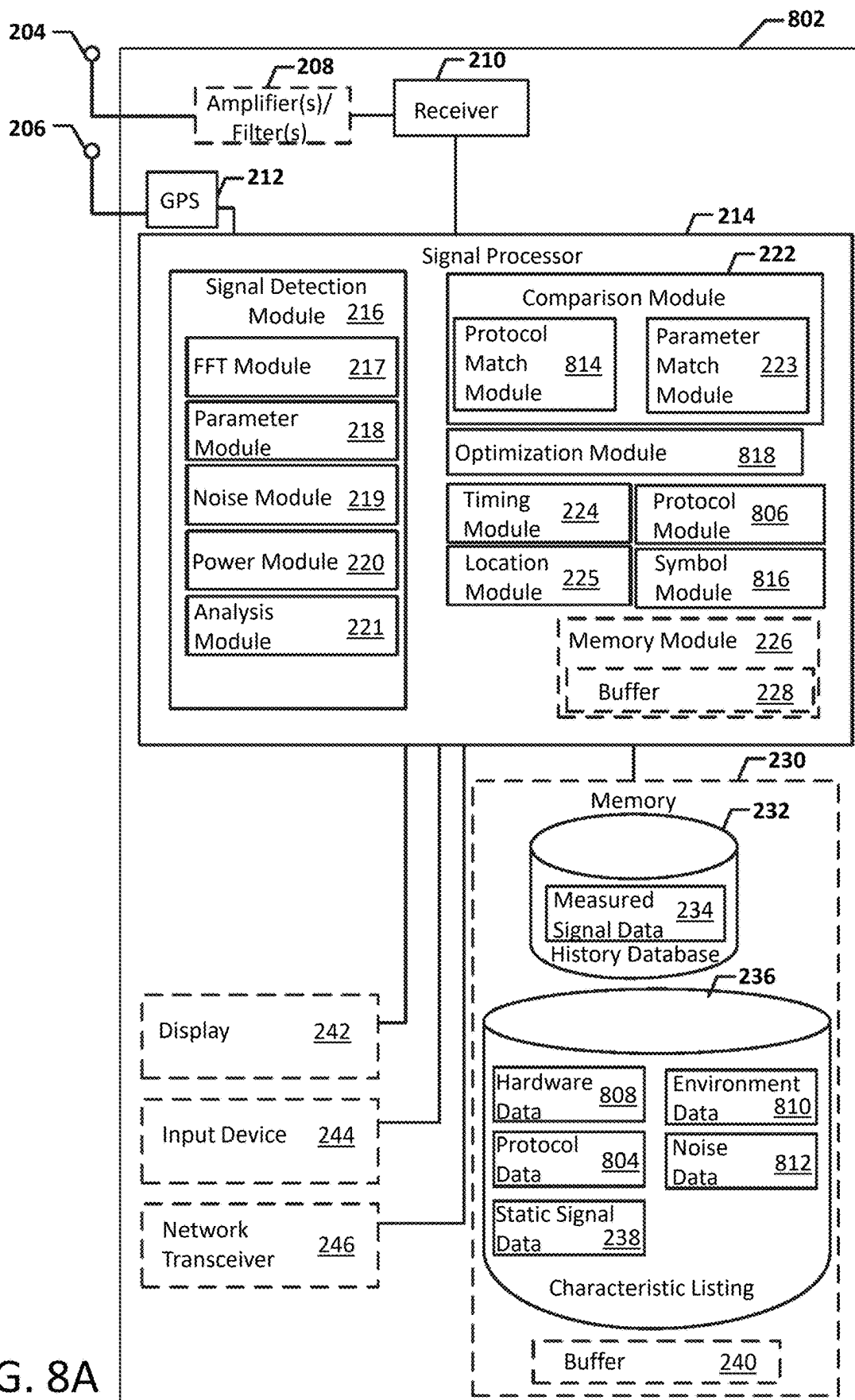


FIG. 8A

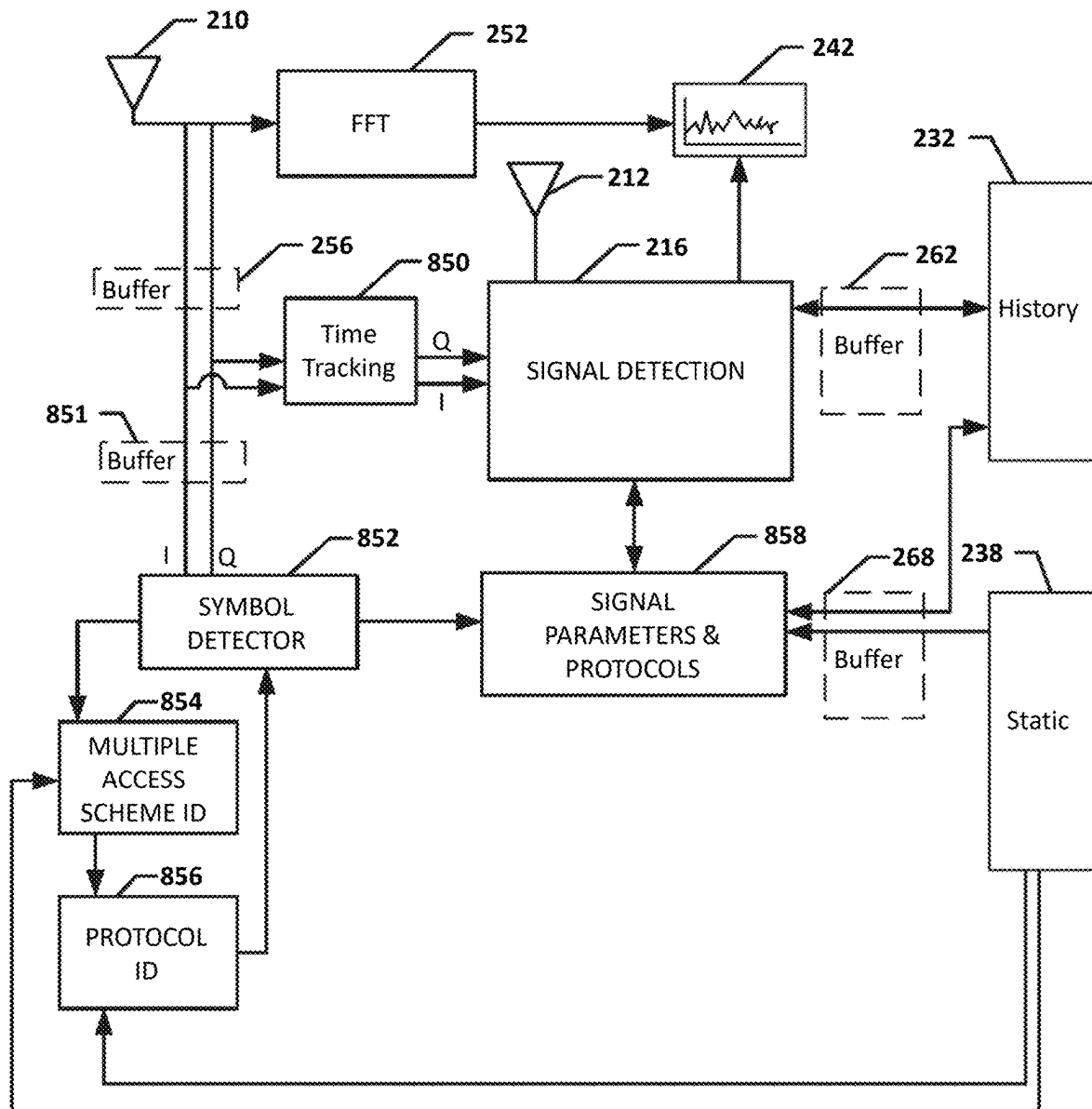
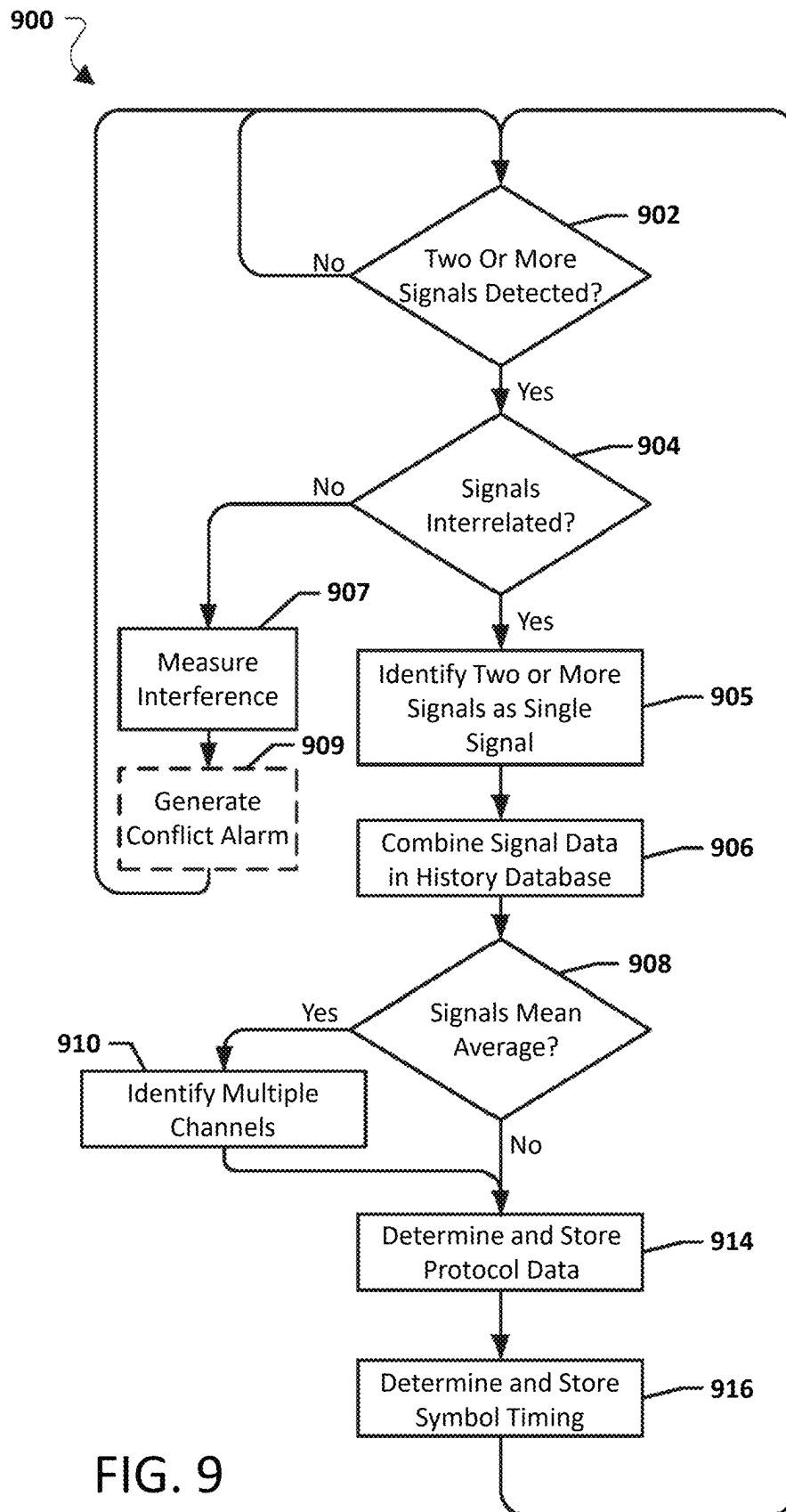


FIG. 8B



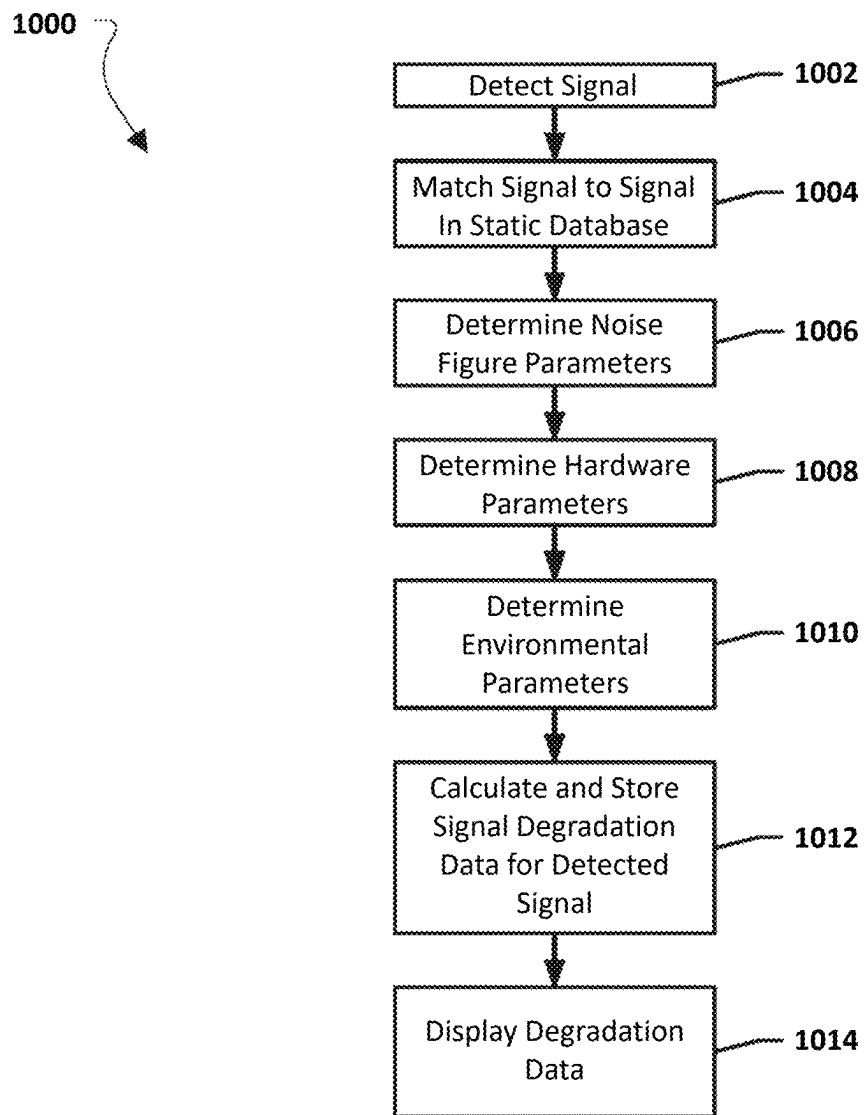


FIG. 10

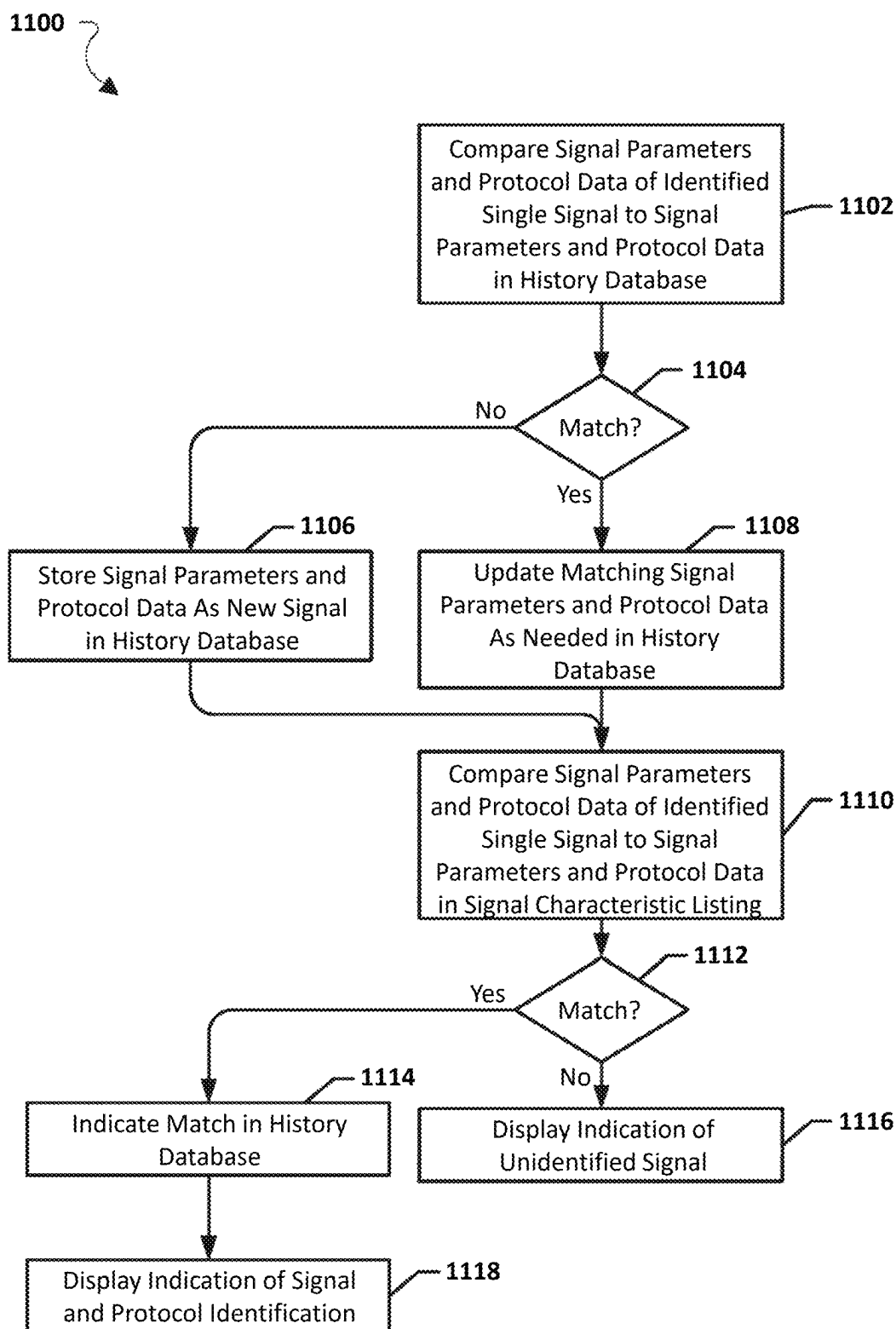


FIG. 11

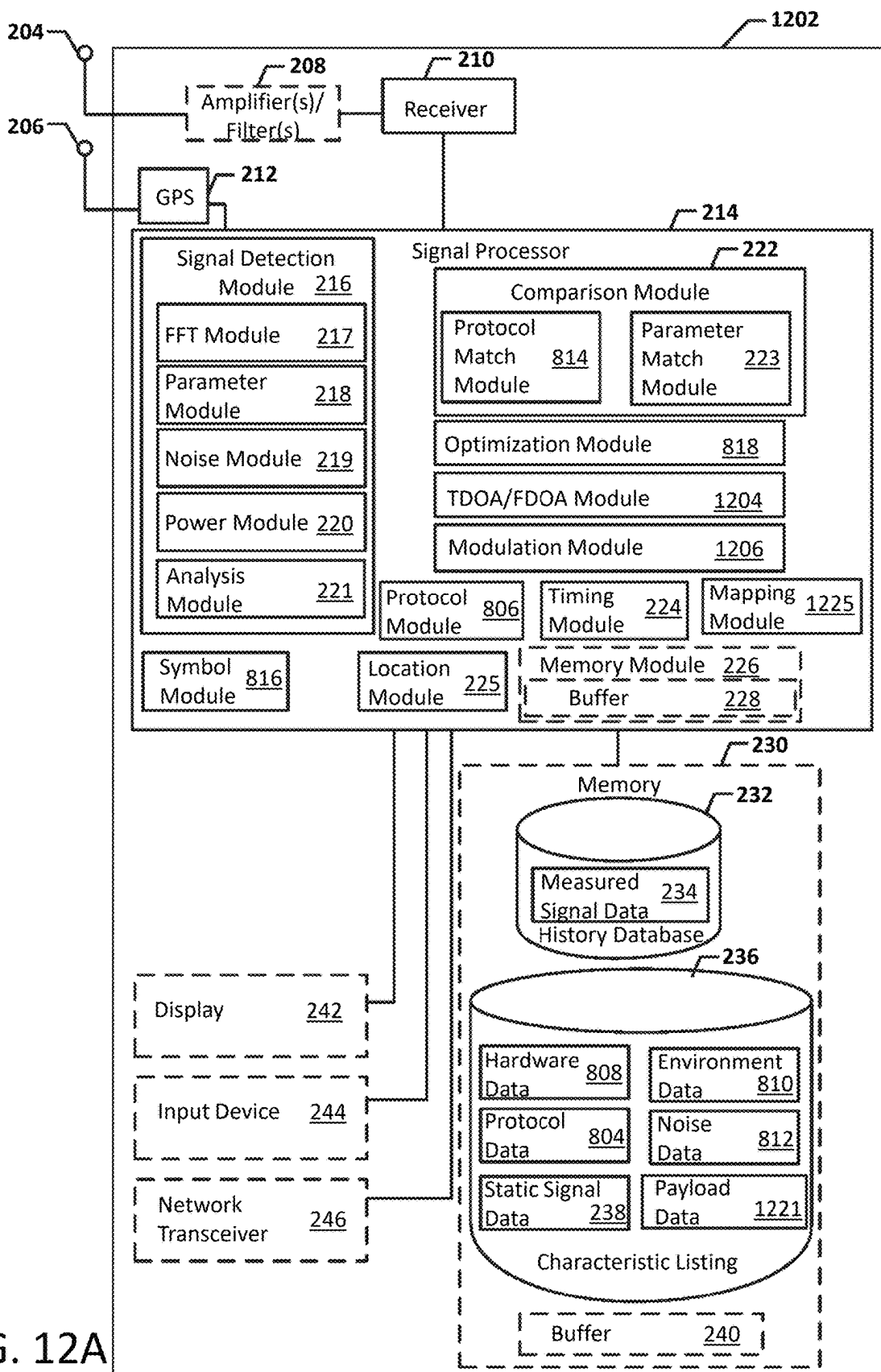


FIG. 12A

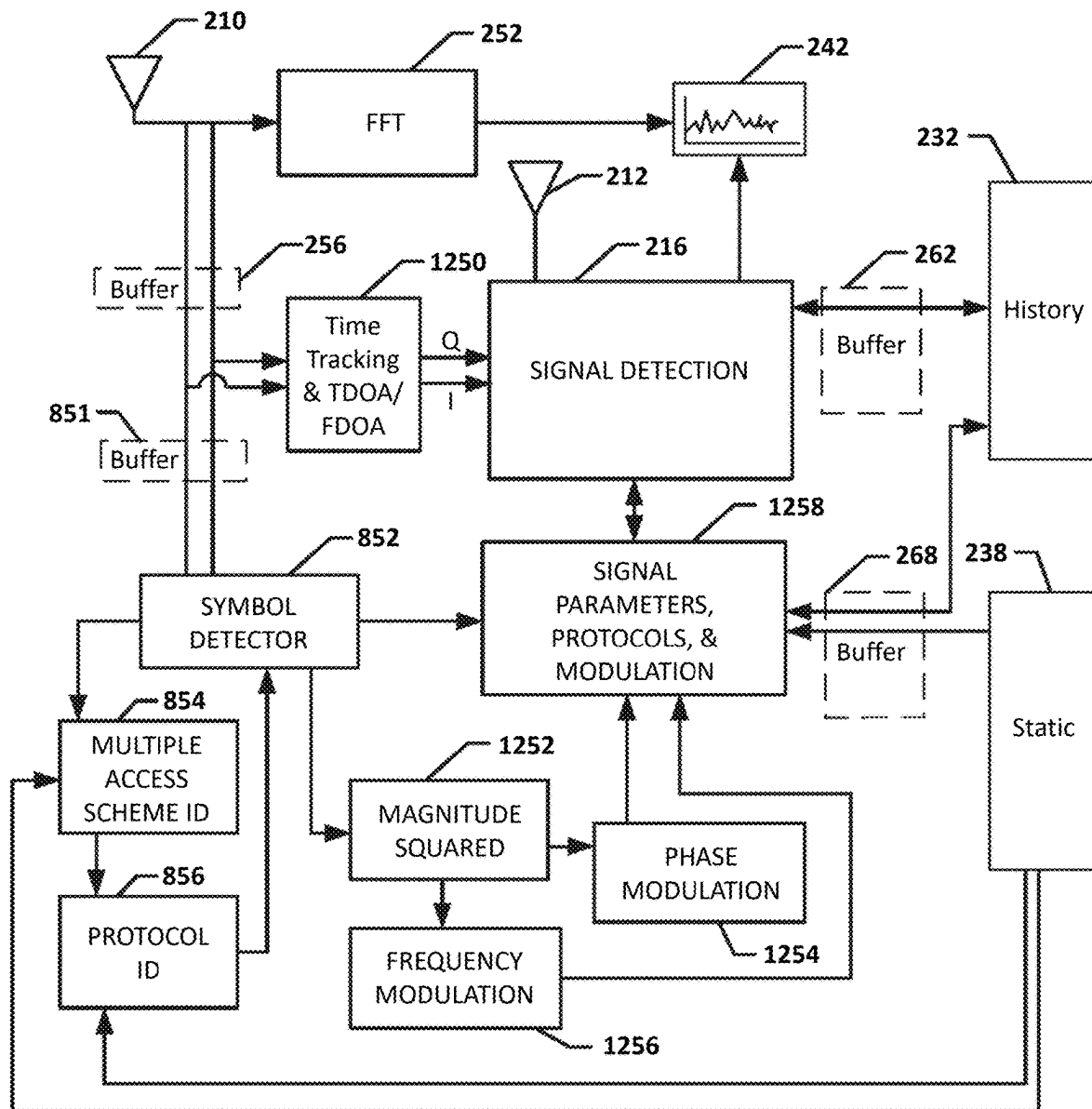


FIG. 12B

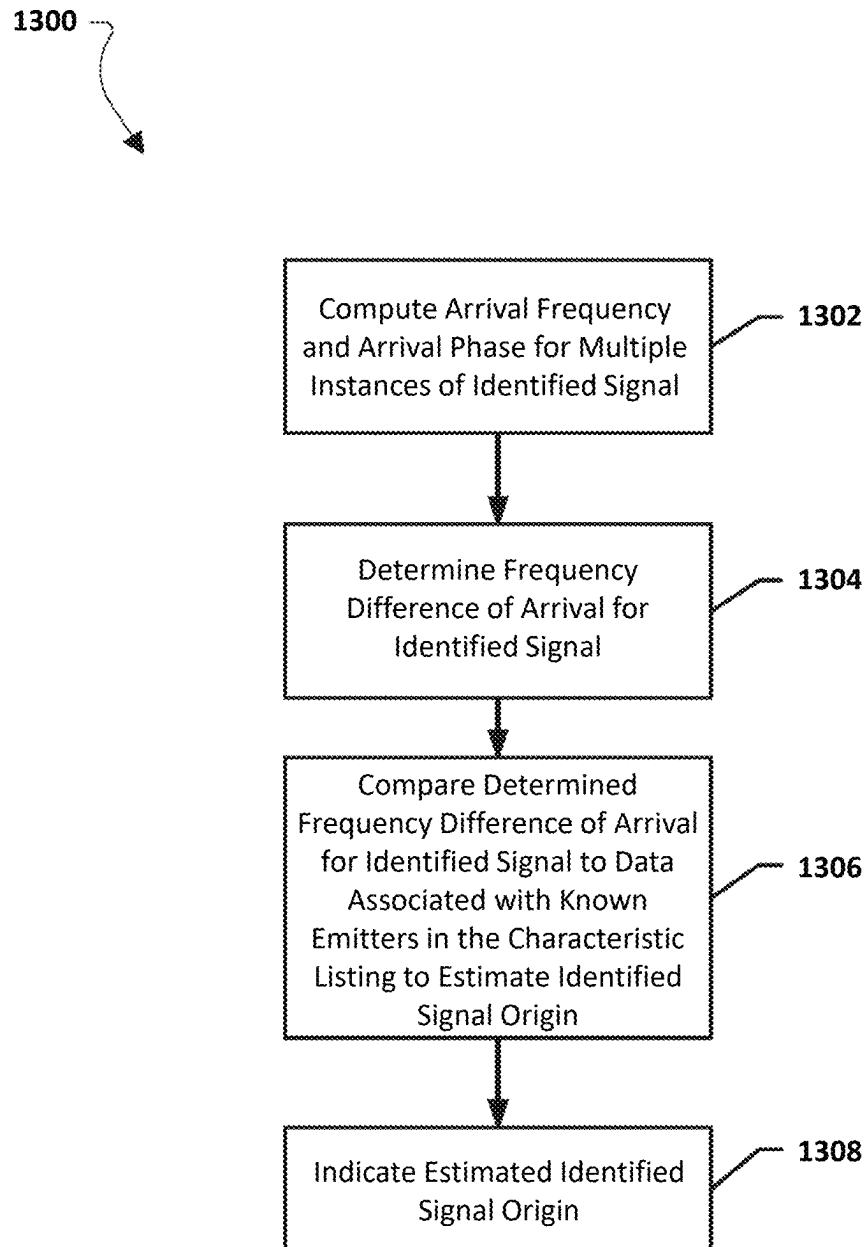


FIG. 13

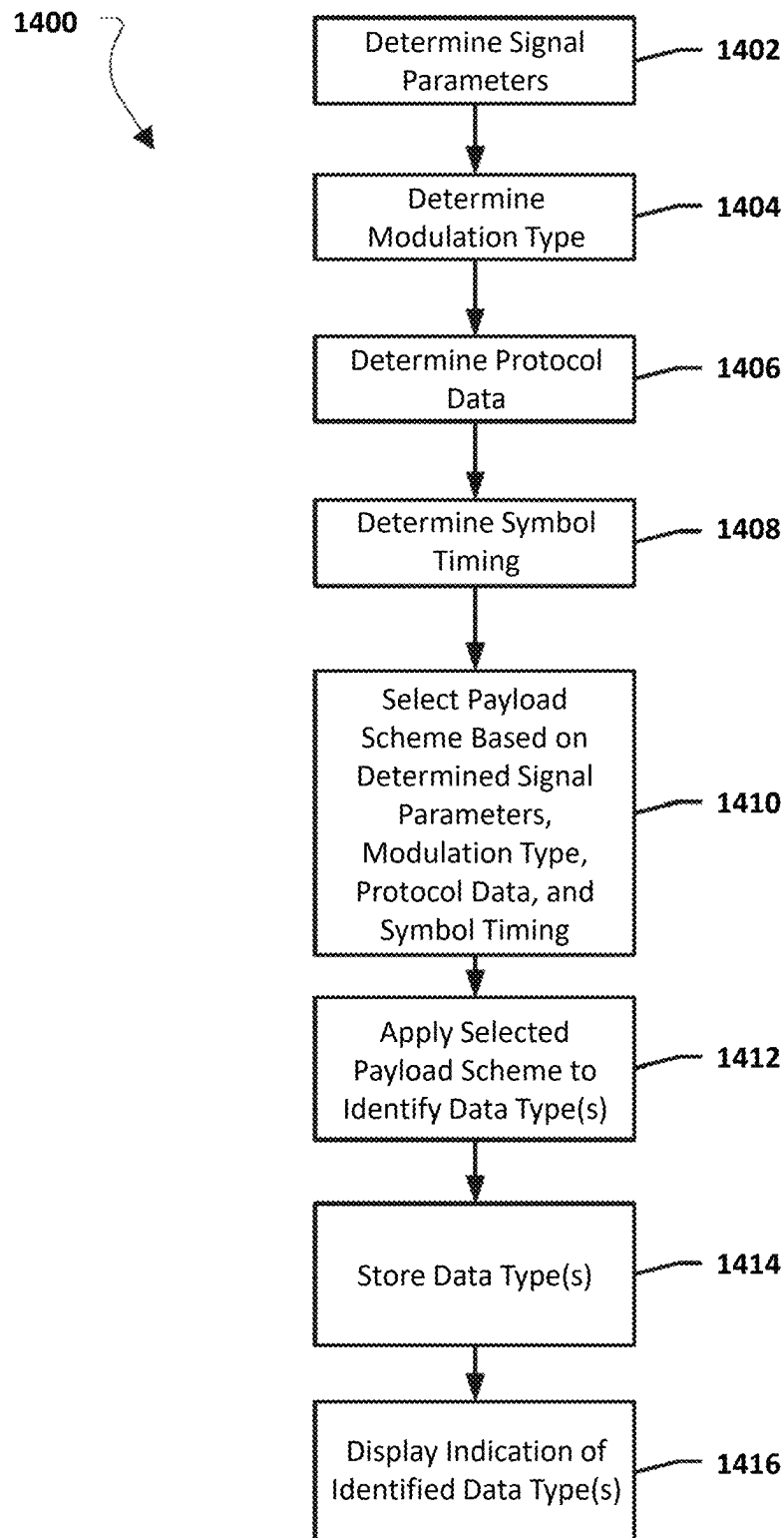
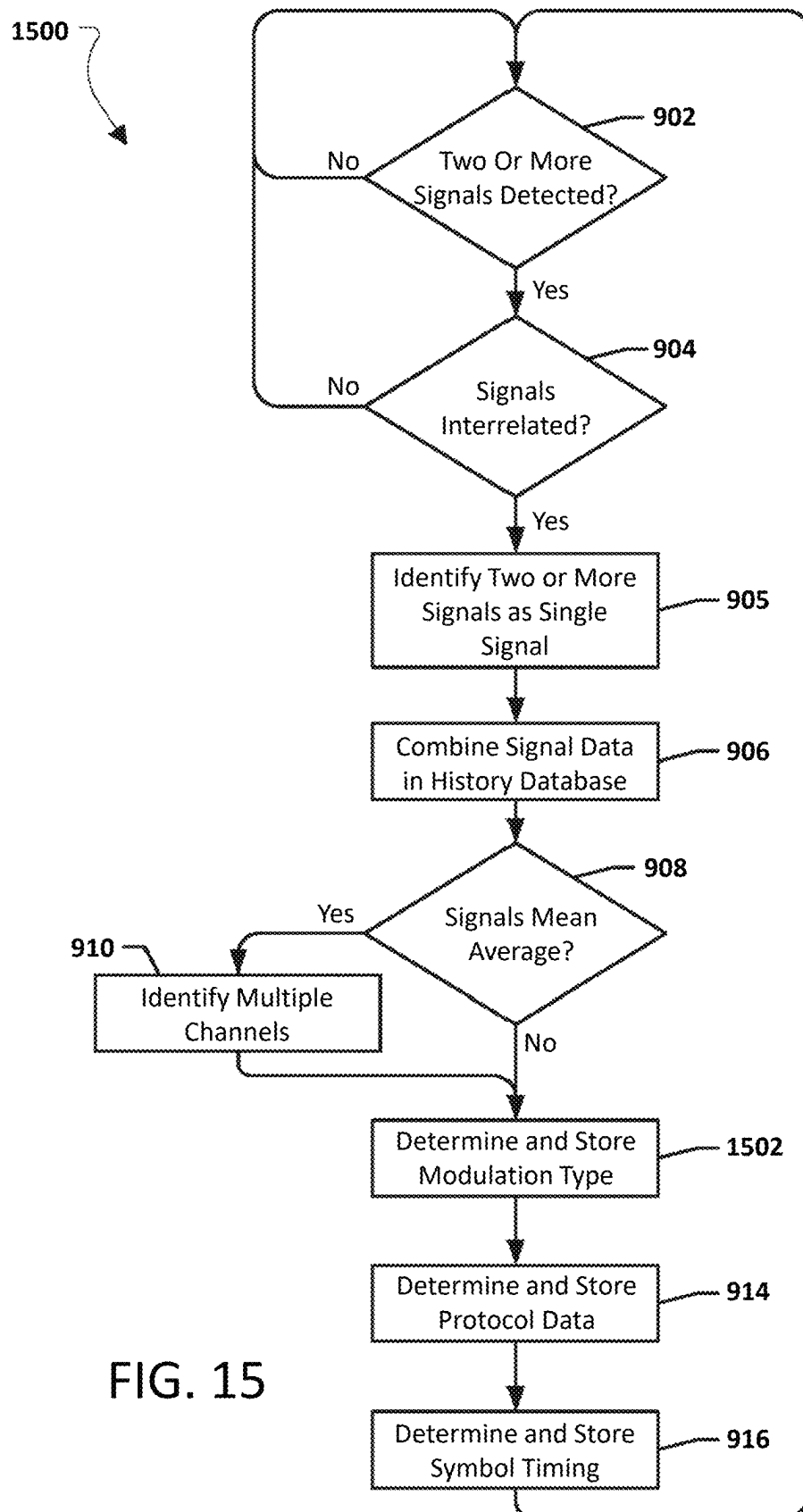
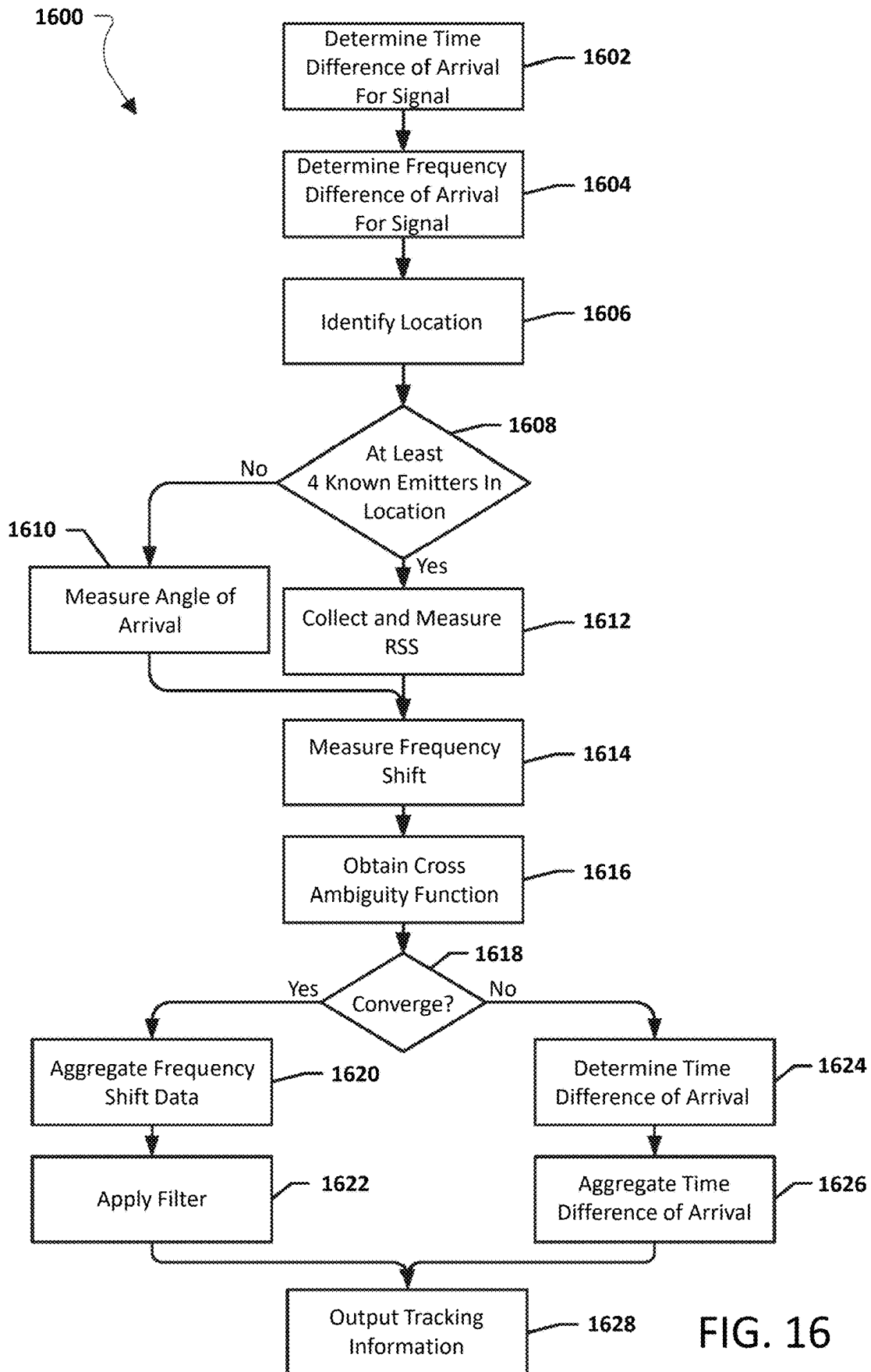


FIG. 14





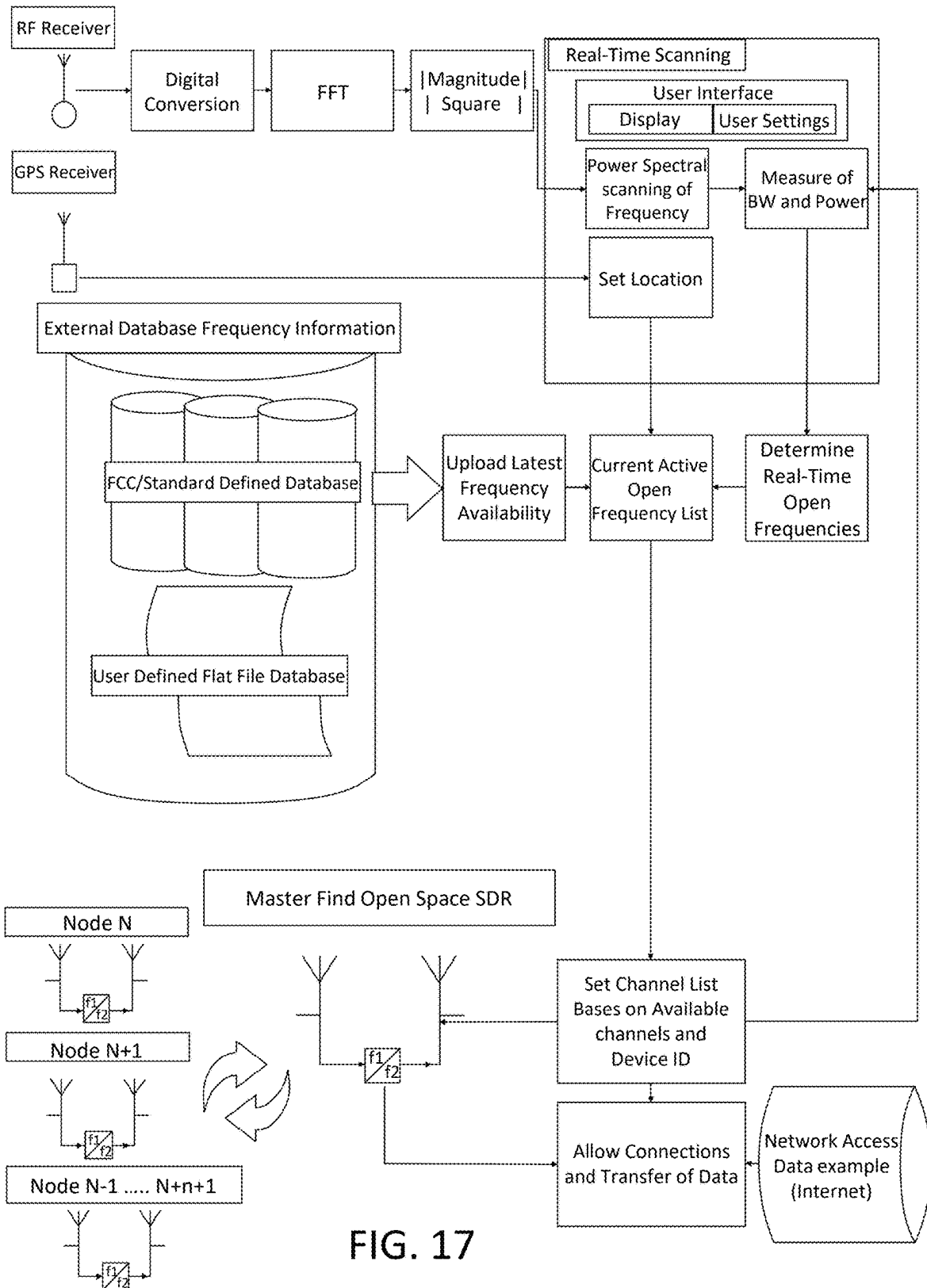


FIG. 17

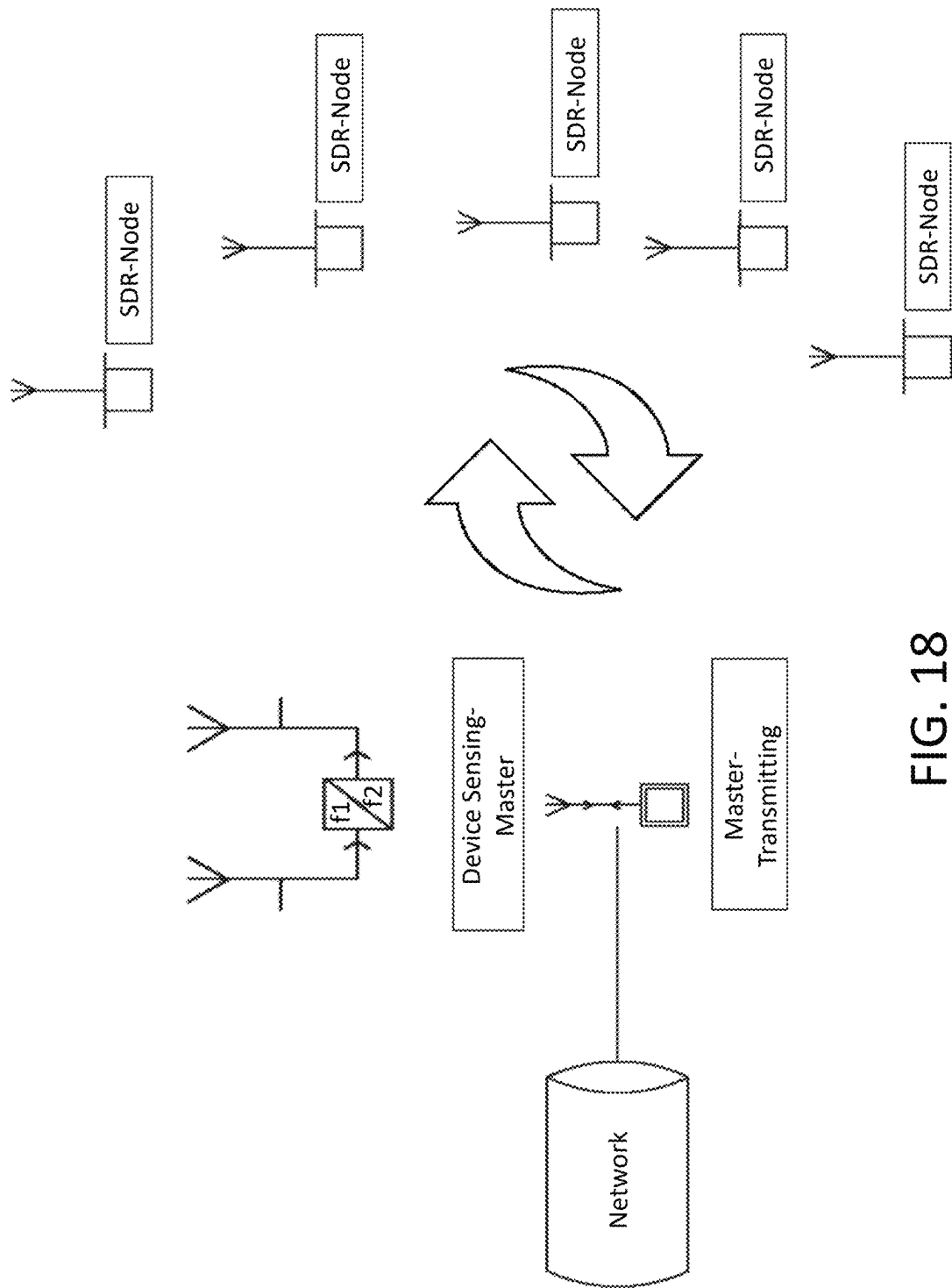
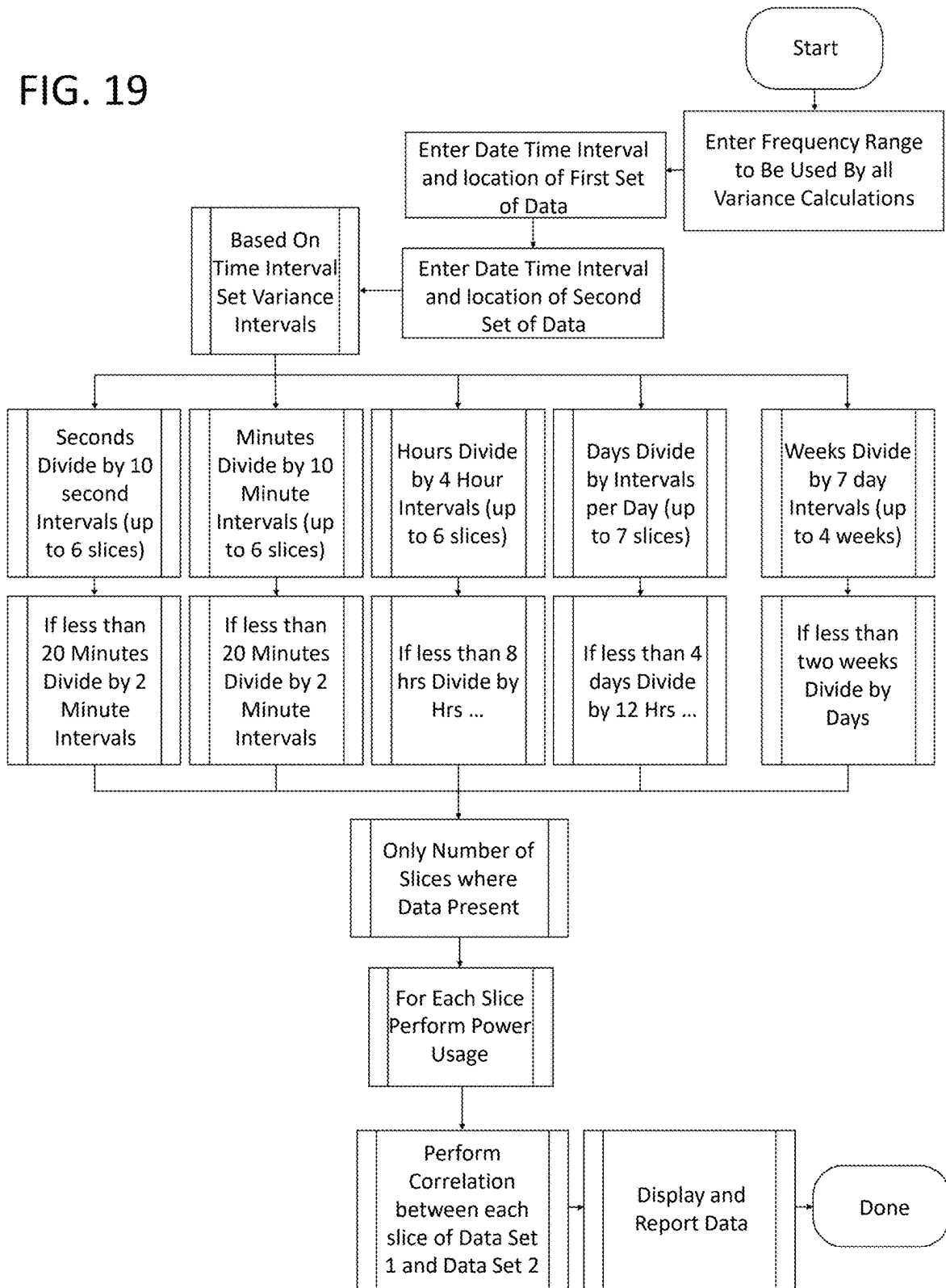


FIG. 19



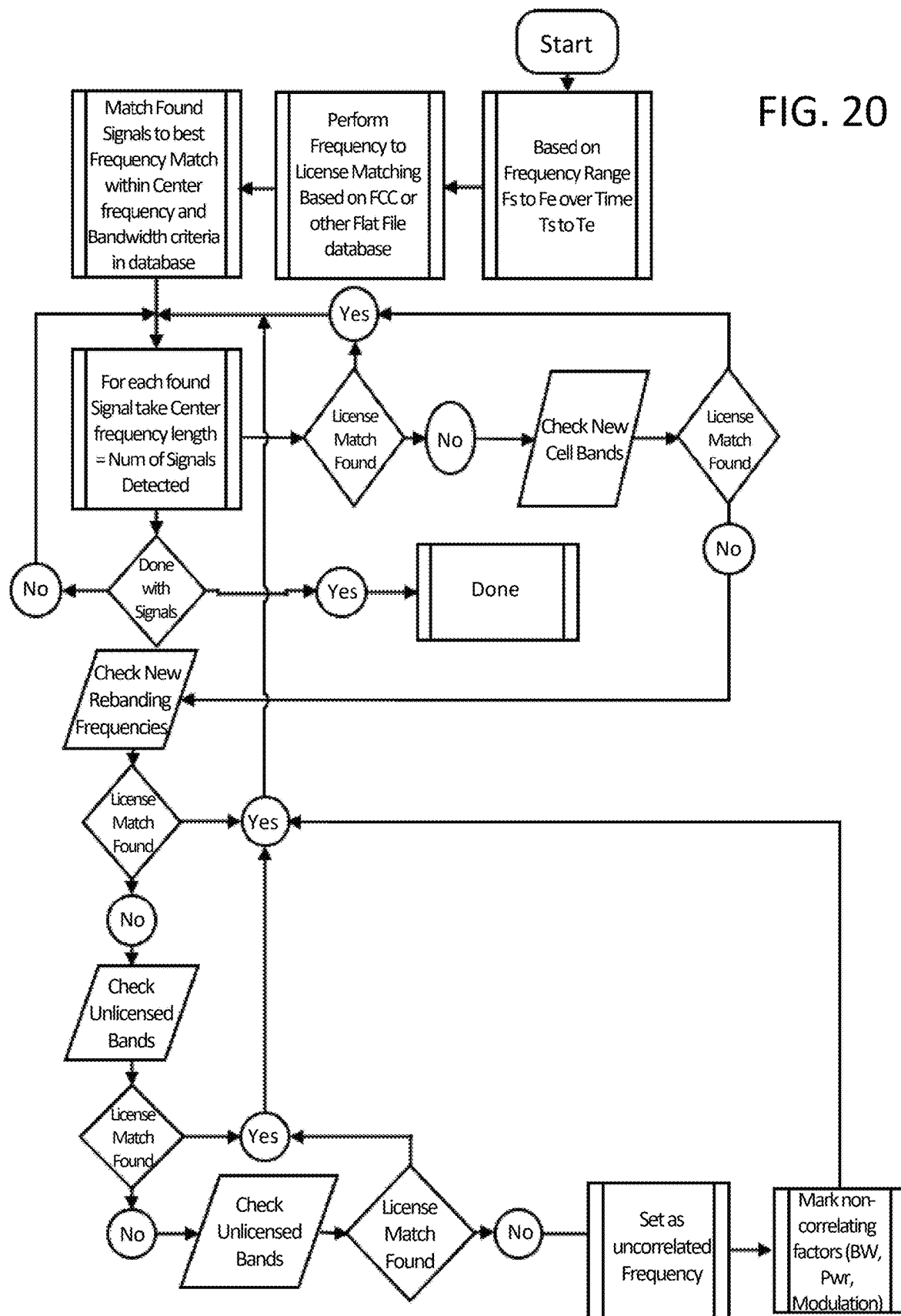


FIG. 21

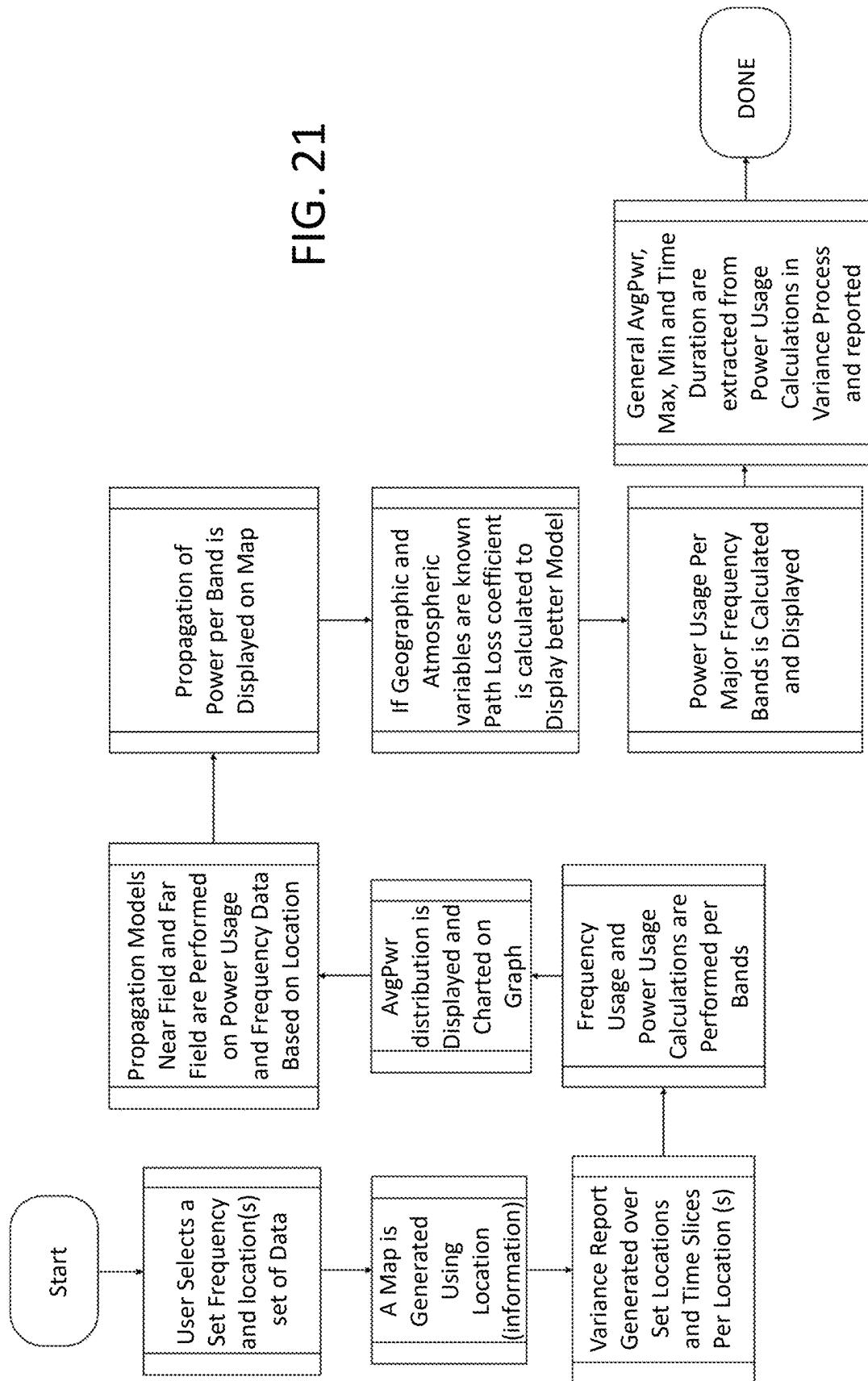
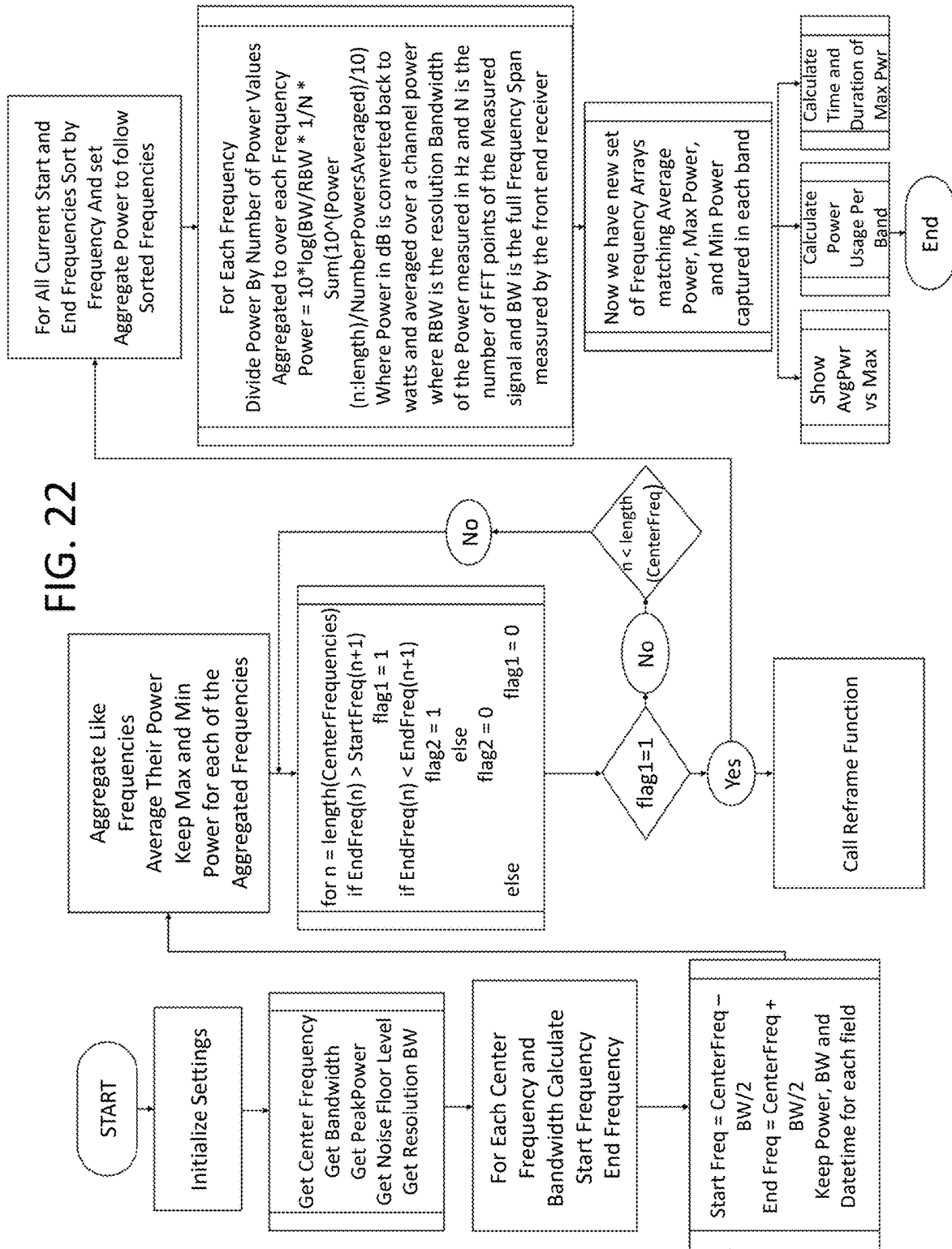


FIG. 22



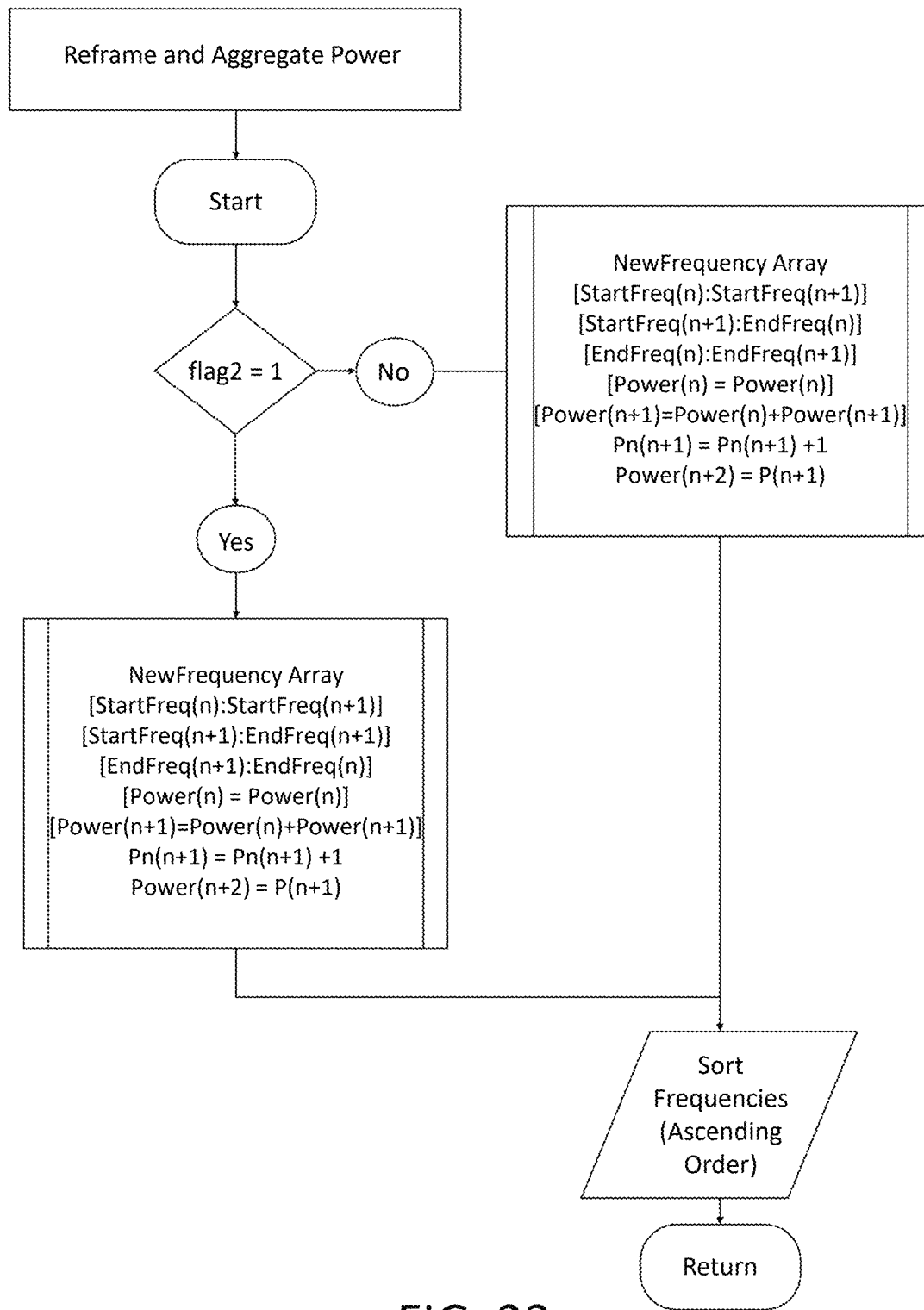


FIG. 23

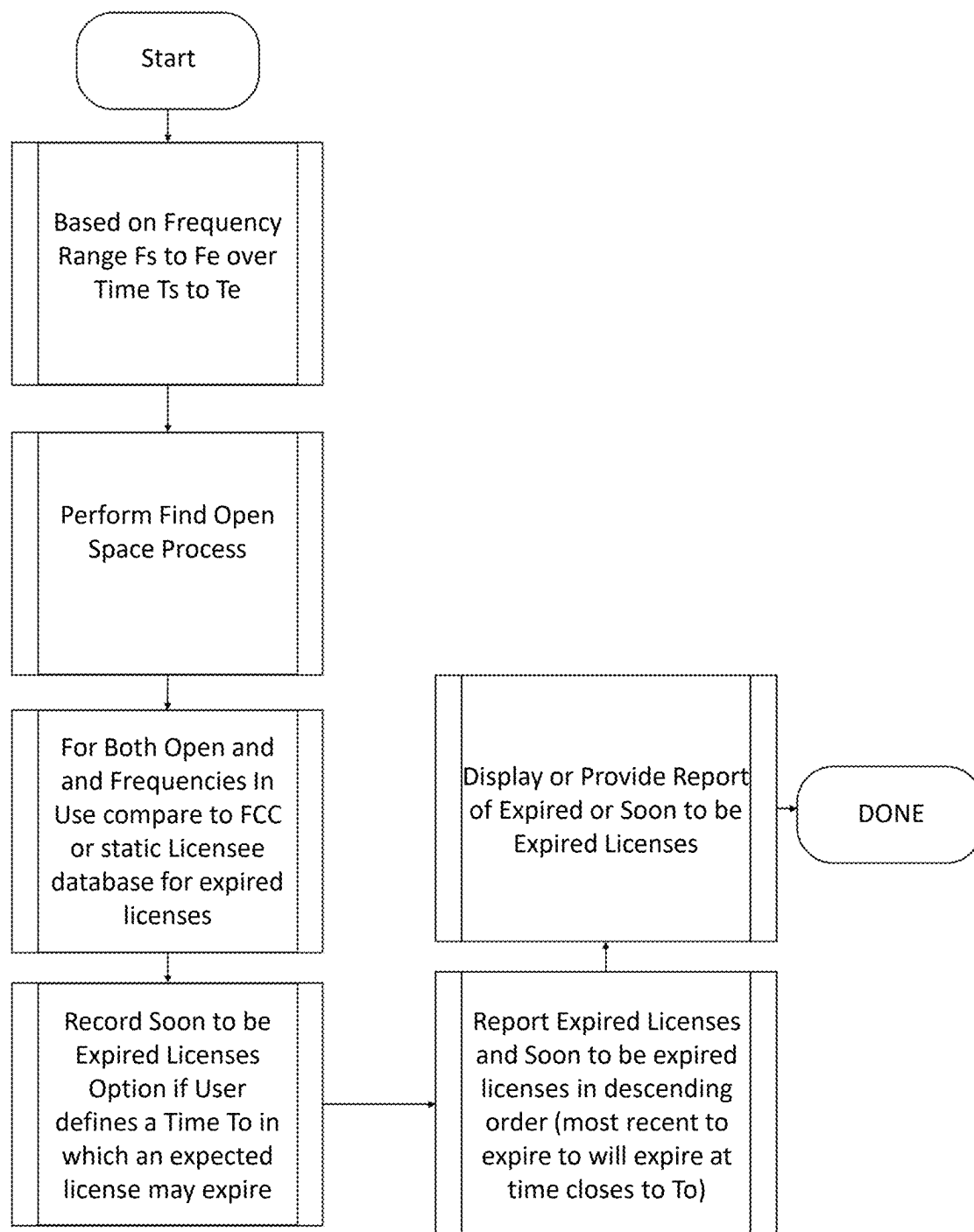
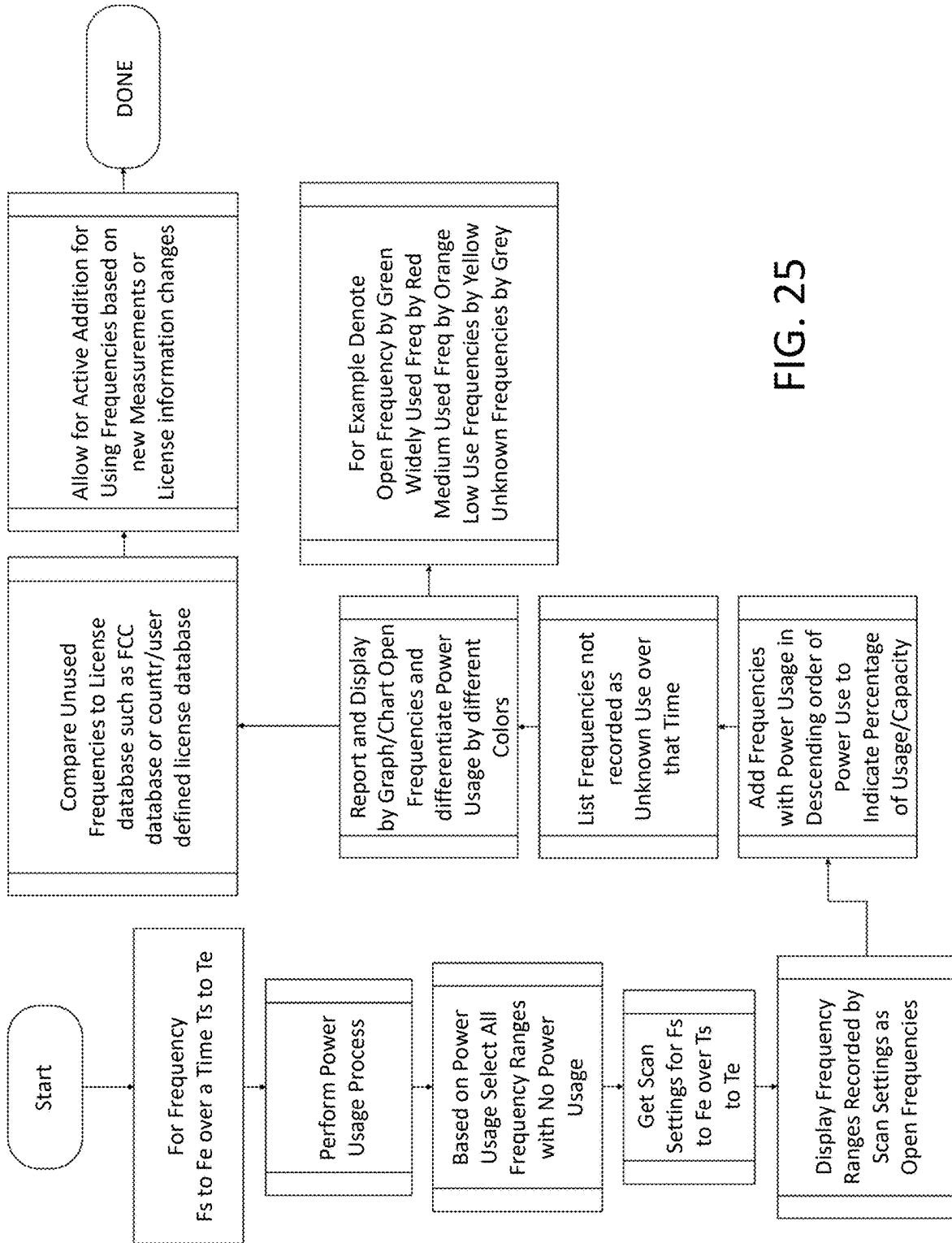
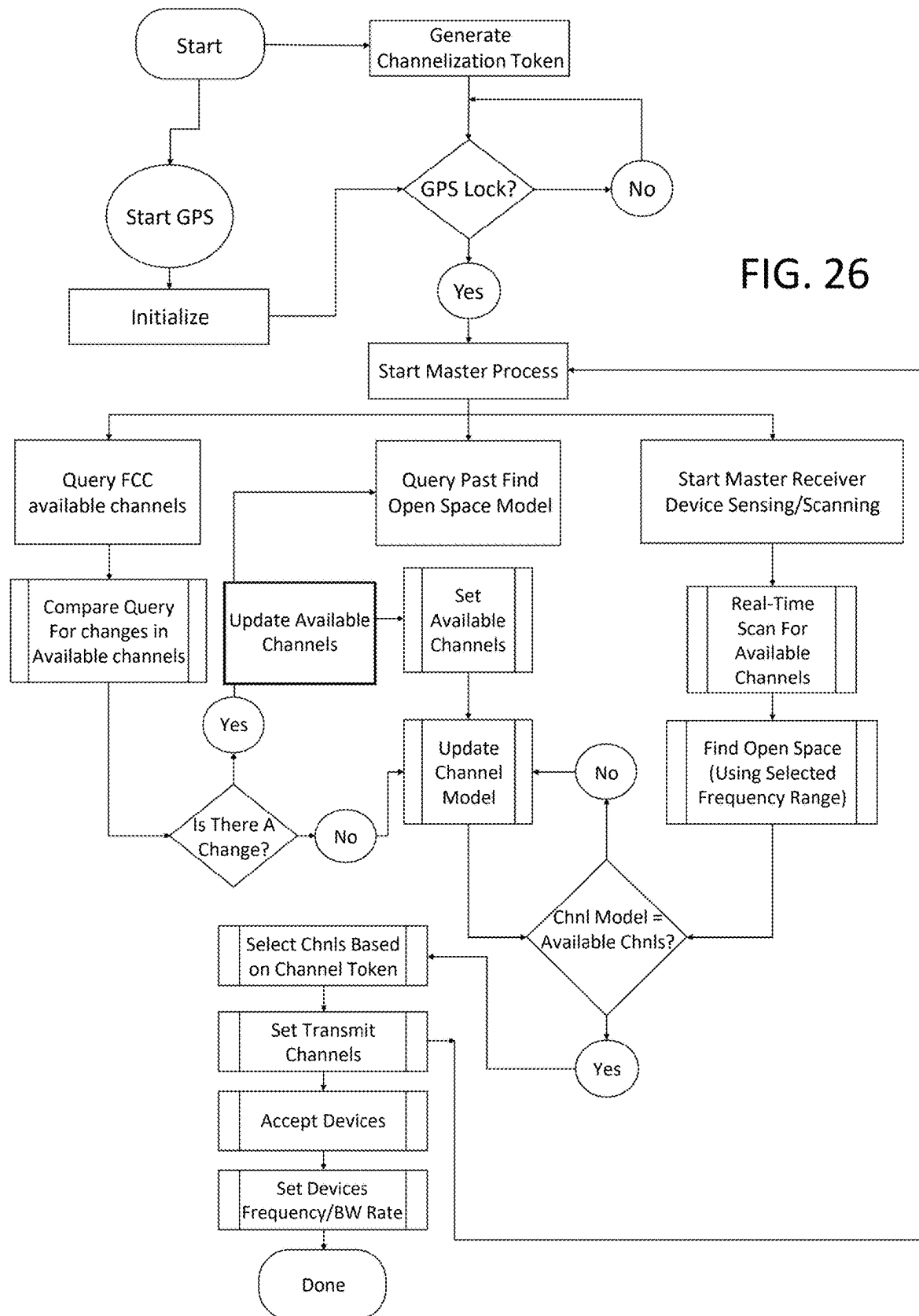


FIG. 24





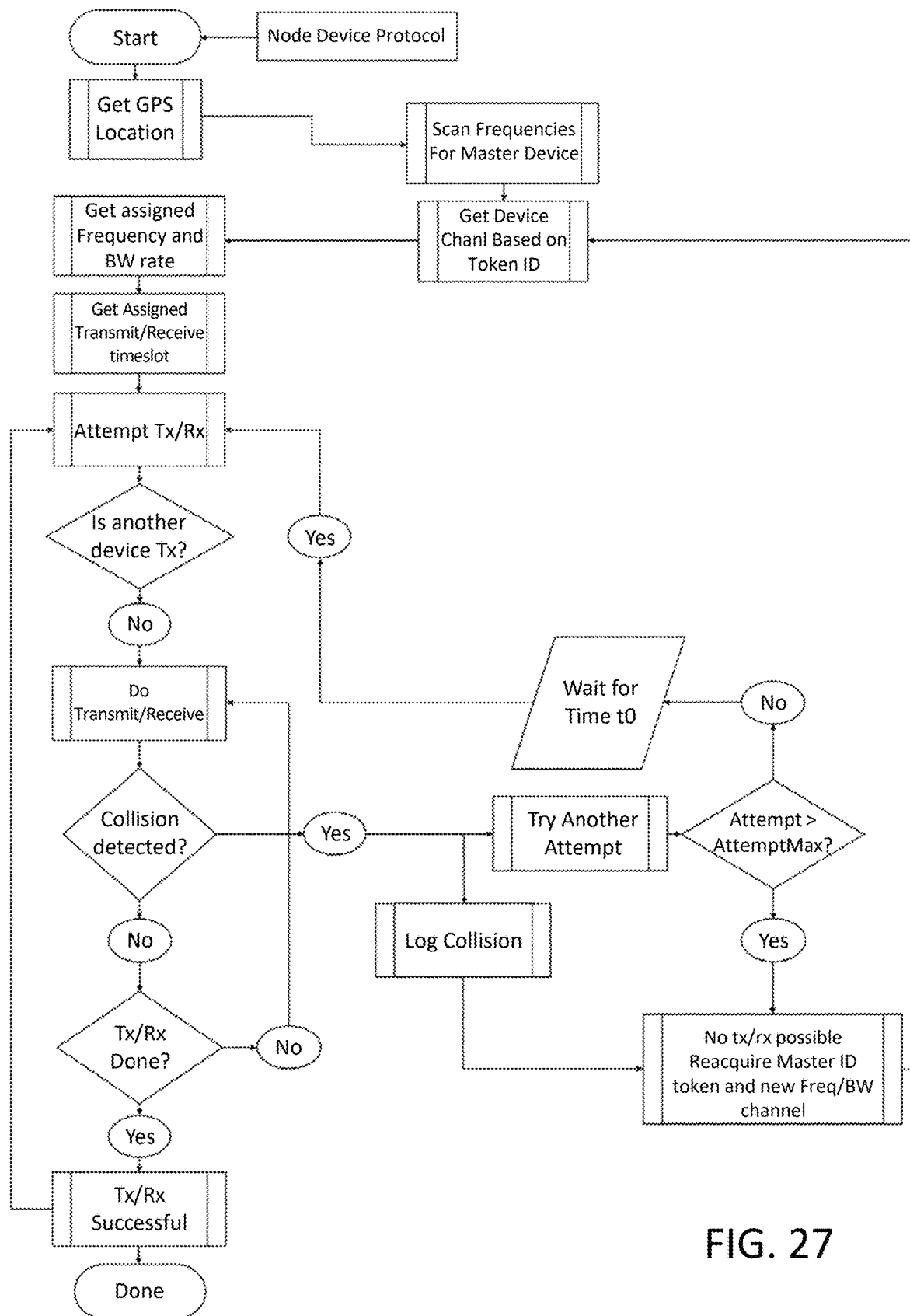
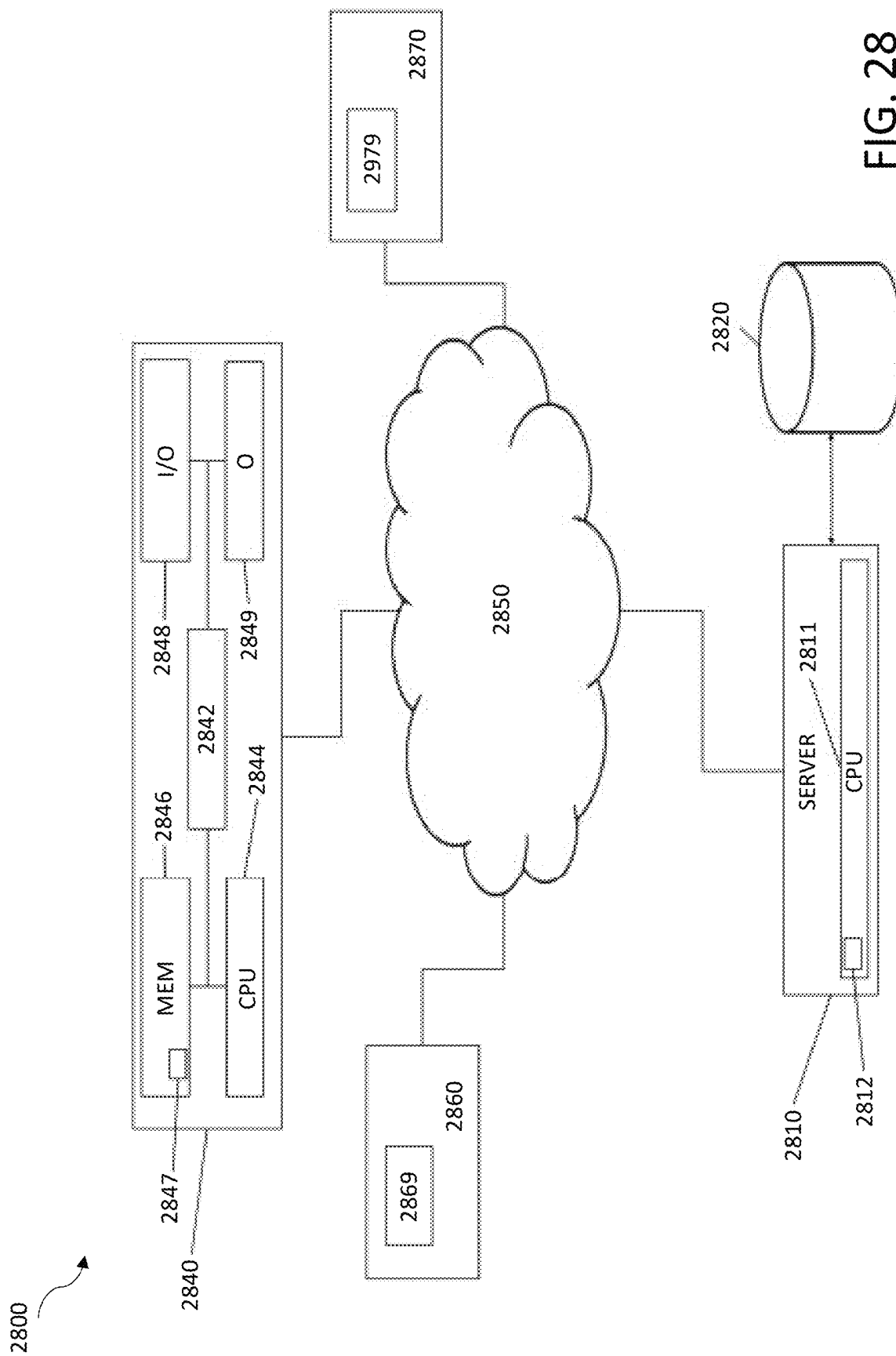


FIG. 27



SYSTEMS, METHODS, AND DEVICES FOR ELECTRONIC SPECTRUM MANAGEMENT FOR IDENTIFYING OPEN SPACE

CROSS REFERENCES TO RELATED APPLICATIONS

This application relates to and claims priority from the following U.S. patents and patent applications. This application is a continuation of U.S. application Ser. No. 18/825,699, filed Sep. 5, 2024, which is a continuation of U.S. application Ser. No. 18/638,122, filed Apr. 17, 2024, which is a continuation of U.S. application Ser. No. 18/525,040, filed Nov. 30, 2023, which is a continuation of U.S. application Ser. No. 18/142,917, filed May 3, 2023, which is a continuation of U.S. application Ser. No. 17/989,265, filed Nov. 17, 2022, which is a continuation of U.S. application Ser. No. 17/062,163, filed Oct. 2, 2020, which is a continuation of U.S. application Ser. No. 16/382,991 filed Apr. 12, 2019, which is a continuation of U.S. application Ser. No. 16/354,353 filed Mar. 15, 2019, which is a continuation of U.S. application Ser. No. 15/987,508 filed May 23, 2018, which is a continuation of U.S. application Ser. No. 15/492,475 filed Apr. 20, 2017, which is a continuation of U.S. application Ser. No. 14/504,819 filed Oct. 2, 2014, which is a continuation of U.S. application Ser. No. 14/329,838 filed Jul. 11, 2014, which is a continuation of U.S. application Ser. No. 14/273,157 filed May 8, 2014, which is a continuation of U.S. application Ser. No. 14/086,871 filed Nov. 21, 2013, which is a continuation-in-part of U.S. application Ser. No. 14/082,873 filed Nov. 18, 2013, which is a continuation of U.S. application Ser. No. 13/912,683 filed Jun. 7, 2013, which claims the benefit of U.S. Provisional Application No. 61/789,758 filed Mar. 15, 2013, each of which is hereby incorporated herein by reference in its entirety. U.S. application Ser. No. 14/086,871 is also a continuation-in-part of U.S. application Ser. No. 14/082,916 filed Nov. 18, 2013, which is a continuation of U.S. application Ser. No. 13/912,893 filed Jun. 7, 2013, which claims the benefit of U.S. Provisional Application No. 61/789,758 filed Mar. 15, 2013, each of which is hereby incorporated herein by reference in its entirety. U.S. application Ser. No. 14/086,871 is also a continuation-in-part of U.S. application Ser. No. 14/082,930 filed Nov. 18, 2013, which is a continuation of U.S. application Ser. No. 13/913,013 filed Jun. 7, 2013, which claims the benefit of U.S. Provisional Application No. 61/789,758 filed Mar. 15, 2013, each of which is hereby incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to spectrum analysis and management for radio frequency signals, and more particularly for automatically identifying open space in a wireless communications spectrum.

2. Description of the Prior Art

Generally, it is known in the prior art to provide wireless communications spectrum management for detecting devices for managing the space. Spectrum management includes the process of regulating the use of radio frequencies to promote efficient use and gain net social benefit. A problem faced in effective spectrum management is the various numbers of devices emanating wireless signal

propagations at different frequencies and across different technological standards. Coupled with the different regulations relating to spectrum usage around the globe effective spectrum management becomes difficult to obtain and at best can only be reached over a long period of time.

Another problem facing effective spectrum management is the growing need from spectrum despite the finite amount of spectrum available. Wireless technologies have exponentially grown in recent years. Consequently, available spectrum has become a valuable resource that must be efficiently utilized. Therefore, systems and methods are needed to effectively manage and optimize the available spectrum that is being used.

Most spectrum management devices may be categorized into two primary types. The first type is a spectral analyzer where a device is specifically fitted to run a 'scanner' type receiver that is tailored to provide spectral information for a narrow window of frequencies related to a specific and limited type of communications standard, such as cellular communication standard. Problems arise with these narrowly tailored devices as cellular standards change and/or spectrum use changes impact the spectrum space of these technologies. Changes to the software and hardware for these narrowly tailored devices become too complicated, thus necessitating the need to purchase a totally different and new device. Unfortunately, this type of device is only for a specific use and cannot be used to alleviate the entire needs of the spectrum management community.

The second type of spectral management device employs a methodology that requires bulky, extremely difficult to use processes, and expensive equipment. In order to attain a broad spectrum management view and complete all the necessary tasks, the device ends up becoming a conglomerate of software and hardware devices that is both hard to use and difficult to maneuver from one location to another.

While there may be several additional problems associated with current spectrum management devices, at least four major problems exist overall: 1) most devices are built to inherently only handle specific spectrum technologies such as 900 MHz cellular spectrum while not being able to mitigate other technologies that may be interfering or competing with that spectrum, 2) the other spectrum management devices consist of large spectrum analyzers, database systems, and spectrum management software that is expensive, too bulky, and too difficult to manage for a user's basic needs, 3) other spectrum management devices in the prior art require external connectivity to remote databases to perform analysis and provide results or reports with analytics to aid in management of spectrum and/or devices, and 4) other devices of the prior art do not function to provide real-time or near real-time data and analysis to allow for efficient management of the space and/or devices and signals therein.

Examples of relevant prior art documents include the following:

U.S. Pat. No. RE43,066 for "System and method for reuse of communications spectrum for fixed and mobile applications with efficient method to mitigate interference" by inventor Mark Allen McHenry, filed Dec. 2, 2008, describes a communications system network enabling secondary use of spectrum on a non-interference basis. The system uses a modulation method to measure the background signals that eliminates self-generated interference and also identifies the secondary signal to all primary users via on/off amplitude modulation, allowing easy resolution of interference claims. The system uses high-processing gain probe waveforms that enable propagation measurements to be made with minimal interference to the primary users. The system measures

background signals and identifies the types of nearby receivers and modifies the local frequency assignments to minimize interference caused by a secondary system due to non-linear mixing interference and interference caused by out-of-band transmitted signals (phase noise, harmonics, and spurs). The system infers a secondary node's elevation and mobility (thus, its probability to cause interference) by analysis of the amplitude of background signals. Elevated or mobile nodes are given more conservative frequency assignments than stationary nodes.

U.S. Pat. No. 7,424,268 for "System and Method for Management of a Shared Frequency Band" by inventors Diener, et al., filed Apr. 22, 2003, discloses a system, method, software and related functions for managing activity in an unlicensed radio frequency band that is shared, both in frequency and time, by signals of multiple types. Signal pulse energy in the band is detected and is used to classify signals according to signal type. Using knowledge of the types of signals occurring in the frequency band and other spectrum activity related statistics (referred to as spectrum intelligence), actions can be taken in a device or network of devices to avoid interfering with other signals, and in general to optimize simultaneous use of the frequency band with the other signals. The spectrum intelligence may be used to suggest actions to a device user or network administrator, or to automatically invoke actions in a device or network of devices to maintain desirable performance.

U.S. Pat. No. 8,249,631 for "Transmission power allocation/control method, communication device and program" by inventor Ryo Sawai, filed Jul. 21, 2010, teaches a method for allocating transmission power to a second communication service making secondary usage of a spectrum assigned to a first communication service, in a node which is able to communicate with a secondary usage node. The method determines an interference power acceptable for two or more second communication services when the two or more second communication services are operated and allocates the transmission powers to the two or more second communication services.

U.S. Pat. No. 8,094,610 for "Dynamic cellular cognitive system" by inventors Wang, et al., filed Feb. 25, 2009, discloses permitting high quality communications among a diverse set of cognitive radio nodes while minimizing interference to primary and other secondary users by employing dynamic spectrum access in a dynamic cellular cognitive system. Diverse device types interoperate, cooperate, and communicate with high spectrum efficiency and do not require infrastructure to form the network. The dynamic cellular cognitive system can expand to a wider geographical distribution via linking to existing infrastructure.

U.S. Pat. No. 8,565,811 for "Software-defined radio using multi-core processor" by inventors Tan, et al., discloses a radio control board passing a plurality of digital samples between a memory of a computing device and a radio frequency (RF) transceiver coupled to a system bus of the computing device. Processing of the digital samples is carried out by one or more cores of a multi-core processor to implement a software-defined radio.

U.S. Pat. No. 8,064,840 for "Method and system for determining spectrum availability within a network" by inventors McHenry, et al., filed Jun. 18, 2009, discloses an invention which determines spectrum holes for a communication network by accumulating the information obtained from previous received signals to determine the presence of a larger spectrum hole that allows a reduced listening period, higher transmit power and a reduced probability of interference with other networks and transmitters.

U.S. Pat. No. 8,326,313 for "Method and system for dynamic spectrum access using detection periods" by inventors McHenry, et al., filed Aug. 14, 2009, discloses methods and systems for dynamic spectrum access (DSA) in a wireless network. A DSA-enabled device may sense spectrum use in a region and, based on the detected spectrum use, select one or more communication channels for use. The devices also may detect one or more other DSA-enabled devices with which they can form DSA networks. A DSA network may monitor spectrum use by cooperative and non-cooperative devices, to dynamically select one or more channels to use for communication while avoiding or reducing interference with other devices.

U.S. Publication No. 2009/0143019 for "Method and apparatus for distributed spectrum sensing for wireless communication" by inventor Stephen J. Shellhammer, filed Jan. 4, 2008, discloses methods and apparatus for determining if a licensed signal having or exceeding a predetermined field strength is present in a wireless spectrum. The signal of interest maybe a television signal or a wireless microphone signal using licensed television spectrum.

U.S. Publication No. 2013/0090071 for "Systems and methods for communication in a white space" by inventors Abraham, et al., filed Apr. 3, 2012, discloses systems, methods, and devices to communicate in a white space. In some aspects, wireless communication transmitted in the white space authorizes an initial transmission by a device. The wireless communication may include power information for determining a power at which to transmit the initial transmission. The initial transmission may be used to request information identifying one or more channels in the white space available for transmitting data.

U.S. Publication No. 2012/0072986 for "Methods for detecting and classifying signals transmitted over a radio frequency spectrum" by inventors Livsics, et al., filed Nov. 1, 2011, discloses a method to classify a signal as non-cooperative (NC) or a target signal. The percentage of power above a first threshold is computed for a channel. Based on the percentage, a signal is classified as a narrowband signal. If the percentage indicates the absence of a narrowband signal, then a lower second threshold is applied to confirm the absence according to the percentage of power above the second threshold. The signal is classified as a narrowband signal or pre-classified as a wideband signal based on the percentage. Pre-classified wideband signals are classified as a wideband NC signal or target signal using spectrum masks.

U.S. Pat. No. 8,326,240 for "System for specific emitter identification" by inventors Kadambe, et al., filed Sep. 27, 2010, describes an apparatus for identifying a specific emitter in the presence of noise and/or interference including (a) a sensor configured to sense radio frequency signal and noise data, (b) a reference estimation unit configured to estimate a reference signal relating to the signal transmitted by one emitter, (c) a feature estimation unit configured to generate one or more estimates of one or more feature from the reference signal and the signal transmitted by that particular emitter, and (d) an emitter identifier configured to identify the signal transmitted by that particular emitter as belonging to a specific device (e.g., devices using Gaussian Mixture Models and the Bayesian decision engine). The apparatus may also include an SINR enhancement unit configured to enhance the SINR of the data before the reference estimation unit estimates the reference signal.

U.S. Pat. No. 7,835,319 for "System and method for identifying wireless devices using pulse fingerprinting and sequence analysis" by inventor Sugar, filed May 9, 2007, discloses methods for identifying devices that are sources of

wireless signals from received radio frequency (RF) energy, and, particularly, sources emitting frequency hopping spread spectrum (FHSS). Pulse metric data is generated from the received RF energy and represents characteristics associated thereto. The pulses are partitioned into groups based on their pulse metric data such that a group comprises pulses having similarities for at least one item of pulse metric data. Sources of the wireless signals are identified based on the partitioning process. The partitioning process involves iteratively subdividing each group into subgroups until all resulting subgroups contain pulses determined to be from a single source. At each iteration, subdividing is performed based on different pulse metric data than at a prior iteration. Ultimately, output data is generated (e.g., a device name for display) that identifies a source of wireless signals for any subgroup that is determined to contain pulses from a single source.

U.S. Pat. No. 8,131,239 for “Method and apparatus for remote detection of radio-frequency devices” by inventors Walker, et al., filed Aug. 21, 2007, describes methods and apparatus for detecting the presence of electronic communications devices, such as cellular phones, including a complex RF stimulus is transmitted into a target area, and nonlinear reflection signals received from the target area are processed to obtain a response measurement. The response measurement is compared to a pre-determined filter response profile to detect the presence of a radio device having a corresponding filter response characteristic. In some embodiments, the pre-determined filter response profile comprises a pre-determined band-edge profile, so that comparing the response measurement to a pre-determined filter response profile comprises comparing the response measurement to the pre-determined band-edge profile to detect the presence of a radio device having a corresponding band-edge characteristic. Invention aims to be useful in detecting hidden electronic devices.

U.S. Pat. No. 8,369,305 for “Correlating multiple detections of wireless devices without a unique identifier” by inventors Diener, et al., filed Jun. 30, 2008, describes at a plurality of first devices, wireless transmissions are received at different locations in a region where multiple target devices may be emitting, and identifier data is subsequently generated. Similar identifier data associated with received emissions at multiple first devices are grouped together into a cluster record that potentially represents the same target device detected by multiple first devices. Data is stored that represents a plurality of cluster records from identifier data associated with received emissions made over time by multiple first devices. The cluster records are analyzed over time to correlate detections of target devices across multiple first devices. It aims to lessen disruptions caused by devices using the same frequency and to protect data.

U.S. Pat. No. 8,155,649 for “Method and system for classifying communication signals in a dynamic spectrum access system” by inventors McHenry, et al., filed Aug. 14, 2009, discloses methods and systems for dynamic spectrum access (DSA) in a wireless network wherein a DSA-enabled device may sense spectrum use in a region and, based on the detected spectrum use, select one or more communication channels for use. The devices also may detect one or more other DSA-enabled devices with which they can form DSA networks. A DSA network may monitor spectrum use by cooperative and non-cooperative devices, to dynamically select one or more channels to use for communication while avoiding or reducing interference with other devices. A DSA network may include detectors such as a narrow-band detec-

tor, wide-band detector, TV detector, radar detector, a wireless microphone detector, or any combination thereof.

U.S. Pat. No. 8,494,464 for “Cognitive networked electronic warfare” by inventors Kadambe, et al., filed Sep. 8, 2010, describes an apparatus for sensing and classifying radio communications including sensor units configured to detect RF signals, a signal classifier configured to classify the detected RF signals into a classification, the classification including at least one known signal type and an unknown signal type, a clustering learning algorithm capable of finding clusters of common signals among the previously seen unknown signals; it is then further configured to use these clusters to retrain the signal classifier to recognize these signals as a new signal type, aiming to provide signal identification to better enable electronic attacks and jamming signals.

U.S. Publication No. 2011/0059747 for “Sensing Wireless Transmissions From a Licensed User of a Licensed Spectral Resource” by inventors Lindoff, et al., filed Sep. 7, 2009, describes sensing wireless transmissions from a licensed user of a licensed spectral resource includes obtaining information indicating a number of adjacent sensors that are concurrently sensing wireless transmissions from the licensed user of the licensed spectral resource. Such information can be obtained from a main node controlling the sensor and its adjacent sensors, or by the sensor itself (e.g., by means of short-range communication equipment targeting any such adjacent sensors). A sensing rate is then determined as a function, at least in part, of the information indicating the number of adjacent sensors that are concurrently sensing wireless transmissions from the licensed user of the licensed spectral resource. Receiver equipment is then periodically operated at the determined sensing rate, wherein the receiver equipment is configured to detect wireless transmissions from the licensed user of the licensed spectral resource.

U.S. Pat. No. 8,463,195 for “Methods and apparatus for spectrum sensing of signal features in a wireless channel” by inventor Shellhammer, filed Nov. 13, 2009, discloses methods and apparatus for sensing features of a signal in a wireless communication system are disclosed. The disclosed methods and apparatus sense signal features by determining a number of spectral density estimates, where each estimate is derived based on reception of the signal by a respective antenna in a system with multiple sensing antennas. The spectral density estimates are then combined, and the signal features are sensed based on the combination of the spectral density estimates. Invention aims to increase sensing performance by addressing problems associated with Rayleigh fading, which causes signals to be less detectable.

U.S. Pat. No. 8,151,311 for “System and method of detecting potential video traffic interference” by inventors Huffman, et al., filed Nov. 30, 2007, describes a method of detecting potential video traffic interference at a video head-end of a video distribution network is disclosed and includes detecting, at a video head-end, a signal populating an ultra-high frequency (UHF) white space frequency. The method also includes determining that a strength of the signal is equal to or greater than a threshold signal strength. Further, the method includes sending an alert from the video head-end to a network management system. The alert indicates that the UHF white space frequency is populated by a signal having a potential to interfere with video traffic delivered via the video head-end. Cognitive radio technology, various sensing mechanisms (energy sensing, National

Television System Committee signal sensing, Advanced Television Systems Committee sensing), filtering, and signal reconstruction are disclosed.

U.S. Pat. No. 8,311,509 for "Detection, communication and control in multimode cellular, TDMA, GSM, spread spectrum, CDMA, OFDM, WiLAN, and WiFi systems" by inventor Feher, filed Oct. 31, 2007, teaches a device for detection of signals, with location finder or location tracker or navigation signal and with Modulation Demodulation (Modem) Format Selectable (MFS) communication signal. Processor for processing a digital signal into cross-correlated in-phase and quadrature-phase filtered signal and for processing a voice signal into Orthogonal Frequency Division Multiplexed (OFDM) or Orthogonal Frequency Division Multiple Access (OFDMA) signal. Each is used in a Wireless Local Area Network (WLAN) and in Voice over Internet Protocol (VOIP) network. Device and location finder with Time Division Multiple Access (TDMA), Global Mobile System (GSM) and spread spectrum Code Division Multiple Access (CDMA) is used in a cellular network. Polar and quadrature modulator and two antenna transmitter for transmission of provided processed signal. Transmitter with two amplifiers operated in separate radio frequency (RF) bands. One transmitter is operated as a Non-Linearly Amplified (NLA) transmitter and the other transmitter is operated as a linearly amplified or linearized amplifier transmitter.

U.S. Pat. No. 8,514,729 for "Method and system for analyzing RF signals in order to detect and classify actively transmitting RF devices" by inventor Blackwell, filed Apr. 3, 2009, discloses methods and apparatuses to analyze RF signals in order to detect and classify RF devices in wireless networks are described. The method includes detecting one or more radio frequency (RF) samples; determining burst data by identifying start and stop points of the one or more RF samples; comparing time domain values for an individual burst with time domain values of one or more predetermined RF device profiles; generating a human-readable result indicating whether the individual burst should be assigned to one of the predetermined RF device profiles; and, classifying the individual burst if assigned to one of the predetermined RF device profiles as being a WiFi device or a non-WiFi device with the non-WiFi device being a RF interference source to a wireless network.

However, none of the prior art references provide solutions to the limitations and longstanding unmet needs existing in this area for automatically identifying open space in a wireless communications spectrum. Thus, there remains a need for automated identification of open space in a wireless communications spectrum in near real-time.

SUMMARY OF THE INVENTION

The present invention addresses the longstanding, unmet needs existing in the prior art and commercial sectors to provide solutions to the at least four major problems existing before the present invention. The present invention relates to systems, methods, and devices of the various embodiments enable spectrum management by identifying, classifying, and cataloging signals of interest based on radio frequency measurements. In an embodiment, signals and the parameters of the signals may be identified and indications of available frequencies may be presented to a user. In another embodiment, the protocols of signals may also be identified. In a further embodiment, the modulation of signals, data types carried by the signals, and estimated signal origins may be identified.

It is an object of this invention is to provide an apparatus for identifying signal emitting devices including: a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device, and then to identify open space available for wireless communications, based upon the information about the signal emitting device(s) operating in the predetermined spectrum.

The present invention further provides systems for identifying white space in wireless communications spectrum by detecting and analyzing signals from any signal emitting devices including at least one apparatus, wherein the at least one apparatus is operable for network-based communication with at least one server computer including a database, and/or with at least one other apparatus, but does not require a connection to the at least one server computer to be operable for identifying signal emitting devices; wherein each of the apparatus is operable for identifying signal emitting devices including: a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device, and then to identify open space available for wireless communications, based upon the information about the signal emitting device(s) operating in the predetermined spectrum.

The present invention is further directed to a method for identifying open space in a wireless communications spectrum including the steps of: providing a device for measuring characteristics of signals from signal emitting devices in a spectrum associated with wireless communications, with measured data characteristics including frequency, power, bandwidth, duration, modulation, and combinations thereof; the device including a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals within the spectrum; and further including the following steps performed within the device housing: assessing whether the measured data includes analog and/or digital signal(s); determining a best fit based on frequency, if the measured power spectrum is designated in an historical or a reference database(s) for frequency ranges; automatically determining a category for either analog or digital signals, based on power and sideband combined with frequency allocation; determining a TDM/FDM/CDM signal, based on duration and bandwidth; identifying at least one signal emitting device from the composite results of the foregoing steps; and then automatically identifying the open space available for wireless communications, based upon the information about the signal emitting device(s) operating in the predetermined spectrum.

Additionally, the present invention provides systems, apparatus, and methods for identifying open space in a wireless communications spectrum using an apparatus having a multiplicity of processors and memory, sensors, and communications transmitters and receivers, all constructed and configured within a housing for automated analysis of detected signals from signal emitting devices, determination

of signal duration and other signal characteristics, and automatically generating information relating to open space within the spectrum for wireless communication.

These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate exemplary embodiments of the invention, and together with the general description given above and the detailed description given below, serve to explain the features of the invention.

FIG. 1 is a system block diagram of a wireless environment suitable for use with the various embodiments.

FIG. 2A is a block diagram of a spectrum management device according to an embodiment.

FIG. 2B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to an embodiment.

FIG. 3 is a process flow diagram illustrating an embodiment method for identifying a signal.

FIG. 4 is a process flow diagram illustrating an embodiment method for measuring sample blocks of a radio frequency scan.

FIGS. 5A-5C are a process flow diagram illustrating an embodiment method for determining signal parameters.

FIG. 6 is a process flow diagram illustrating an embodiment method for displaying signal identifications.

FIG. 7 is a process flow diagram illustrating an embodiment method for displaying one or more open frequency.

FIG. 8A is a block diagram of a spectrum management device according to another embodiment.

FIG. 8B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to another embodiment.

FIG. 9 is a process flow diagram illustrating an embodiment method for determining protocol data and symbol timing data.

FIG. 10 is a process flow diagram illustrating an embodiment method for calculating signal degradation data.

FIG. 11 is a process flow diagram illustrating an embodiment method for displaying signal and protocol identification information.

FIG. 12A is a block diagram of a spectrum management device according to a further embodiment.

FIG. 12B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to a further embodiment.

FIG. 13 is a process flow diagram illustrating an embodiment method for estimating a signal origin based on a frequency difference of arrival.

FIG. 14 is a process flow diagram illustrating an embodiment method for displaying an indication of an identified data type within a signal.

FIG. 15 is a process flow diagram illustrating an embodiment method for determining modulation type, protocol data, and symbol timing data.

FIG. 16 is a process flow diagram illustrating an embodiment method for tracking a signal origin.

FIG. 17 is a schematic diagram illustrating an embodiment for scanning and finding open space.

FIG. 18 is a diagram of an embodiment wherein software defined radio nodes are in communication with a master transmitter and device sensing master.

FIG. 19 is a process flow diagram of an embodiment method of temporally dividing up data into intervals for power usage analysis.

FIG. 20 is a flow diagram illustrating an embodiment wherein frequency to license matching occurs.

FIG. 21 is a flow diagram illustrating an embodiment method for reporting power usage information.

FIG. 22 is a flow diagram illustrating an embodiment method for creating frequency arrays.

FIG. 23 is a flow diagram illustrating an embodiment method for reframe and aggregating power when producing frequency arrays.

FIG. 24 is a flow diagram illustrating an embodiment method of reporting license expirations.

FIG. 25 is a flow diagram illustrating an embodiment method of reporting frequency power use.

FIG. 26 is a flow diagram illustrating an embodiment method of connecting devices.

FIG. 27 is a flow diagram illustrating an embodiment method of addressing collisions.

FIG. 28 is a schematic diagram of an embodiment of the invention illustrating a virtualized computing network and a plurality of distributed devices.

DETAILED DESCRIPTION

Priority U.S. Patent Applications 61/789,758, Ser. Nos. 13/912,683, 13/912,893, 13/913,013, 14/028,873, 14/082,916, 14/082,930, 14/086,871, 14/273,157 and 14/329,838 each are herein incorporated by reference in their entirety.

Referring now to the drawings in general, the illustrations are for the purpose of describing at least one preferred embodiment and/or examples of the invention and are not intended to limit the invention thereto. Various embodiments are described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers are used throughout the drawings to refer to the same or like parts. References made to particular examples and implementations are for illustrative purposes, and are not intended to limit the scope of the invention or the claims.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

The present invention provides systems, methods, and devices for spectrum analysis and management by identifying, classifying, and cataloging at least one or a multiplicity of signals of interest based on radio frequency measurements and location and other measurements, and using near real-time parallel processing of signals and their corresponding parameters and characteristics in the context of historical and static data for a given spectrum.

The systems, methods and apparatus according to the present invention preferably have the ability to detect in near real-time, and more preferably to detect, sense, measure, and/or analyze in near real-time, and more preferably to perform any near real-time operations within about 1 second or less. Advantageously, the present invention and its real-time functionality described herein uniquely provide and enable the apparatus units to compare to historical data, to update data and/or information, and/or to provide more data

and/or information on the open space, on the device that may be occupying the open space, and combinations, in the near real-time compared with the historically scanned (15 min to 30 days) data, or historical database information.

The systems, methods, and devices of the various embodiments enable spectrum management by identifying, classifying, and cataloging signals of interest based on radio frequency measurements. In an embodiment, signals and the parameters of the signals may be identified and indications of available frequencies may be presented to a user. In another embodiment, the protocols of signals may also be identified. In a further embodiment, the modulation of signals, data types carried by the signals, and estimated signal origins may be identified.

Embodiments are directed to a spectrum management device that may be configurable to obtain spectrum data over a wide range of wireless communication protocols. Embodiments may also provide for the ability to acquire data from and sending data to database depositories that may be used by a plurality of spectrum management customers.

In one embodiment, a spectrum management device may include a signal spectrum analyzer that may be coupled with a database system and spectrum management interface. The device may be portable or may be a stationary installation and may be updated with data to allow the device to manage different spectrum information based on frequency, bandwidth, signal power, time, and location of signal propagation, as well as modulation type and format and to provide signal identification, classification, and geo-location. A processor may enable the device to process spectrum power density data as received and to process raw I/Q complex data that may be used for further signal processing, signal identification, and data extraction.

In an embodiment, a spectrum management device may comprise a low noise amplifier that receives a radio frequency (RF) energy from an antenna. The antenna may be any antenna structure that is capable of receiving RF energy in a spectrum of interest. The low noise amplifier may filter and amplify the RF energy. The RF energy may be provided to an RF translator. The RF translator may perform a fast Fourier transform (FFT) and either a square magnitude or a fast convolution spectral periodogram function to convert the RF measurements into a spectral representation. In an embodiment, the RF translator may also store a timestamp to facilitate calculation of a time of arrival and an angle of arrival. The In-Phase and Quadrature (I/Q) data may be provided to a spectral analysis receiver or it may be provided to a sample data store where it may be stored without being processed by a spectral analysis receiver. The input RF energy may also be directly digital down-converted and sampled by an analog to digital converter (ADC) to generate complex I/Q data. The complex I/Q data may be equalized to remove multipath, fading, white noise and interference from other signaling systems by fast parallel adaptive filter processes. This data may then be used to calculate modulation type and baud rate. Complex sampled I/Q data may also be used to measure the signal angle of arrival and time of arrival. Such information as angle of arrival and time of arrival may be used to compute more complex and precise direction finding. In addition, they may be used to apply geo-location techniques. Data may be collected from known signals or unknown signals and time spaced in order to provide expedient information. I/Q sampled data may contain raw signal data that may be used to demodulate and translate signals by streaming them to a signal analyzer or to a real-time demodulator software defined radio that may have the newly identified signal parameters for the signal of

interest. The inherent nature of the input RF allows for any type of signal to be analyzed and demodulated based on the reconfiguration of the software defined radio interfaces.

A spectral analysis receiver may be configured to read raw In-Phase (I) and Quadrature (Q) data and either translate directly to spectral data or down convert to an intermediate frequency (IF) up to half the Nyquist sampling rate to analyze the incoming bandwidth of a signal. The translated spectral data may include measured values of signal energy, frequency, and time. The measured values provide attributes of the signal under review that may confirm the detection of a particular signal of interest within a spectrum of interest. In an embodiment, a spectral analysis receiver may have a referenced spectrum input of 0 Hz to 12.4 GHz with capability of fiber optic input for spectrum input up to 60 GHz.

In an embodiment, the spectral analysis receiver may be configured to sample the input RF data by fast analog down-conversion of the RF signal. The down-converted signal may then be digitally converted and processed by fast convolution filters to obtain a power spectrum. This process may also provide spectrum measurements including the signal power, the bandwidth, the center frequency of the signal as well as a Time of Arrival (TOA) measurement. The TOA measurement may be used to create a timestamp of the detected signal and/or to generate a time difference of arrival iterative process for direction finding and fast triangulation of signals. In an embodiment, the sample data may be provided to a spectrum analysis module. In an embodiment, the spectrum analysis module may evaluate the sample data to obtain the spectral components of the signal.

In an embodiment, the spectral components of the signal may be obtained by the spectrum analysis module from the raw I/Q data as provided by an RF translator. The I/Q data analysis performed by the spectrum analysis module may operate to extract more detailed information about the signal, including by way of example, modulation type (e.g., FM, AM, QPSK, 16QAM, etc.) and/or protocol (e.g., GSM, CDMA, OFDM, LTE, etc.). In an embodiment, the spectrum analysis module may be configured by a user to obtain specific information about a signal of interest. In an alternate embodiment, the spectral components of the signal may be obtained from power spectral component data produced by the spectral analysis receiver.

In an embodiment, the spectrum analysis module may provide the spectral components of the signal to a data extraction module. The data extraction module may provide the classification and categorization of signals detected in the RF spectrum. The data extraction module may also acquire additional information regarding the signal from the spectral components of the signal. For example, the data extraction module may provide modulation type, bandwidth, and possible system in use information. In another embodiment, the data extraction module may select and organize the extracted spectral components in a format selected by a user.

The information from the data extraction module may be provided to a spectrum management module. The spectrum management module may generate a query to a static database to classify a signal based on its components. For example, the information stored in static database may be used to determine the spectral density, center frequency, bandwidth, baud rate, modulation type, protocol (e.g., GSM, CDMA, OFDM, LTE, etc.), system or carrier using licensed spectrum, location of the signal source, and a timestamp of the signal of interest. These data points may be provided to a data store for export. In an embodiment and as more fully

13

described below, the data store may be configured to access mapping software to provide the user with information on the location of the transmission source of the signal of interest. In an embodiment, the static database includes frequency information gathered from various sources including, but not limited to, the Federal Communication Commission, the International Telecommunication Union, and data from users. As an example, the static database may be an SQL database. The data store may be updated, downloaded or merged with other devices or with its main relational database. Software API applications may be included to allow database merging with third-party spectrum databases that may only be accessed securely.

In the various embodiments, the spectrum management device may be configured in different ways. In an embodiment, the front end of system may comprise various hardware receivers that may provide In-Phase and Quadrature complex data. The front end receiver may include API set commands via which the system software may be configured to interface (i.e., communicate) with a third party receiver. In an embodiment, the front end receiver may perform the spectral computations using FFT (Fast Fourier Transform) and other DSP (Digital Signal Processing) to generate a fast convolution periodogram that may be re-sampled and averaged to quickly compute the spectral density of the RF environment.

In an embodiment, cyclic processes may be used to average and correlate signal information by extracting the changes inside the signal to better identify the signal of interest that is present in the RF space. A combination of amplitude and frequency changes may be measured and averaged over the bandwidth time to compute the modulation type and other internal changes, such as changes in frequency offsets, orthogonal frequency division modulation, changes in time (e.g., Time Division Multiplexing), and/or changes in I/Q phase rotation used to compute the baud rate and the modulation type. In an embodiment, the spectrum management device may have the ability to compute several processes in parallel by use of a multi-core processor and along with several embedded field programmable gate arrays (FPGA). Such multi-core processing may allow the system to quickly analyze several signal parameters in the RF environment at one time in order to reduce the amount of time it takes to process the signals. The amount of signals computed at once may be determined by their bandwidth requirements. Thus, the capability of the system may be based on a maximum frequency $F_s/2$. The number of signals to be processed may be allocated based on their respective bandwidths. In another embodiment, the signal spectrum may be measured to determine its power density, center frequency, bandwidth and location from which the signal is emanating and a best match may be determined based on the signal parameters based on information criteria of the frequency.

In another embodiment, a GPS and direction finding location (DF) system may be incorporated into the spectrum management device and/or available to the spectrum management device. Adding GPS and DF ability may enable the user to provide a location vector using the National Marine Electronics Association's (NMEA) standard form. In an embodiment, location functionality is incorporated into a specific type of GPS unit, such as a U.S. government issued receiver. The information may be derived from the location presented by the database internal to the device, a database imported into the device, or by the user inputting geolocation parameters of longitude and latitude which may be derived as degrees, minutes and seconds, decimal minutes,

14

or decimal form and translated to the necessary format with the default being 'decimal' form. This functionality may be incorporated into a GPS unit. The signal information and the signal classification may then be used to locate the signaling device as well as to provide a direction finding capability.

A type of triangulation using three units as a group antenna configuration performs direction finding by using multilateration. Commonly used in civil and military surveillance applications, multilateration is able to accurately locate an aircraft, vehicle, or stationary emitter by measuring the "Time Difference of Arrival" (TDOA) of a signal from the emitter at three or more receiver sites. If a pulse is emitted from a platform, it will arrive at slightly different times at two spatially separated receiver sites, the TDOA being due to the different distances of each receiver from the platform. This location information may then be supplied to a mapping process that utilizes a database of mapping images that are extracted from the database based on the latitude and longitude provided by the geo-location or direction finding device. The mapping images may be scanned in to show the points of interest where a signal is either expected to be emanating from based on the database information or from an average taken from the database information and the geo-location calculation performed prior to the mapping software being called. The user can control the map to maximize or minimize the mapping screen to get a better view which is more fit to provide information of the signal transmissions. In an embodiment, the mapping process does not rely on outside mapping software. The mapping capability has the ability to generate the map image and to populate a mapping database that may include information from third party maps to meet specific user requirements.

In an embodiment, triangulation and multilateration may utilize a Bayesian type filter that may predict possible movement and future location and operation of devices based on input collected from the TDOA and geolocation processes and the variables from the static database pertaining to the specified signal of interest. The Bayesian filter takes the input changes in time difference and its inverse function (i.e., frequency difference) and takes an average changes in signal variation to detect and predict the movement of the signals. The signal changes are measured within 1 ns time difference and the filter may also adapt its gradient error calculation to remove unwanted signals that may cause errors due to signal multipath, inter-symbol interference, and other signal noise.

In an embodiment the changes within a 1 ns time difference for each sample for each unique signal may be recorded. The spectrum management device may then perform the inverse and compute and record the frequency difference and phase difference between each sample for each unique signal. The spectrum management device may take the same signal and calculates an error based on other input signals coming in within the 1 ns time and may average and filter out the computed error to equalize the signal. The spectrum management device may determine the time difference and frequency difference of arrival for that signal and compute the odds of where the signal is emanating from based on the frequency band parameters presented from the spectral analysis and processor computations, and determines the best position from which the signal is transmitted (i.e., origin of the signal).

FIG. 1 illustrates a wireless environment 100 suitable for use with the various embodiments. The wireless environment 100 may include various sources 104, 106, 108, 110, 112, and 114 generating various radio frequency (RF) sig-

15

nals **116**, **118**, **120**, **122**, **124**, **126**. As an example, mobile devices **104** may generate cellular RF signals **116**, such as CDMA, GSM, 3G signals, etc. As another example, wireless access devices **106**, such as Wi-Fi® routers, may generate RF signals **118**, such as Wi-Fi® signals. As a further example, satellites **108**, such as communication satellites or GPS satellites, may generate RF signals **120**, such as satellite radio, television, or GPS signals. As a still further example, base stations **110**, such as a cellular base station, may generate RF signals **122**, such as CDMA, GSM, 3G signals, etc. As another example, radio towers **112**, such as local AM or FM radio stations, may generate RF signals **124**, such as AM or FM radio signals. As another example, government service provides **114**, such as police units, fire fighters, military units, air traffic control towers, etc. may generate RF signals **126**, such as radio communications, tracking signals, etc. The various RF signals **116**, **118**, **120**, **122**, **124**, **126** may be generated at different frequencies, power levels, in different protocols, with different modulations, and at different times. The various sources **104**, **106**, **108**, **110**, **112**, and **114** may be assigned frequency bands, power limitations, or other restrictions, requirements, and/or licenses by a government spectrum control entity, such as the FCC. However, with so many different sources **104**, **106**, **108**, **110**, **112**, and **114** generating so many different RF signals **116**, **118**, **120**, **122**, **124**, **126**, overlaps, interference, and/or other problems may occur. A spectrum management device **102** in the wireless environment **100** may measure the RF energy in the wireless environment **100** across a wide spectrum and identify the different RF signals **116**, **118**, **120**, **122**, **124**, **126** which may be present in the wireless environment **100**. The identification and cataloging of the different RF signals **116**, **118**, **120**, **122**, **124**, **126** which may be present in the wireless environment **100** may enable the spectrum management device **102** to determine available frequencies for use in the wireless environment **100**. In addition, the spectrum management device **102** may be able to determine if there are available frequencies for use in the wireless environment **100** under certain conditions (i.e., day of week, time of day, power level, frequency band, etc.). In this manner, the RF spectrum in the wireless environment **100** may be managed.

FIG. 2A is a block diagram of a spectrum management device **202** according to an embodiment. The spectrum management device **202** may include an antenna structure **204** configured to receive RF energy expressed in a wireless environment. The antenna structure **204** may be any type antenna, and may be configured to optimize the receipt of RF energy across a wide frequency spectrum. The antenna structure **204** may be connected to one or more optional amplifiers and/or filters **208** which may boost, smooth, and/or filter the RF energy received by antenna structure **204** before the RF energy is passed to an RF receiver **210** connected to the antenna structure **204**. In an embodiment, the RF receiver **210** may be configured to measure the RF energy received from the antenna structure **204** and/or optional amplifiers and/or filters **208**. In an embodiment, the RF receiver **210** may be configured to measure RF energy in the time domain and may convert the RF energy measurements to the frequency domain. In an embodiment, the RF receiver **210** may be configured to generate spectral representation data of the received RF energy. The RF receiver **210** may be any type RF receiver, and may be configured to generate RF energy measurements over a range of frequencies, such as 0 kHz to 24 GHz, 9 kHz to 6 GHz, etc. In an embodiment, the frequency scanned by the RF receiver **210** may be user selectable. In an embodiment, the RF receiver

16

210 may be connected to a signal processor **214** and may be configured to output RF energy measurements to the signal processor **214**. As an example, the RF receiver **210** may output raw In-Phase (I) and Quadrature (Q) data to the signal processor **214**. As another example, the RF receiver **210** may apply signals processing techniques to output complex In-Phase (I) and Quadrature (Q) data to the signal processor **214**. In an embodiment, the spectrum management device may also include an antenna **206** connected to a location receiver **212**, such as a GPS receiver, which may be connected to the signal processor **214**. The location receiver **212** may provide location inputs to the signal processor **214**.

The signal processor **214** may include a signal detection module **216**, a comparison module **222**, a timing module **224**, and a location module **225**. Additionally, the signal processor **214** may include an optional memory module **226** which may include one or more optional buffers **228** for storing data generated by the other modules of the signal processor **214**.

In an embodiment, the signal detection module **216** may operate to identify signals based on the RF energy measurements received from the RF receiver **210**. The signal detection module **216** may include a Fast Fourier Transform (FFT) module **217** which may convert the received RF energy measurements into spectral representation data. The signal detection module **216** may include an analysis module **221** which may analyze the spectral representation data to identify one or more signals above a power threshold. A power module **220** of the signal detection module **216** may control the power threshold at which signals may be identified. In an embodiment, the power threshold may be a default power setting or may be a user selectable power setting. A noise module **219** of the signal detection module **216** may control a signal threshold, such as a noise threshold, at or above which signals may be identified. The signal detection module **216** may include a parameter module **218** which may determine one or more signal parameters for any identified signals, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, etc. In an embodiment, the signal processor **214** may include a timing module **224** which may record time information and provide the time information to the signal detection module **216**. Additionally, the signal processor **214** may include a location module **225** which may receive location inputs from the location receiver **212** and determine a location of the spectrum management device **202**. The location of the spectrum management device **202** may be provided to the signal detection module **216**.

In an embodiment, the signal processor **214** may be connected to one or more memory **230**. The memory **230** may include multiple databases, such as a history or historical database **232** and characteristics listing **236**, and one or more buffers **240** storing data generated by signal processor **214**. While illustrated as connected to the signal processor **214** the memory **230** may also be on chip memory residing on the signal processor **214** itself. In an embodiment, the history or historical database **232** may include measured signal data **234** for signals that have been previously identified by the spectrum management device **202**. The measured signal data **234** may include the raw RF energy measurements, time stamps, location information, one or more signal parameters for any identified signals, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, etc., and identifying information determined from the characteristics listing **236**. In an embodiment, the history

or historical database 232 may be updated as signals are identified by the spectrum management device 202. In an embodiment, the characteristic listing 236 may be a database of static signal data 238. The static signal data 238 may include data gathered from various sources including by way of example and not by way of limitation the Federal Communication Commission, the International Telecommunication Union, telecom providers, manufacture data, and data from spectrum management device users. Static signal data 238 may include known signal parameters of transmitting devices, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, geographic information for transmitting devices, and any other data that may be useful in identifying a signal. In an embodiment, the static signal data 238 and the characteristic listing 236 may correlate signal parameters and signal identifications. As an example, the static signal data 238 and characteristic listing 236 may list the parameters of the local fire and emergency communication channel correlated with a signal identification indicating that signal is the local fire and emergency communication channel.

In an embodiment, the signal processor 214 may include a comparison module 222 which may match data generated by the signal detection module 216 with data in the history or historical database 232 and/or characteristic listing 236. In an embodiment the comparison module 222 may receive signal parameters from the signal detection module 216, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, and/or receive parameter from the timing module 224 and/or location module 225. The parameter match module 223 may retrieve data from the history or historical database 232 and/or the characteristic listing 236 and compare the retrieved data to any received parameters to identify matches. Based on the matches the comparison module may identify the signal. In an embodiment, the signal processor 214 may be optionally connected to a display 242, an input device 244, and/or network transceiver 246. The display 242 may be controlled by the signal processor 214 to output spectral representations of received signals, signal characteristic information, and/or indications of signal identifications on the display 242. In an embodiment, the input device 244 may be any input device, such as a keyboard and/or knob, mouse, virtual keyboard or even voice recognition, enabling the user of the spectrum management device 202 to input information for use by the signal processor 214. In an embodiment, the network transceiver 246 may enable the spectrum management device 202 to exchange data with wired and/or wireless networks, such as to update the characteristic listing 236 and/or upload information from the history or historical database 232.

FIG. 2B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device 202 according to an embodiment. A receiver 210 may output RF energy measurements, such as I and Q data to a FFT module 252 which may generate a spectral representation of the RF energy measurements which may be output on a display 242. The I and Q data may also be buffered in a buffer 256 and sent to a signal detection module 216. The signal detection module 216 may receive location inputs from a location receiver 212 and use the received I and Q data to detect signals. Data from the signal detection module 216 may be buffered in a buffer 262 and written into a history or historical database 232. Additionally, data from the historical database may be used to aid in the detection of signals by the signal detection

module 216. The signal parameters of the detected signals may be determined by a signal parameters module 218 using information from the history or historical database 232 and/or a static database 238 listing signal characteristics. Data from the signal parameters module 218 may be buffered in a buffer 268, stored in the history or historical database 232, and/or sent to the signal detection module 216 and/or display 242. In this manner, signals may be detected and indications of the signal identification may be displayed to a user of the spectrum management device.

FIG. 3 illustrates a process flow of an embodiment method 300 for identifying a signal. In an embodiment the operations of method 300 may be performed by the processor 214 of a spectrum management device 202. In block 302 the processor 214 may determine the location of the spectrum management device 202. In an embodiment, the processor 214 may determine the location of the spectrum management device 202 based on a location input, such as GPS coordinates, received from a location receiver, such as a GPS receiver 212. In block 304 the processor 214 may determine the time. As an example, the time may be the current clock time as determined by the processor 214 and may be a time associated with receiving RF measurements. In block 306 the processor 214 may receive RF energy measurements. In an embodiment, the processor 214 may receive RF energy measurements from an RF receiver 210. In block 308 the processor 214 may convert the RF energy measurements to spectral representation data. As an example, the processor may apply a Fast Fourier Transform (FFT) to the RF energy measurements to convert them to spectral representation data. In optional block 310 the processor 214 may display the spectral representation data on a display 242 of the spectrum management device 202, such as in a graph illustrating amplitudes across a frequency spectrum.

In block 312 the processor 214 may identify one or more signal above a threshold. In an embodiment, the processor 214 may analyze the spectral representation data to identify a signal above a power threshold. A power threshold may be an amplitude measure selected to distinguish RF energies associated with actual signals from noise. In an embodiment, the power threshold may be a default value. In another embodiment, the power threshold may be a user selectable value. In block 314 the processor 214 may determine signal parameters of any identified signal or signals of interest. As examples, the processor 214 may determine signal parameters such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration for the identified signals. In block 316 the processor 214 may store the signal parameters of each identified signal, a location indication, and time indication for each identified signal in a history database 232. In an embodiment, a history database 232 may be a database resident in a memory 230 of the spectrum management device 202 which may include data associated with signals actually identified by the spectrum management device.

In block 318 the processor 214 may compare the signal parameters of each identified signal to signal parameters in a signal characteristic listing. In an embodiment, the signal characteristic listing may be a static database 238 stored in the memory 230 of the spectrum management device 202 which may correlate signal parameters and signal identifications. In determination block 320 the processor 214 may determine whether the signal parameters of the identified signal or signals match signal parameters in the characteristic listing 236. In an embodiment, a match may be determined based on the signal parameters being within a speci-

fied tolerance of one another. As an example, a center frequency match may be determined when the center frequencies are within plus or minus 1 kHz of each other. In this manner, differences between real world measured conditions of an identified signal and ideal conditions listed in a characteristics listing may be accounted for in identifying matches. If the signal parameters do not match (i.e., determination block 320="No"), in block 326 the processor 214 may display an indication that the signal is unidentified on a display 242 of the spectrum management device 202. In this manner, the user of the spectrum management device may be notified that a signal is detected, but has not been positively identified. If the signal parameters do match (i.e., determination block 320="Yes"), in block 324 the processor 214 may display an indication of the signal identification on the display 242. In an embodiment, the signal identification displayed may be the signal identification correlated to the signal parameter in the signal characteristic listing which matched the signal parameter for the identified signal. Upon displaying the indications in blocks 324 or 326 the processor 214 may return to block 302 and cyclically measure and identify further signals of interest.

FIG. 4 illustrates an embodiment method 400 for measuring sample blocks of a radio frequency scan. In an embodiment the operations of method 400 may be performed by the processor 214 of a spectrum management device 202. As discussed above, in blocks 306 and 308 the processor 214 may receive RF energy measurements and convert the RF energy measurements to spectral representation data. In block 402 the processor 214 may determine a frequency range at which to sample the RF spectrum for signals of interest. In an embodiment, a frequency range may be a frequency range of each sample block to be analyzed for potential signals. As an example, the frequency range may be 240 kHz. In an embodiment, the frequency range may be a default value. In another embodiment, the frequency range may be a user selectable value. In block 404 the processor 214 may determine a number (N) of sample blocks to measure. In an embodiment, each sample block may be sized to the determined of default frequency range, and the number of sample blocks may be determined by dividing the spectrum of the measured RF energy by the frequency range. In block 406 the processor 214 may assign each sample block a respective frequency range. As an example, if the determined frequency range is 240 kHz, the first sample block may be assigned a frequency range from 0 kHz to 240 kHz, the second sample block may be assigned a frequency range from 240 kHz to 480 kHz, etc. In block 408 the processor 214 may set the lowest frequency range sample block as the current sample block. In block 409 the processor 214 may measure the amplitude across the set frequency range for the current sample block. As an example, at each frequency interval (such as 1 Hz) within the frequency range of the sample block the processor 214 may measure the received signal amplitude. In block 410 the processor 214 may store the amplitude measurements and corresponding frequencies for the current sample block. In determination block 414 the processor 214 may determine if all sample blocks have been measured. If all sample blocks have not been measured (i.e., determination block 414="No"), in block 416 the processor 214 may set the next highest frequency range sample block as the current sample block. As discussed above, in blocks 409, 410, and 414 the processor 214 may measure and store amplitudes and determine whether all blocks are sampled. If all blocks have been

sampled (i.e., determination block 414="Yes"), the processor 214 may return to block 306 and cyclically measure further sample blocks.

FIGS. 5A, 5B, and 5C illustrate the process flow for an embodiment method 500 for determining signal parameters. In an embodiment the operations of method 500 may be performed by the processor 214 of a spectrum management device 202. Referring to FIG. 5A, in block 502 the processor 214 may receive a noise floor average setting. In an embodiment, the noise floor average setting may be an average noise level for the environment in which the spectrum management device 202 is operating. In an embodiment, the noise floor average setting may be a default setting and/or may be user selectable setting. In block 504 the processor 214 may receive the signal power threshold setting. In an embodiment, the signal power threshold setting may be an amplitude measure selected to distinguish RF energies associated with actual signals from noise. In an embodiment the signal power threshold may be a default value and/or may be a user selectable setting. In block 506 the processor 214 may load the next available sample block. In an embodiment, the sample blocks may be assembled according to the operations of method 400 described above with reference to FIG. 4. In an embodiment, the next available sample block may be an oldest in time sample block which has not been analyzed to determine whether signals of interest are present in the sample block. In block 508 the processor 214 may average the amplitude measurements in the sample block. In determination block 510 the processor 214 may determine whether the average for the sample block is greater than or equal to the noise floor average set in block 502. In this manner, sample blocks including potential signals may be quickly distinguished from sample blocks which may not include potential signals reducing processing time by enabling sample blocks without potential signals to be identified and ignored. If the average for the sample block is lower than the noise floor average (i.e., determination block 510="No"), no signals of interest may be present in the current sample block. In determination block 514 the processor 214 may determine whether a cross block flag is set. If the cross block flag is not set (i.e., determination block 514="No"), in block 506 the processor 214 may load the next available sample block and in block 508 average the sample block 508.

If the average of the sample block is equal to or greater than the noise floor average (i.e., determination block 510="Yes"), the sample block may potentially include a signal of interest and in block 512 the processor 214 may reset a measurement counter (C) to 1. The measurement counter value indicating which sample within a sample block is under analysis. In determination block 516 the processor 214 may determine whether the RF measurement of the next frequency sample (C) is greater than the signal power threshold. In this manner, the value of the measurement counter (C) may be used to control which sample RF measurement in the sample block is compared to the signal power threshold. As an example, when the counter (C) equals 1, the first RF measurement may be checked against the signal power threshold and when the counter (C) equals 2 the second RF measurement in the sample block may be checked, etc. If the C RF measurement is less than or equal to the signal power threshold (i.e., determination block 516="No"), in determination block 517 the processor 214 may determine whether the cross block flag is set. If the cross block flag is not set (i.e., determination block 517="No"), in determination block 522 the processor 214 may determine whether the end of the sample block is

21

reached. If the end of the sample block is reached (i.e., determination block 522="Yes"), in block 506 the processor 214 may load the next available sample block and proceed in blocks 508, 510, 514, and 512 as discussed above. If the end of the sample block is not reached (i.e., determination block 522="No"), in block 524 the processor 214 may increment the measurement counter (C) so that the next sample in the sample block is analyzed.

If the C RF measurement is greater than the signal power threshold (i.e., determination block 516="Yes"), in block 518 the processor 214 may check the status of the cross block flag to determine whether the cross block flag is set. If the cross block flag is not set (i.e., determination block 518="No"), in block 520 the processor 214 may set a sample start. As an example, the processor 214 may set a sample start by indicating a potential signal of interest may be discovered in a memory by assigning a memory location for RF measurements associated with the sample start. Referring to FIG. 5B, in block 526 the processor 214 may store the C RF measurement in a memory location for the sample currently under analysis. In block 528 the processor 214 may increment the measurement counter (C) value.

In determination block 530 the processor 214 may determine whether the C RF measurement (e.g., the next RF measurement because the value of the RF measurement counter was incremented) is greater than the signal power threshold. If the C RF measurement is greater than the signal power threshold (i.e., determination block 530="Yes"), in determination block 532 the processor 214 may determine whether the end of the sample block is reached. If the end of the sample block is not reached (i.e., determination block 532="No"), there may be further RF measurements available in the sample block and in block 526 the processor 214 may store the C RF measurement in the memory location for the sample. In block 528 the processor may increment the measurement counter (C) and in determination block 530 determine whether the C RF measurement is above the signal power threshold and in block 532 determine whether the end of the sample block is reached. In this manner, successive sample RF measurements may be checked against the signal power threshold and stored until the end of the sample block is reached and/or until a sample RF measurement falls below the signal power threshold. If the end of the sample block is reached (i.e., determination block 532="Yes"), in block 534 the processor 214 may set the cross block flag. In an embodiment, the cross block flag may be a flag in a memory available to the processor 214 indicating the signal potential spans across two or more sample blocks. In a further embodiment, prior to setting the cross block flag in block 534, the slope of a line drawn between the last two RF measurement samples may be used to determine whether the next sample block likely contains further potential signal samples. A negative slope may indicate that the signal of interest is fading and may indicate the last sample was the final sample of the signal of interest. In another embodiment, the slope may not be computed and the next sample block may be analyzed regardless of the slope.

If the end of the sample block is reached (i.e., determination block 532="Yes") and in block 534 the cross block flag is set, referring to FIG. 5A, in block 506 the processor 214 may load the next available sample block, in block 508 may average the sample block, and in block 510 determine whether the average of the sample block is greater than or equal to the noise floor average. If the average is equal to or greater than the noise floor average (i.e., determination block 510="Yes"), in block 512 the processor 214 may reset the

22

measurement counter (C) to 1. In determination block 516 the processor 214 may determine whether the C RF measurement for the current sample block is greater than the signal power threshold. If the C RF measurement is greater than the signal power threshold (i.e., determination block 516="Yes"), in determination block 518 the processor 214 may determine whether the cross block flag is set. If the cross block flag is set (i.e., determination block 518="Yes"), referring to FIG. 5B, in block 526 the processor 214 may store the C RF measurement in the memory location for the sample and in block 528 the processor may increment the measurement counter (C). As discussed above, in blocks 530 and 532 the processor 214 may perform operations to determine whether the C RF measurement is greater than the signal power threshold and whether the end of the sample block is reached until the C RF measurement is less than or equal to the signal power threshold (i.e., determination block 530="No") or the end of the sample block is reached (i.e., determination block 532="Yes"). If the end of the sample block is reached (i.e., determination block 532="Yes"), as discussed above in block 534 the cross block flag may be set (or verified and remain set if already set) and in block 535 the C RF measurement may be stored in the sample.

If the end of the sample block is reached (i.e., determination block 532="Yes") and in block 534 the cross block flag is set, referring to FIG. 5A, the processor may perform operations of blocks 506, 508, 510, 512, 516, and 518 as discussed above. If the average of the sample block is less than the noise floor average (i.e., determination block 510="No") and the cross block flag is set (i.e., determination block 514="Yes"), the C RF measurement is less than or equal to the signal power threshold (i.e., determination block 516="No") and the cross block flag is set (i.e., determination block 517="Yes"), or the C RF measurement is less than or equal to the signal power threshold (i.e., determination block 516="No"), referring to FIG. 5B, in block 538 the processor 214 may set the sample stop. As an example, the processor 214 may indicate that a sample end is reached in a memory and/or that a sample is complete in a memory. In block 540 the processor 214 may compute and store complex I and Q data for the stored measurements in the sample. In block 542 the processor 214 may determine a mean of the complex I and Q data. Referring to FIG. 5C, in determination block 544 the processor 214 may determine whether the mean of the complex I and Q data is greater than a signal threshold. If the mean of the complex I and Q data is less than or equal to the signal threshold (i.e., determination block 544="No"), in block 550 the processor 214 may indicate the sample is noise and discard data associated with the sample from memory.

If the mean is greater than the signal threshold (i.e., determination block 544="Yes"), in block 546 the processor 214 may identify the sample as a signal of interest. In an embodiment, the processor 214 may identify the sample as a signal of interest by assigning a signal identifier to the signal, such as a signal number or sample number. In block 548 the processor 214 may determine and store signal parameters for the signal. As an example, the processor 214 may determine and store a frequency peak of the identified signal, a peak power of the identified signal, an average power of the identified signal, a signal bandwidth of the identified signal, and/or a signal duration of the identified signal. In block 552 the processor 214 may clear the cross block flag (or verify that the cross block flag is unset). In block 556 the processor 214 may determine whether the end of the sample block is reached. If the end of the sample block is not reached (i.e., determination block 556="No") in block

23

558 the processor 214 may increment the measurement counter (C), and referring to FIG. 5A in determination block 516 may determine whether the C RF measurement is greater than the signal power threshold. Referring to FIG. 5C, if the end of the sample block is reached (i.e., determination block 556="Yes"), referring to FIG. 5A, in block 506 the processor 214 may load the next available sample block.

FIG. 6 illustrates a process flow for an embodiment method 600 for displaying signal identifications. In an embodiment, the operations of method 600 may be performed by a processor 214 of a spectrum management device 202. In determination block 602 the processor 214 may determine whether a signal is identified. If a signal is not identified (i.e., determination block 602="No"), in block 604 the processor 214 may wait for the next scan. If a signal is identified (i.e., determination block 602="Yes"), in block 606 the processor 214 may compare the signal parameters of an identified signal to signal parameters in a history database 232. In determination block 608 the processor 214 may determine whether signal parameters of the identified signal match signal parameters in the history database 232. If there is no match (i.e., determination block 608="No"), in block 610 the processor 214 may store the signal parameters as a new signal in the history database 232. If there is a match (i.e., determination block 608="Yes"), in block 612 the processor 214 may update the matching signal parameters as needed in the history database 232.

In block 614 the processor 214 may compare the signal parameters of the identified signal to signal parameters in a signal characteristic listing 236. In an embodiment, the characteristic listing 236 may be a static database separate from the history database 232, and the characteristic listing 236 may correlate signal parameters with signal identifications. In determination block 616 the processor 214 may determine whether the signal parameters of the identified signal match any signal parameters in the signal characteristic listing 236. In an embodiment, the match in determination 616 may be a match based on a tolerance between the signal parameters of the identified signal and the parameters in the characteristic listing 236. If there is a match (i.e., determination block 616="Yes"), in block 618 the processor 214 may indicate a match in the history database 232 and in block 622 may display an indication of the signal identification on a display 242. As an example, the indication of the signal identification may be a display of the radio call sign of an identified FM radio station signal. If there is not a match (i.e., determination block 616="No"), in block 620 the processor 214 may display an indication that the signal is an unidentified signal. In this manner, the user may be notified a signal is present in the environment, but that the signal does not match to a signal in the characteristic listing.

FIG. 7 illustrates a process flow of an embodiment method 700 for displaying one or more open frequency. In an embodiment, the operations of method 700 may be performed by the processor 214 of a spectrum management device 202. In block 702 the processor 214 may determine a current location of the spectrum management device 202. In an embodiment, the processor 214 may determine the current location of the spectrum management device 202 based on location inputs received from a location receiver 212, such as GPS coordinates received from a GPS receiver 212. In block 704 the processor 214 may compare the current location to the stored location value in the historical database 232. As discussed above, the historical or history database 232 may be a database storing information about signals previously actually identified by the spectrum management device 202. In determination block 706 the processor

24

processor 214 may determine whether there are any matches between the location information in the historical database 232 and the current location. If there are no matches (i.e., determination block 706="No"), in block 710 the processor 214 may indicate incomplete data is available. In other words the spectrum data for the current location has not previously been recorded.

If there are matches (i.e., determination block 706="Yes"), in optional block 708 the processor 214 may display a plot of one or more of the signals matching the current location. As an example, the processor 214 may compute the average frequency over frequency intervals across a given spectrum and may display a plot of the average frequency over each interval. In block 712 the processor 214 may determine one or more open frequencies at the current location. As an example, the processor 214 may determine one or more open frequencies by determining frequency ranges in which no signals fall or at which the average is below a threshold. In block 714 the processor 214 may display an indication of one or more open frequency on a display 242 of the spectrum management device 202.

FIG. 8A is a block diagram of a spectrum management device 802 according to an embodiment. Spectrum management device 802 is similar to spectrum management device 202 described above with reference to FIG. 2A, except that spectrum management device 802 may include symbol module 816 and protocol module 806 enabling the spectrum management device 802 to identify the protocol and symbol information associated with an identified signal as well as protocol match module 814 to match protocol information. Additionally, the characteristic listing 236 of spectrum management device 802 may include protocol data 804, hardware data 808, environment data 810, and noise data 812 and an optimization module 818 may enable the signal processor 214 to provide signal optimization parameters.

The protocol module 806 may identify the communication protocol (e.g., LTE, CDMA, etc.) associated with a signal of interest. In an embodiment, the protocol module 806 may use data retrieved from the characteristic listing, such as protocol data 804 to help identify the communication protocol. The symbol detector module 816 may determine symbol timing information, such as a symbol rate for a signal of interest. The protocol module 806 and/or symbol module 816 may provide data to the comparison module 222. The comparison module 222 may include a protocol match module 814 which may attempt to match protocol information for a signal of interest to protocol data 804 in the characteristic listing to identify a signal of interest. Additionally, the protocol module 806 and/or symbol module 816 may store data in the memory module 226 and/or history database 232. In an embodiment, the protocol module 806 and/or symbol module 816 may use protocol data 804 and/or other data from the characteristic listing 236 to help identify protocols and/or symbol information in signals of interest.

The optimization module 818 may gather information from the characteristic listing, such as noise figure parameters, antenna hardware parameters, and environmental parameters correlated with an identified signal of interest to calculate a degradation value for the identified signal of interest. The optimization module 818 may further control the display 242 to output degradation data enabling a user of the spectrum management device 802 to optimize a signal of interest.

FIG. 8B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to an embodiment. Only those logical operations illustrated in FIG. 8B different

25

from those described above with reference to FIG. 2B will be discussed. As illustrated in FIG. 8B, as received time tracking 850 may be applied to the I and Q data from the receiver 210. An additional buffer 851 may further store the I and Q data received and a symbol detector 852 may identify the symbols of a signal of interest and determine the symbol rate. A multiple access scheme identifier module 854 may identify whether a signal is part of a multiple access scheme (e.g., CDMA), and a protocol identifier module 856 may attempt to identify the protocol the signal of interest is associated with. The multiple access scheme identifier module 854 and protocol identifier module 856 may retrieve data from the static database 238 to aid in the identification of the access scheme and/or protocol. The symbol detector module 852 may pass data to the signal parameter and protocol module 858 which may store protocol and symbol information in addition to signal parameter information for signals of interest.

FIG. 9 illustrates a process flow of an embodiment method 900 for determining protocol data and symbol timing data. In an embodiment, the operations of method 900 may be performed by the processor 214 of a spectrum management device 802. In determination block 902 the processor 214 may determine whether two or more signals are detected. If two or more signals are not detected (i.e., determination block 902="No"), in determination block 902 the processor 214 may continue to determine whether two or more signals are detected. If two or more signals are detected (i.e., determination block 902="Yes"), in determination block 904 the processor 214 may determine whether the two or more signals are interrelated. In an embodiment, a mean correlation value of the spectral decomposition of each signal may indicate the two or more signals are interrelated. As an example, a mean correlation of each signal may generate a value between 0.0 and 1, and the processor 214 may compare the mean correlation value to a threshold, such as a threshold of 0.75. In such an example, a mean correlation value at or above the threshold may indicate the signals are interrelated while a mean correlation value below the threshold may indicate the signals are not interrelated and may be different signals. In an embodiment, the mean correlation value may be generated by running a full energy bandwidth correlation of each signal, measuring the values of signal transition for each signal, and for each signal transition running a spectral correlation between signals to generate the mean correlation value. If the signals are not interrelated (i.e., determination block 904="No"), the signals may be two or more different signals, and in block 907 processor 214 may measure the interference between the two or more signals. In an optional embodiment, in optional block 909 the processor 214 may generate a conflict alarm indicating the two or more different signals interfere. In an embodiment, the conflict alarm may be sent to the history database and/or a display. In determination block 902 the processor 214 may continue to determine whether two or more signals are detected. If the two signal are interrelated (i.e., determination block 904="Yes"), in block 905 the processor 214 may identify the two or more signals as a single signal. In block 906 the processor 214 may combine signal data for the two or more signals into a signal single entry in the history database. In determination block 908 the processor 214 may determine whether the signals mean averages. If the mean averages (i.e., determination block 908="Yes"), the processor 214 may identify the signal as having multiple channels in determination block 910. If the mean does not average (i.e., determination block 908="No") or after identifying the signal as having multiple

26

channels in determination block 910, in block 914 the processor 214 may determine and store protocol data for the signal. In block 916 the processor 214 may determine and store symbol timing data for the signal, and the method 900 may return to block 902.

FIG. 10 illustrates a process flow of an embodiment method 1000 for calculating signal degradation data. In an embodiment, the operations of method 1000 may be performed by the processor 214 of a spectrum management device 202. In block 1002 the processor may detect a signal. In block 1004 the processor 214 may match the signal to a signal in a static database. In block 1006 the processor 214 may determine noise figure parameters based on data in the static database 236 associated with the signal. As an example, the processor 214 may determine the noise figure of the signal based on parameters of a transmitter outputting the signal according to the static database 236. In block 1008 the processor 214 may determine hardware parameters associated with the signal in the static database 236. As an example, the processor 214 may determine hardware parameters such as antenna position, power settings, antenna type, orientation, azimuth, location, gain, and equivalent isotropically radiated power (EIRP) for the transmitter associated with the signal from the static database 236. In block 1010 processor 214 may determine environment parameters associated with the signal in the static database 236. As an example, the processor 214 may determine environment parameters such as rain, fog, and/or haze based on a delta correction factor table stored in the static database and a provided precipitation rate (e.g., mm/hr). In block 1012 the processor 214 may calculate and store signal degradation data for the detected signal based at least in part on the noise figure parameters, hardware parameters, and environmental parameters. As an example, based on the noise figure parameters, hardware parameters, and environmental parameters free space losses of the signal may be determined. In block 1014 the processor 214 may display the degradation data on a display 242 of the spectrum management device 202. In a further embodiment, the degradation data may be used with measured terrain data of geographic locations stored in the static database to perform pattern distortion, generate propagation and/or next neighbor interference models, determine interference variables, and perform best fit modeling to aide in signal and/or system optimization.

FIG. 11 illustrates a process flow of an embodiment method 1100 for displaying signal and protocol identification information. In an embodiment, the operations of method 1100 may be performed by a processor 214 of a spectrum management device 202. In block 1102 the processor 214 may compare the signal parameters and protocol data of an identified signal to signal parameters and protocol data in a history database 232. In an embodiment, a history database 232 may be a database storing signal parameters and protocol data for previously identified signals. In block 1104 the processor 214 may determine whether there is a match between the signal parameters and protocol data of the identified signal and the signal parameters and protocol data in the history database 232. If there is not a match (i.e., determination block 1104="No"), in block 1106 the processor 214 may store the signal parameters and protocol data as a new signal in the history database 232. If there is a match (i.e., determination block 1104="Yes"), in block 1108 the processor 214 may update the matching signal parameters and protocol data as needed in the history database 232.

In block 1110 the processor 214 may compare the signal parameters and protocol data of the identified signal to

27

signal parameters and protocol data in the signal characteristic listing 236. In determination block 1112 the processor 214 may determine whether the signal parameters and protocol data of the identified signal match any signal parameters and protocol data in the signal characteristic listing 236. If there is a match (i.e., determination block 1112="Yes"), in block 1114 the processor 214 may indicate a match in the history database and in block 1118 may display an indication of the signal identification and protocol on a display. If there is not a match (i.e., determination block 1112="No"), in block 1116 the processor 214 may display an indication that the signal is an unidentified signal. In this manner, the user may be notified a signal is present in the environment, but that the signal does not match to a signal in the characteristic listing.

FIG. 12A is a block diagram of a spectrum management device 1202 according to an embodiment. Spectrum management device 1202 is similar to spectrum management device 802 described above with reference to FIG. 8A, except that spectrum management device 1202 may include TDOA/FDOA module 1204 and modulation module 1206 enabling the spectrum management device 1202 to identify the modulation type employed by a signal of interest and calculate signal origins. The modulation module 1206 may enable the signal processor to determine the modulation applied to signal, such as frequency modulation (e.g., FSK, MSK, etc.) or phase modulation (e.g., BPSK, QPSK, QAM, etc.) as well as to demodulate the signal to identify payload data carried in the signal. The modulation module 1206 may use payload data 1221 from the characteristic listing to identify the data types carried in a signal. As examples, upon demodulating a portion of the signal the payload data may enable the processor 214 to determine whether voice data, video data, and/or text based data is present in the signal. The TDOA/FDOA module 1204 may enable the signal processor 214 to determine time difference of arrival for signals of interest and/or frequency difference of arrival for signals of interest. Using the TDOA/FDOA information estimates of the origin of a signal may be made and passed to a mapping module 1225 which may control the display 242 to output estimates of a position and/or direction of movement of a signal.

FIG. 12B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to an embodiment. Only those logical operations illustrated in FIG. 12B different from those described above with reference to FIG. 8B will be discussed. A magnitude squared 1252 operation may be performed on data from the symbol detector 852 to identify whether frequency or phase modulation is present in the signal. Phase modulated signals may be identified by the phase modulation 1254 processes and frequency modulated signals may be identified by the frequency modulation 1256 processes. The modulation information may be passed to a signal parameters, protocols, and modulation module 1258.

FIG. 13 illustrates a process flow of an embodiment method 1300 for estimating a signal origin based on a frequency difference of arrival. In an embodiment, the operations of method 1300 may be performed by a processor 214 of a spectrum management device 1202. In block 1302 the processor 214 may compute frequency arrivals and phase arrivals for multiple instances of an identified signal. In block 1304 the processor 214 may determine frequency difference of arrival for the identified signal based on the computed frequency difference and phase difference. In block 1306 the processor may compare the determined frequency difference of arrival for the identified signal to

28

data associated with known emitters in the characteristic listing to estimate an identified signal origin. In block 1308 the processor 214 may indicate the estimated identified signal origin on a display of the spectrum management device. As an example, the processor 214 may overlay the estimated origin on a map displayed by the spectrum management device.

FIG. 14 illustrates a process flow of an embodiment method for displaying an indication of an identified data type within a signal. In an embodiment, the operations of method 1400 may be performed by a processor 214 of a spectrum management device 1202. In block 1402 the processor 214 may determine the signal parameters for an identified signal of interest. In block 1404 the processor 214 may determine the modulation type for the signal of interest. In block 1406 the processor 214 may determine the protocol data for the signal of interest. In block 1408 the processor 214 may determine the symbol timing for the signal of interest. In block 1410 the processor 214 may select a payload scheme based on the determined signal parameters, modulation type, protocol data, and symbol timing. As an example, the payload scheme may indicate how data is transported in a signal. For example, data in over the air television broadcasts may be transported differently than data in cellular communications and the signal parameters, modulation type, protocol data, and symbol timing may identify the applicable payload scheme to apply to the signal. In block 1412 the processor 214 may apply the selected payload scheme to identify the data type or types within the signal of interest. In this manner, the processor 214 may determine what type of data is being transported in the signal, such as voice data, video data, and/or text based data. In block 1414 the processor may store the data type or types. In block 1416 the processor 214 may display an indication of the identified data types.

FIG. 15 illustrates a process flow of an embodiment method 1500 for determining modulation type, protocol data, and symbol timing data. Method 1500 is similar to method 900 described above with reference to FIG. 9, except that modulation type may also be determined. In an embodiment, the operations of method 1500 may be performed by a processor 214 of a spectrum management device 1202. In blocks 902, 904, 905, 906, 908, and 910 the processor 214 may perform operations of like numbered blocks of method 900 described above with reference to FIG. 9. In block 1502 the processor may determine and store a modulation type. As an example, a modulation type may be an indication that the signal is frequency modulated (e.g., FSK, MSK, etc.) or phase modulated (e.g., BPSK, QPSK, QAM, etc.). As discussed above, in block 914 the processor may determine and store protocol data and in block 916 the processor may determine and store timing data.

In an embodiment, based on signal detection, a time tracking module, such as a TDOA/FDOA module 1204, may track the frequency repetition interval at which the signal is changing. The frequency repetition interval may also be tracked for a burst signal. In an embodiment, the spectrum management device may measure the signal environment and set anchors based on information stored in the historic or static database about known transmitter sources and locations. In an embodiment, the phase information about a signal be extracted using a spectral decomposition correlation equation to measure the angle of arrival ("AOA") of the signal. In an embodiment, the processor of the spectrum management device may determine the received power as the Received Signal Strength ("RSS") and based on the AOA and RSS may measure the frequency difference of

arrival. In an embodiment, the frequency shift of the received signal may be measured and aggregated over time. In an embodiment, after an initial sample of a signal, known transmitted signals may be measured and compared to the RSS to determine frequency shift error. In an embodiment, the processor of the spectrum management device may compute a cross ambiguity function of aggregated changes in arrival time and frequency of arrival. In an additional embodiment, the processor of the spectrum management device may retrieve FFT data for a measured signal and aggregate the data to determine changes in time of arrival and frequency of arrival. In an embodiment, the signal components of change in frequency of arrival may be averaged through a Kalman filter with a weighted tap filter from 2 to 256 weights to remove measurement error such as noise, multipath interference, etc. In an embodiment, frequency difference of arrival techniques may be applied when either the emitter of the signal or the spectrum management device are moving or when then emitter of the signal and the spectrum management device are both stationary. When the emitter of the signal and the spectrum management device are both stationary the determination of the position of the emitter may be made when at least four known other known signal emitters positions are known and signal characteristics may be available. In an embodiment, a user may provide the four other known emitters and/or may use already in place known emitters, and may use the frequency, bandwidth, power, and distance values of the known emitters and their respective signals. In an embodiment, where the emitter of the signal or spectrum management device may be moving, frequency difference of arrival techniques may be performed using two known emitters.

FIG. 16 illustrates an embodiment method for tracking a signal origin. In an embodiment, the operations of method 1600 may be performed by a processor 214 of a spectrum management device 1202. In block 1602 the processor 214 may determine a time difference of arrival for a signal of interest. In block 1604 the processor 214 may determine a frequency difference of arrival for the signal of interest. As an example, the processor 214 may take the inverse of the time difference of arrival to determine the frequency difference of arrival of the signal of interest. In block 1606 the processor 214 may identify the location. As an example, the processor 214 may determine the location based on coordinates provided from a GPS receiver. In determination block 1608 the processor 214 may determine whether there are at least four known emitters present in the identified location. As an example, the processor 214 may compare the geographic coordinates for the identified location to a static database and/or historical database to determine whether at least four known signals are within an area associated with the geographic coordinates. If at least four known emitters are present (i.e., determination block 1608="Yes"), in block 1612 the processor 214 may collect and measure the RSS of the known emitters and the signal of interest. As an example, the processor 214 may use the frequency, bandwidth, power, and distance values of the known emitters and their respective signals and the signal of interest. If less than four known emitters are present (i.e., determination block 1608="No"), in block 1610 the processor 214 may measure the angle of arrival for the signal of interest and the known emitter. Using the RSS or angle or arrival, in block 1614 the processor 214 may measure the frequency shift and in block 1616 the processor 214 may obtain the cross ambiguity function. In determination block 1618 the processor 214 may determine whether the cross ambiguity function converges to a solution. If the cross ambiguity function does converge to a

solution (i.e., determination block 1618="Yes"), in block 1620 the processor 214 may aggregate the frequency shift data. In block 1622 the processor 214 may apply one or more filter to the aggregated data, such as a Kalman filter. Additionally, the processor 214 may apply equations, such as weighted least squares equations and maximum likelihood equations, and additional filters, such as a non-line-of-sight ("NLOS") filters to the aggregated data. In an embodiment, the cross ambiguity function may resolve the position of the emitter of the signal of interest to within 3 meters. If the cross ambiguity function does not converge to a solution (i.e., determination block 1618="No"), in block 1624 the processor 214 may determine the time difference of arrival for the signal and in block 1626 the processor 214 may aggregate the time shift data. Additionally, the processor may filter the data to reduce interference. Whether based on frequency difference of arrival or time difference of arrival, the aggregated and filtered data may indicate a position of the emitter of the signal of interest, and in block 1628 the processor 214 may output the tracking information for the position of the emitter of the signal of interest to a display of the spectrum management device and/or the historical database. In an additional embodiment, location of emitters, time and duration of transmission at a location may be stored in the history database such that historical information may be used to perform and predict movement of signal transmission. In a further embodiment, the environmental factors may be considered to further reduce the measured error and generate a more accurate measurement of the location of the emitter of the signal of interest.

The processor 214 of spectrum management devices 202, 802 and 1202 may be any programmable microprocessor, microcomputer or multiple processor chip or chips that can be configured by software instructions (applications) to perform a variety of functions, including the functions of the various embodiments described above. In some devices, multiple processors may be provided, such as one processor dedicated to wireless communication functions and one processor dedicated to running other applications. Typically, software applications may be stored in the internal memory 226 or 230 before they are accessed and loaded into the processor 214. The processor 214 may include internal memory sufficient to store the application software instructions. In many devices the internal memory may be a volatile or nonvolatile memory, such as flash memory, or a mixture of both. For the purposes of this description, a general reference to memory refers to memory accessible by the processor 214 including internal memory or removable memory plugged into the device and memory within the processor 214 itself.

The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skill in the art the order of steps in the foregoing embodiments may be performed in any order. Words such as "thereafter," "then," "next," etc. are not intended to limit the order of the steps; these words are simply used to guide the reader through the description of the methods. Further, any reference to claim elements in the singular, for example, using the articles "a," "an" or "the" is not to be construed as limiting the element to the singular.

The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of

both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

The hardware used to implement the various illustrative logics, logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but, in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Alternatively, some steps or methods may be performed by circuitry that is specific to a given function.

In one or more exemplary aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable medium or non-transitory processor-readable medium. The steps of a method or algorithm disclosed herein may be embodied in a processor-executable software module which may reside on a non-transitory computer-readable or processor-readable storage medium. Non-transitory computer-readable or processor-readable storage media may be any storage media that may be accessed by a computer or a processor. By way of example but not limitation, such non-transitory computer-readable or processor-readable media may include RAM, ROM, EEPROM, FLASH memory, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of non-transitory computer-readable and processor-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product. Identifying Devices in White Space.

The present invention provides for systems, methods, and apparatus solutions for device sensing in white space, which improves upon the prior art by identifying sources of signal emission by automatically detecting signals and creating unique signal profiles. Device sensing has an important function and applications in military and other intelligence

sectors, where identifying the emitter device is crucial for monitoring and surveillance, including specific emitter identification (SEI).

At least two key functions are provided by the present invention: signal isolation and device sensing. Signal Isolation according to the present invention is a process whereby a signal is detected, isolated through filtering and amplification, amongst other methods, and key characteristics extracted. Device Sensing according to the present invention is a process whereby the detected signals are matched to a device through comparison to device signal profiles and may include applying a confidence level and/or rating to the signal-profile matching. Further, device sensing covers technologies that permit storage of profile comparisons such that future matching can be done with increased efficiency and/or accuracy. The present invention systems, methods, and apparatus are constructed and configured functionally to identify any signal emitting device, including by way of example and not limitation, a radio, a cell phone, etc.

Regarding signal isolation, the following functions are included in the present invention: amplifying, filtering, detecting signals through energy detection, waveform-based, spectral correlation-based, radio identification-based, or matched filter method, identifying interference, identifying environmental baseline(s), and/or identify signal characteristics.

Regarding device sensing, the following functions are included in the present invention: using signal profiling and/or comparison with known database(s) and previously recorded profile(s), identifying the expected device or emitter, stating the level of confidence for the identification, and/or storing profiling and sensing information for improved algorithms and matching. In preferred embodiments of the present invention, the identification of the at least one signal emitting device is accurate to a predetermined degree of confidence between about 80 and about 95 percent, and more preferably between about 80 and about 100 percent. The confidence level or degree of confidence is based upon the amount of matching measured data compared with historical data and/or reference data for predetermined frequency and other characteristics.

The present invention provides for wireless signal-emitting device sensing in the white space based upon a measured signal, and considers the basis of license(s) provided in at least one reference database, preferably the federal communication commission (FCC) and/or other defined database including license listings. The methods include the steps of providing a device for measuring characteristics of signals from signal emitting devices in a spectrum associated with wireless communications, the characteristics of the measured data from the signal emitting devices including frequency, power, bandwidth, duration, modulation, and combinations thereof; making an assessment or categorization on analog and/or digital signal(s); determining the best fit based on frequency if the measured power spectrum is designated in historical and/or reference data, including but not limited to the FCC or other database(s) for select frequency ranges; determining analog or digital, based on power and sideband combined with frequency allocation; determining a TDM/FDM/CDM signal, based on duration and bandwidth; determining best modulation fit for the desired signal, if the bandwidth and duration match the signal database(s); adding modulation identification to the database; listing possible modulations with best percentage fit, based on the power, bandwidth, frequency, duration, database allocation, and combinations thereof; and identifying at least one signal emitting device from the composite

33

results of the foregoing steps. Additionally, the present invention provides that the phase measurement of the signal is calculated between the difference of the end frequency of the bandwidth and the peak center frequency and the start frequency of the bandwidth and the peak center frequency to get a better measurement of the sideband drop off rate of the signal to help determine the modulation of the signal.

In embodiments of the present invention, an apparatus is provided for automatically identifying devices in a spectrum, the apparatus including a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device. The characteristics of signals and measured data from the signal emitting devices include frequency, power, bandwidth, duration, modulation, and combinations thereof.

The present invention systems including at least one apparatus, wherein the at least one apparatus is operable for network-based communication with at least one server computer including a database, and/or with at least one other apparatus, but does not require a connection to the at least one server computer to be operable for identifying signal emitting devices; wherein each of the apparatus is operable for identifying signal emitting devices including: a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device.

Identifying Open Space in a Wireless Communication Spectrum

The present invention provides for systems, methods, and apparatus solutions for automatically identifying open space, including open space in the white space of a wireless communication spectrum. Importantly, the present invention identifies the open space as the space that is unused and/or seldom used (and identifies the owner of the licenses for the seldom used space, if applicable), including unlicensed spectrum, white space, guard bands, and combinations thereof. Method steps of the present invention include: automatically obtaining a listing or report of all frequencies in the frequency range; plotting a line and/or graph chart showing power and bandwidth activity; setting frequencies based on a frequency step and/or resolution so that only user-defined frequencies are plotted; generating files, such as by way of example and not limitation, .csv or .pdf files, showing average and/or aggregated values of power, bandwidth and frequency for each derived frequency step; and showing an activity report over time, over day vs. night, over frequency bands if more than one, in white space if requested, in Industrial, Scientific, and Medical (ISM) band or space if requested; and if frequency space is seldom in that area, then identify and list frequencies and license holders.

Additional steps include: automatically scanning the frequency span, wherein a default scan includes a frequency span between about 54 MHz and about 804 MHz; an ISM scan between about 900 MHz and about 2.5 GHz; an ISM scan between about 5 GHz and about 5.8 GHz; and/or a

34

frequency range based upon inputs provided by a user. Also, method steps include scanning for an allotted amount of time between a minimum of about 15 minutes up to about 30 days; preferably scanning for allotted times selected from the following: a minimum of about 15 minutes; about 30 minutes; about 1 hour increments; about 5 hour increments; about 10 hour increments; about 24 hours; about 1 day; and about up to 30 days; and combinations thereof. In preferred embodiments, if the apparatus is configured for automatically scanning for more than about 15 minutes, then the apparatus is preferably set for updating results, including updating graphs and/or reports for an approximately equal amount of time (e.g., every 15 minutes).

The systems, methods, and apparatus also provide for automatically calculating a percent activity associated with the identified open space on predetermined frequencies and/or ISM bands.

Automated Reports and Visualization of Analytics

Various reports for describing and illustrating with visualization the data and analysis of the device, system and method results from spectrum management activities include at least reports on power usage, RF survey, and variance.

The systems, methods, and devices of the various embodiments enable spectrum management by identifying, classifying, and cataloging signals of interest based on radio frequency measurements. In an embodiment, signals and the parameters of the signals may be identified and indications of available frequencies may be presented to a user. In another embodiment, the protocols of signals may also be identified. In a further embodiment, the modulation of signals, devices or device types emitting signals, data types carried by the signals, and estimated signal origins may be identified.

Referring again to the drawings, FIG. 17 is a schematic diagram illustrating an embodiment for scanning and finding open space. A plurality of nodes are in wireless or wired communication with a software defined radio, which receives information concerning open channels following real-time scanning and access to external database frequency information.

FIG. 18 is a diagram of an embodiment of the invention wherein software defined radio nodes are in wireless or wired communication with a master transmitter and device sensing master.

FIG. 19 is a process flow diagram of an embodiment method of temporally dividing up data into intervals for power usage analysis and comparison. The data intervals are initially set to seconds, minutes, hours, days and weeks, but can be adjusted to account for varying time periods (e.g., if an overall interval of data is only a week, the data interval divisions would not be weeks). In one embodiment, the interval slicing of data is used to produce power variance information and reports.

FIG. 20 is a flow diagram illustrating an embodiment wherein frequency to license matching occurs. In such an embodiment the center frequency and bandwidth criteria can be checked against a database to check for a license match. Both licensed and unlicensed bands can be checked against the frequencies, and, if necessary, non-correlating factors can be marked when a frequency is uncorrelated.

FIG. 21 is a flow diagram illustrating an embodiment method for reporting power usage information, including locational data, data broken down by time intervals, frequency and power usage information per band, average power distribution, propagation models, atmospheric fac-

35

tors, which is capable of being represented graphical, quantitatively, qualitatively, and overlaid onto a geographic or topographic map.

FIG. 22 is a flow diagram illustrating an embodiment method for creating frequency arrays. For each initialization, an embodiment of the invention will determine a center frequency, bandwidth, peak power, noise floor level, resolution bandwidth, power and date/time. Start and end frequencies are calculated using the bandwidth and center frequency and like frequencies are aggregated and sorted in order to produce a set of frequency arrays matching power measurements captured in each band.

FIG. 23 is a flow diagram illustrating an embodiment method for reframe and aggregating power when producing frequency arrays.

FIG. 24 is a flow diagram illustrating an embodiment method of reporting license expirations by accessing static or FCC databases.

FIG. 25 is a flow diagram illustrating an embodiment method of reporting frequency power use in graphical, chart, or report format, with the option of adding frequencies from FCC or other databases.

FIG. 26 is a flow diagram illustrating an embodiment method of connecting devices. After acquiring a GPS location, static and FCC databases are accessed to update license information, if available. A frequency scan will find open spaces and detect interferences and/or collisions. Based on the master device ID, set a random generated token to select channel from available channel model and continually transmit ID channel token. If node device reads ID, it will set itself to channel based on token and device will connect to master device. Master device will then set frequency and bandwidth channel. For each device connected to master, a frequency, bandwidth, and time slot in which to transmit is set. In one embodiment, these steps can be repeated until the max number of devices is connected. As new devices are connected, the device list is updated with channel model and the device is set as active. Disconnected devices are set as inactive. If collision occurs, update channel model and get new token channel. Active scans will search for new or lost devices and update devices list, channel model, and status accordingly. Channel model IDs are actively sent out for new or lost devices.

FIG. 27 is a flow diagram illustrating an embodiment method of addressing collisions.

FIG. 28 is a schematic diagram of an embodiment of the invention illustrating a virtualized computing network and a plurality of distributed devices. FIG. 28 is a schematic diagram of one embodiment of the present invention, illustrating components of a cloud-based computing system and network for distributed communication therewith by mobile communication devices. FIG. 28 illustrates an exemplary virtualized computing system for embodiments of the present invention loyalty and rewards platform. As illustrated in FIG. 28, a basic schematic of some of the key components of a virtualized computing (or cloud-based) system according to the present invention is shown. The system 2800 comprises at least one remote server computer 2810 with a processing unit 2811 and memory. The server 2810 is constructed, configured and coupled to enable communication over a network 2850. The server provides for user interconnection with the server over the network with the at least one apparatus as described hereinabove 2840 positioned remotely from the server. Furthermore, the system is operable for a multiplicity of devices or apparatus embodiments 2860, 2870 for example, in a client/server architecture, as shown. Alternatively, interconnection through the

36

network 2850 using the at least one device or apparatus for measuring signal emitting devices, each of the at least one apparatus is operable for network-based communication. Also, alternative architectures may be used instead of the client/server architecture. For example, a computer communications network, or other suitable architecture may be used. The network 2850 may be the Internet, an intranet, or any other network suitable for searching, obtaining, and/or using information and/or communications. The system of the present invention further includes an operating system 2812 installed and running on the at least one remote server 2810, enabling the server 2810 to communicate through network 2850 with the remote, distributed devices or apparatus embodiments as described hereinabove. The operating system may be any operating system known in the art that is suitable for network communication.

Certain modifications and improvements will occur to those skilled in the art upon a reading of the foregoing description. The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

The invention claimed is:

1. An apparatus for identifying open space in an electromagnetic spectrum comprising:

at least one processor; and

at least one memory;

wherein the apparatus is operable to receive signal data of at least one signal measured by a sensor; and wherein the apparatus is operable to calculate a percent activity associated with open space on predetermined frequencies, user-defined frequencies, and/or Industrial, Scientific, and Medical (ISM) bands based on the received signal data.

2. The apparatus of claim 1, wherein the open space is determined based on a dynamically adjustable interval.

3. The apparatus of claim 1, wherein identification of the open space is automated.

4. The apparatus of claim 1, wherein identification of the open space is based upon the predetermined frequencies, the user-defined frequencies, and/or the ISM bands.

5. The apparatus of claim 1, wherein identification of the open space occurs in near-real time or real time.

6. The apparatus of claim 1, wherein the apparatus is further configured to detect interference in the electromagnetic spectrum based on the signal data.

7. An apparatus for identifying open space in an electromagnetic spectrum comprising:

at least one processor; and

at least one memory;

wherein the apparatus receives measured signal data of at least one signal;

wherein the measured signal data includes energy measurements of the at least one signal, at least one parameter of the at least one signal, and/or at least one change for the at least one signal; and

wherein the apparatus is operable to calculate a percent activity associated with at least one open space on predetermined frequencies, user-defined frequencies, and/or Industrial, Scientific, and Medical (ISM) bands based on the measured signal data.

8. The apparatus of claim 7, wherein the apparatus is further configured to detect interference in the electromagnetic spectrum based on the measured signal data.

37

9. The apparatus of claim 7, wherein identification of the at least one open space occurs in near-real time or real time and/or is automated.

10. The apparatus of claim 7, wherein the at least one open space is determined based on a dynamically adjustable interval.

11. A method for identifying open space in an electromagnetic spectrum comprising:

at least one apparatus including a processor and a memory receiving signal data from at least one remote sensor over a network or the at least one apparatus creating the signal data using at least one integrated sensor; and calculating a percent activity associated with at least one open space on predetermined frequencies, user-defined frequencies, and/or Industrial, Scientific, and Medical (ISM) bands based on the signal data;

wherein the signal data includes energy measurements of at least one signal, at least one parameter of the at least one signal, and/or at least one change for the at least one signal.

12. The method of claim 11, wherein the signal data is generated by the at least one apparatus sensing and measuring signals transmitted from at least one signal emitting device in the electromagnetic spectrum.

38

13. The method of claim 11 wherein identification of the at least one open space is automated.

14. The method of claim 11 wherein the at least one open space is determined based on a dynamically adjustable interval.

15. The method of claim 11, wherein identification of the at least one open space is based upon the predetermined frequencies, the user-defined frequencies, and/or the ISM bands.

16. The method of claim 11, wherein identification of the at least one open space is displayed or visually indicated, in whole or in part, by the at least one apparatus.

17. The method of claim 11, wherein identification of the at least one open space occurs in near-real time or real time.

18. The method of claim 11, further comprising detecting interference in the electromagnetic spectrum based on the signal data.

19. The method of claim 11, wherein the at least one open space includes unused space or unlicensed spectrum.

20. The method of claim 11, wherein the at least one open space includes white space.

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